TENDER ID : 2024_MES_688696_1
PROVN OF HRDC BLDG AT INF SCHOOL, MHOW
CONTENTS

Srl No.	Description	Page Nos	
		From	**To**
1.	Contents	1	1
2.	Forwarding letter including instructions to tenderers	2	5
3.	Notice of tender (IAFW 2162) including Appx 'A' to Notice of tender and Intigrity Pact	6	20
4.	Lump Sum Tender and contract for works required IAFW 2159(Revised-1947) including BoQ (Quotation Bid)	21	66
5.	General Conditions of contracts IAFW-2249 (1989 Print) including Errata and Amendments to IAFW-2249	67	67
6.	*Schedule of minimum wages	68	68
7.	Special conditions including Appx 'A' & 'B' and Yardstick.	69	86
8.	Particular specifications including Annexure 'A'	87	184
9.	Errata/Amendments to tender documents		
10.	Relevant correspondence		
11.	Acceptance letter		
		Total pages	

Drawing –____Sheets

_____ _____

(Signature of Contractor)

Dy Dir / Asst Dir (Contracts)
For Accepting Officer

Tele No: 0761-2603027
Fax No : 0761-2601027

CHIEF ENGINEER JABALPUR ZONE
MILITARY ENGINEER SERVICES
RIDGE ROAD
JABALPUR CANTT – 482001

752955/A/ 07 /E8 30 Jan 2025

M/S……
……

PROVN OF HRDC BLDG AT INF SCHOOL, MHOW

Dear Sir,

1. Tender documents in respect of above work are uploaded on the site www.defproc.gov.in. The tender is based on single stage two cover cover e-tendering system. The contents of Cover I & Cover II are specified in NOTICE OF TENDER.

2. Bids will be received online by ACCEPTING OFFICER upto the date and time mentioned in the **NOTICE INVITING TENDER (NIT).** No tender/bid will be received in physical form and any tender/bid received in such manner will be treated as non bona fide.

3. Bid will be opened online on a due date fixed for opening as per critical dates given in the portal. Cover 1 will be opened first. Tenderers may see the result of the opening of Cover 1 on the tender portal. Date of opening of Cover 2 shall be decided after technical evaluation of Cover 1 keeping in view the various eligibility criteria given in the NIT and the same will be intimated to the tenderers while uploading the technical evaluation on the tender portal.

4. Your attention is also drawn to instruction on filling and submission of tender attached herewith. You may forward your points on tender documents through e-mail and/or depute your technical representative for discussion on tender/drawings and to clarify doubts, if any, before the bid submission starts. You are requested not to write piece meal points and forward your points duly consolidated in one go. You may also attendthe pre bid meeting on the date given in the tender.

5. Unenlisted contractors are required to submit the scanned copies (in pdf file) ofdocuments required as per eligibility criteria mentioned in instructions for filling the tenderdocuments and Appendix ‘A’ to NIT alongwith EARNEST MONEY DEPOSIT (EMD) andtender fee on procurement portal mentioned above and submit the physical documentsin the office of Chief Engineer Delhi Zone within time limit specified in NIT. Inadequacy/deficiency of documents shall make the bid liable for rejection resulting in disqualificationfor opening of finance bid.

6. Enlisted contractors of MES shall submit the scanned copies (pdf file) of enlistment letter, tender fee and such other documents as mentioned in Appx ‘A’ to NIT on e-procurement portal and submit physical documents in the office of Chief Engineer Jabalpur before date & time fixed for this purpose.

7. The contractor must ensure that the tender/bid is uploaded on the tender portal using the authorised DSC in time as the Accepting Officer will take no cognizance of any quotations/offer received in any other electronic or physical form like email/fax/by hand/through post from tenderer/bidder.

FORWARDING LETTER (CONTD…)

8. Keeping in view delays due to system failure or other communication related failures, it is suggested that tender/bid may be uploaded sufficiently in advance of the last due date and time fixed on the portal.

9. General Conditions of Contracts (IAFW-2249) (1989 Print) and errata and amendments thereto, Schedule of Minimum Fair Wages and MES SSR (Part-I and Part-II) are not enclosed with these documents. These are available for perusal in the Office of GE/AGE concerned and this office. Tenderer is also requested to keep the copy of these documents with him as same are available in the market.

10. **PERFORMANCE SECURITY (REFER CONDITION 19 OF IAFW 2249 GENERAL CONDITIONS OF CONTRACT):** After acceptance of the Tender, the contractor will be required to lodge with the Accepting Officer PERFORMANCE SECURITY DEPOSIT @ **05%** of contract amount. The amount is required to be lodged within 28 (Twenty eight) days of the receipt by the contractor of notification of acceptance of tender/bid, failing which action as stipulated in Condition 19 of GCC (i.e IAFW 2249) shall be taken.

11. ANY TENDERER, WHICH PROPOSES ALTERATIONS TO ANY OF THE CONDITIONS, SPECIFICATIONS LAID DOWN IN THE TENDER DOCUMENTS OR ANY NEW CONDITION, WHATSOEVER, IS LIABLE TO BE REJECTED.

_____ _____

(Signature of Contractor) Dy Dir/ Asst Dir (Contracts)
For Accepting Officer

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER

1. EARNEST MONEY DEPOSIT (EMD)

Contractor(s) who are not enlisted with MES/who are enlisted but have not executed the Standing Security Bond shall submit Earnest Money Deposit as detailed in Notice of Tender in one of the following forms, along with their tender/bid:-

(a) Deposit at Call Receipt from a Scheduled Bank in favour of GE/AGE(I) concerned.

(b) Receipted Treasury Challan, the amount being credited to the Revenue Deposit of **GE(MIS) Mhow.**

(c) Insurance Security bond.

It is advisable that Earnest Money is deposited in the form of deposit call receipt from an approved Schedule Bank for easy refund. In case the tenderer/bidder wants to lodge 'Earnest Money Deposit' in any other form allowed by MES, a confirmation about its acceptability will be obtained from the Accepting Officer well in advance of the bid submission end date and time. Earnest Money Deposit shall be submitted in the name of concerned GE/AGE(I).

NOTES:- Earnest Money Deposit (EMD) in the form of cheque/Bank Guarantee etc will not be accepted. **Non-Submission of Earnest Money Deposit (EMD) (scanned copy alongwith Technical Bid & hard copy before the date & time fixed for opening of BOQ) will render the bid disqualified for opening of Cover -II (finance bid).**

2. GENERAL INSTRUCTIONS FOR COMPLIANCE

2.1 The bids received only in the electronic form will be considered. All bids shall be submitted on defproc.gov.in portal. Documents should be scanned and forwarded in 'pdf' format and 'xls' form as indicated. Bids shall be uploaded on 'defproc.gov.in' portal on or before the bid closing date mentioned in the tender. No tender/bid in any other electronic or physical form like date email/fax/by hand/through post will be considered.

2.2 The bid shall be DIGITALLY signed using authorised DSC. All pages of tender documents, drawings, corrections/alterations shall be signed/initialed by the lowest bidder after acceptance of tender for making original and CTC of the contract.

2.3 Digital signatory of the bid/tender shall ensure that he is competent to bind the contractor (through partnership deed, general power of attorney or Memorandum and Article of Association of the Company) in all the matters pertaining to the contract with Union of India including arbitration clause. A scanned copy of the documents in confirmation of such authority shall be attached with the tender/bid in 'pdf' form in Cover 1. It shall be ensured that power of attorney shall be executed in accordance with the constitution of the company as laid down in its Memorandum & Article of Association.

2.4 Drawings, if issued in physical form, must be returned duly initialled by the tenderer/bidder in separate envelope indicating his name and address.

2.5 The tender shall be signed, dated and witnessed at all places provided for in the documents after acceptance. All corrections shall be initialled. The Contractor shall initial every page of tender and shall sign all drawings forming part of the tender. Any tender/bid, which proposes alterations to any of the conditions whatsoever, is liable to be rejected.

3. BOQ:-

3.1 The tenderer shall quote his rates on the BOQ EXCEL FILE only as per guide line of e-procure web portal. No alteration to the format will be accepted and such bid will be disqualified.

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER (CONTD...)

3.2. In case any tenderer wishes to revise/modify the rates quoted in the BOQ file, he can do so only in the BOQ files before uploading the tender through **https://defproc.gov.in** site only before closing date & time.

3.3. After the uploading of tender, Department may upload the errata/ amendment through corrigendum. The tenderer/bidder should submit their offer considering the errata/amendment carried out through corrigendum issued from time to time.

3.4. While uploading the bid, the tenderers/bidders should specifically check whether any **revised BOQ** has been uploaded by department through corrigendum prior to **Bid submission start date**. Tenderers/bidders attention is specifically drawn to the fact that they should submit their offer on revised BOQ only. In case any tenderer/bidder submits offer on pre-revised BOQ in lieu of Revised BOQ, **it will be considered as a willful negligence by the tenderer/bidder and quotation shall be considered non-bonafide.**

4. Tenders/bidders who uploaded their priced tenders/bids and are desirous of being present at the time of opening of the tenders/bids, may do so at the appointed time.

5. Hard copies of all above documents shall be sent by the contractor to the Tender issuing authority well in advance to be received before the date & time fixed for the same.

6. The contractor shall employ Indian Nationals after verifying their antecedents and loyalty. Attention is also drawn to relevant provisions in Special Condition referred here-in-after and also Conditions 24 & 25 of IAFW-2249 (General Conditions of Contract).

_____ _____

(Signature of Contractor) Dy Dir / Asst Dir (Contracts)
For Accepting Officer

MILITARY ENGINEER SERVICES
NOTICE INVITING TENDER (NIT)

1. A tender is invited for the work as mentioned in Appendix 'A' to this NOTICE INVITING TENDER (NIT).

2. The work is estimated to cost as indicated in aforesaid Appendix 'A'. This estimate, however, is not a guarantee and is merely given as a rough guide and if the work costs more or less, a tenderer/bidder will have no claim on that account.

3. The work is to be completed within the period as indicated in aforesaid Appendix 'A' in accordance with the phasing, if any, indicated in the tender from the date of handing over site, which will be on or about two weeks after the date of acceptance of tender.

4. Contractors whose names are on the MES approved list and within whose financial category the estimated amount would fall and unenlisted contractors may submit tender/bid subject to other criteria mentioned in Appendix A. Not more than one tender shall be submitted/uploaded by one contractor/ firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same contract as separate competitors. Two firms shall be deemed to have business dealing if any of the partners/proprietor/director is common among both of them. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

5. The Office of Chief Engineer Jabalpur will be the Accepting Officer here-in-after referred to as such for the purpose of this contract.

6. Not more than one tender/bid shall be submitted/uploaded by one bidder firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same tender as separate competitors. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

7. The Technical Bid and Financial Bid (Cover-1 and Cover-2) shall be uploaded by the contractor on or before the date & time mentioned in **NIT**. A scanned copy of DD with enlistment details and other documents as specified in Appendix A shall be uploaded as Cover-1 (Technical bid) of the tender on e-tendering portal. DD is refundable in case the contractor is not considered eligible in technical evaluation of Cover 1 resulting in non-opening of Cover1. The applicant contractor shall bear the cost of bank charges for procuring and en-cashing the DD including revalidation of DDs and shall not have any claim from Government whatsoever on this account.

8. Tender form and conditions of contract and other necessary documents shall be available on website defproc.gov.in for download and shall form part of contract agreement in case the tender/bid is accepted.

9. In case of MES enlisted contractor who has not executed the Standing Security Bond and un enlisted contractor, the Cover-I shall be accompanied by Earnest Money for the amount mentioned in Appendix 'A' in the form of deposit at call receipt in favour of concerned CCE/GE/GE (I)/AGE (I) (see Appendix 'A') by a Scheduled Bank or in received treasury Challan the amount being credited to the revenue deposit of the concerned CCE/GE/GE(I)/AGE(I) (see Appendix 'A'). The CCE/GE/GE (I)/AGE (I) will return the Earnest Money, wherever applicable, to all unsuccessful tenderers/bidders by endorsing an authority on the deposit at call receipt for it’s refund, on receipt of intimation from the Accepting Officer to do that.

NIT (CONTD...)

10. In case of successful contractor i.e. the lowest contractor having submitted EMD, he shall have the option of converting the EMD instrument into part of the Performance Security to be deposited by him within 28 days from the receipt of intimation of acceptance of tenderfrom Accepting Officer.

11. Sample of materials and stores to be supplied by the contractor will also be available for inspection by the bidder at the office of concerned **GE(MIS) Mhow** during working hours. The bidder is advised to visit the site of work by making prior appointment with **GE(MIS) Mhow**, who is the Executing Agency of the work (see Appendix 'A'). The bidder shall be deemed to have full knowledge of all relevant documents, samples, site etc whether he has inspected them or not.

12. Any bid which proposes any alteration to any of the conditions laid down or proposes any other new condition whatsoever, is liable to be rejected.

13. The uploading of bid by a bidder implies that bidder has read this notice and the conditions of contract and has made himself aware of the scope and specification of work to be done and of the conditions and rates at which stores (as applicable) etc will be issued to him and local conditions and other factors having bearing on the execution of the work.

14. The bidder must be in possession of a copy of the MES Schedule (SSR) (Part-I & Part-II of latest edition) including amendments and errata thereto.

15. Accepting Officer does not bind himself to accept the lowest or any tender/bid or to give any reason for not doing so.

16. The Accepting Officer reserves the right to accept a tender submitted by a Public Undertaking/Small & Medium Enterprises (SMEs), giving a price preference/purchase preference over other tender(s)/bids which may be lower, as are admissible under the Government Policy. No claim for any compensation or otherwise shall be admissible for such tenderer/bidder whose tender/bid is rejected.

17. The **Notice Inviting Tender (NIT)** including Appendix 'A' and Annexures thereto, if any, shall form part of the contract agreement.

_____ _____

(Signature of Contractor) Dy Dir/ Asst Dir (Contracts)
For Accepting Officer

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT)

1.	Name of work	**PROVN OF HRDC BLDG AT INF SCHOOL, MHOW** **TENDER ID : 2024_MES_688696_1**
2	Estimated Cost	**Rs. 510.525 Lakhs** (At par Market)
3	Period of completion	84 weeks
4	Cost of tender documents	**Rs. 3000.00** in the shape of DD / Bankers cheque from any schedule Bank in favour of **GE(MIS) Mhow** and payable at **Mhow**. (Note : In case of retendering, the contractor who had quoted in the previous call is not required to submit the cost of tender.)
5	Website / portal address	www.defproc.gov.in
6	Type of Contract	The tender shall be based on drawing and specifications (IAFW – 2159 and GCC (IAFW-2249) with Sch 'A'(list of items of work) to be priced by MES. The Contractor is required to quote the lump sum amounts for parts of schedule 'A' and quote rates against items of other parts of schedule 'A'.
7	Information Details	
	(a) Bid Publish date	Please refer defproc portal for critical dates.
	(b) Bid submission start date	
	(c) Bid submission end date	
	(d) Date / time for opening of bid (Cover-1)	
8	Eligibility Criteria	
	(A) For MES enlisted Contractors	Contractor shall be enlisted with MES in **Class ' A '** and above and category **'a (i) '**subject to satisfactory remarks wrt performance in respect of works in hand as reflected in Work Load return (WLR) or any other report circulated by competent engineer authority .
	(B) For other contractors	(i) Contractor not enlisted with MES should meet the enlistment criteria of **Class ' A ' & category a(i)** contractor with regard to satisfactorily completion of requisite value works with Central/State Government/ Central/ State PSUs/ AWHO/ AFNHB/ CGEWHO / DGMAP, annual turnover, bank solvency, working capital and other requirements given in Para 1.4 & 1.5 of Section 1 of MES Manual of Contracts 2020 as available in all MES formations as well as MES website **(www.mes.gov.in).** (ii) Not carrying adverse remarks in Work Load Report (WLR) or any other similar report circulated by any competent authority, if already working in MES. (iii) Not suspended/debarred/blacklisted (either permanently or temporarily) from participating in any bid or for business dealings by any Central/State Government Department or any Central/State Government PSU or any Autonomous Bodyunder Central/State Government or any Local Body as on the bid submission end date. (iv) Details of work completed and under progress in MES be submitted in following format. SN | CA No. & Name of Work | Value of CA | Date of Commencement | Date of Completion | Extended date of Completion Unenlisted Contractors who have secured two works in MES should get themselves registered in appropriate designated class with any registering authority, else the firm will not be eligible for participation in the tender unless until the firm is enlisted with the MES.
	(C) For All contractors	Contractor will not be allowed to execute the work by subletting or through power of attorney holder on his behalf to a third party/ another firm **except Sons / Daughters/ Spouse of Proprietor/ Partner/ Director and firm's own employees, Director, Project Manager**.This shall be subject to certain conditions which will be prescribed in the NIT forming part of the tender documents.
9	Tender issuing and Accepting Officer	Chief Engineer Jabalpur Zone 482001
10	Executing agency	**GE(MIS) Mhow**
11	Earnest Money	Rs. 5,80,300.00 in favour of **GE(MIS) Mhow**

APPENDIX 'A' TO NOTICE INVITING TENDER …Contd

NOTES:-

1. In case after opening of Cover-1, the number of MES enlisted contractors of eligible class as well as eligible un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT is less than 7 (Seven), applications in respect of MES contractors of one class or two classes (in case of remote and difficult areas to be decided as per list circulated by CE command/ADG) below the eligible class shall also be considered subject to fulfilment of other eligibility criteria given in the NIT. Therefore MES contractor's one class below (two classes below in case of remote and difficult areas) may also bid for this tender. Such contractors (contractors of one/two classes below the eligible class) shall not be considered in case their present residual work in hand is more than **FIVE TIMES** their present tendering limit. However in case such contractors fulfil the criteria of upgradation to the stipulated eligible class based on past experience of completed works (individual work experience and/or average annual turnover, as applicable) and financial soundness (solvency/financial soundness and working capital), the ceiling of present residual work will not apply and they will be considered for issue of tender. Such bidders shall upload in their Cover-1 bid details related to residual work in hand like details of works in hand showing names of work, names of Accepting Officers, contract amounts, dates of commencement and completion (stipulated) and progress as on bid submission date. Such contractors, if claim to fulfil the criteria of upgradation shall also upload the requisite information/documents in support of upgradation. These details shall be verified by the Tender Issuing Authority from concerned formations in case bids of such contractors are considered for evaluation.

2. In case after opening of Cover 1, the number of MES enlisted contractors of eligible class as well um-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT, are 7 (Seven) or more, applications of only those one class below the eligible class bidders shall be considered, who have previously completed similar works satisfactorily and are meeting the criteria of up gradation in respect of past experience of completed works (individual work experience and/or average annual turnover as applicable) and financial soundness (solvency/financial soundness and working capital) as per details given in Manual on Contracts. Therefore such contractors shall upload the requisite information/ documents in the Cover-1.

3. Unenlisted contractor shall be considered provided he meets the criteria. Foreign firms shall not be eligible for this tender. However Indian firms having foreign national/Indian nationals staying abroad/ Indian national having taken foreign citizenship, as director (s) shall be considered subject to security clearance from the concerned authorities.

4. Contractors enlisted with MES will upload following documents in Cover-1 for checking eligibility:-

(a) Application for tender on Firm's letterhead.
(b) Enlistment letter issued by the Registering Authority duly renewed for the cycle period of vogue.
(c) Scanned copy of DD/Bankers Cheque towards cost of tender and EMD instrument is case SSD bond is not signed at the time of registration.
(d) Scanned copy of EPF code No
(e) Goods &Service tax registration No
(f) Scanned copy of IP duly signed by bidder on each page.

APPENDIX 'A' TO NOTICE INVITING TENDER …Contd

5. Contractors not enlisted with MES will be required to upload following documents in Cover 1 for checking eligibility:-

(a) Application for tender on Firm's letterhead.

(b) Scanned copy of DD/Bankers cheque toward cost of tender and Earnest Money Deposit (EMD) instrument.

(c) Copy of Police Verification Report/Police Clearance Certificate/Character Certificate from the Police Authority of the area where the registered office of the firm is located/notarized copy of valid passport of Proprietor/each Partner/each Director.

(d) All documents required for enlistment in MES for the class mentioned in Para 8 (b) above as per Para 1.5 of Section-I of MES Manual on Contracts 2020.

(e) Details of works being executed in MES if any.

(f) Any other document required as described in this Appendix.

6. Tenders not accompanied by scanned copies of requisite DD/Bankers Cheque towards cost of tender and earnest money (as applicable) in Cover-1 shall not be considered for validation of 'T' bid and their Financial bids will not be opened.

7. Contractors should ensure that their original physical DDs and Earnest Money Deposit (EMD) instruments (as applicable) reach the office of Accepting Officer within 07 (Seven) days of bid submission end date, failing which following action shall be taken:-

(a) In case of tenders from an enlisted contractor of MES, where scanned copies of requisite DD/ Bankers Cheque towards cost of tender have been uploaded in Cover-1 but physical copies are not received within the stipulated period, their financial bids (Cover-2) will be opened. However non-submission of physical copies of cost of tender shall be considered as wilful negligence of the tenderer with ulterior motives and such tenderer shall be banned from bidding for a period of six months commencing from the date of opening of Financial Bid (Cover-2).

(b) In case of tenders from unenlisted contractor, where scanned copies of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover-1 but physical copies are not received within the stipulated period, their financial bids (Cover-2) will not be opened. Name of such contractors along with complete address shall be circulated for not opening of their bids for a period of six months commending from the date of opening of Financial bid (Cover-2).

(c) In case of tenders from enlisted and unenlisted contractors, where scanned copies of instruments for Earnest Money Deposit (as applicable) have been uploaded in Cover-1 but the same are not received in physical form with stipulated period, such tenders shall not qualify for opening of financial bid (Cover-2).

8. Contractor will not be allowed to execute the work by subletting or through power of attorney to a third party/another firm on his behalf. However a contractor can execute the work through power of attorney to sons/daughters/spouse of Proprietor/Partner/Director and firm's own employees, director, project manager **provided they are not having a separate enlisted firm in MES in their name as Proprietor/ Partner/Director**.

9. After opening of Cover 1 and during its technical evaluation, in case any deficiency is noticed in the documents required to be uploaded by the tenderers as per NIT, communication in the form of e-mail/SMS/Speed Post etc shall be sent to the contractor to rectify the deficiency within a period of seven days from date of communication failing which their financial bid (Cover 2) shall not be opened and contractor shall not have any claim on the same.

APPENDIX 'A' TO NOTICE INVITING TENDER …Contd

10. Invitation for e-tender does not constitute any guarantee for validation of Technical bid and subsequent opening of financial bid of any applicant/bidder merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the Technical bid and not to open the financial bid of any applicant/bidder. Technical bid validation shall be decided by the Accepting Officer based on eligibility of the firm as per criteria given in this Appendix. Tenderer/bidder will be informed regarding non-validation of his Technical bid assigning reasons therefore through tender evaluation report which shall be uploaded on the website. Such tenderer, if desires, may appeal to the next higher Engineer Authority (NHEA) viz HQ, Chief Engineer, Central Command, Lucknow on email id **jtdgcontceengrll-mes@nic.in** with a copy to Accepting officer on email id **cezj2-mes@nic.in** before the scheduled date of opening of Cover 2. NHEA shall decide the matter within a period of seven working days from the date of receipt of appeal. The decision of the NHEA shall be final and binding. The tenderer/bidder shall not be entitled for any compensation whatsoever for rejection of his bid.

11. In case an unenlisted contractor is already executing works in MES, he shall notbe considered eligible for the subject tender if the total value of such works is more than two times the tendering limit of the MES Class of contractor for which it is eligible. For this purpose, details of the works being executed by such a contractor shall be uploaded in the Cover -1 of the bid and shall be checked/verified by the Accepting Officer.

12. In case the BOQ is revised through the corrigendum and the bidder has failed to quote on revised BOQ (ie he has quoted on pre revised BOQ), such bid shall be treated as willful negligence by the bidder and his quotation shall be considered non-bonafideIn such cases the lowest tender shall be determined from amongst the valid/bonafide bids only. Accepting Officer may decide whether to re-tender or consider the lowest bonafide tender for acceptance.

13. Revoking the offer or revising the rates upward / offering voluntary reduction by the lowest tenderer after opening of Cover 2 shall be considered as a willful default.Forthis default a penalty of an amount equal to Earnest Money shall be levied. In case of an unenlisted tenderer, Earnest Money deposited by him shall be forfeited. In case of MES enlisted tenderer having deposited the Standing Security Bond, an amount equal to the earnest money stipulated in the NIT, shall be notified to the tenderer for depositing through MRO and consideration of such tenderer in tender evaluation for future works shall remain suspended till the aforementioned amount is deposited in the Government Treasury. No other disciplinary/administrative action shall be taken against such tenderers. In such a situation, the next lowest offer shall not be considered for acceptance. Instead, retendering shall be resorted to in a transparent and fair manner and the defaulting tenderer and his related firm if any, shall not be eligible for this tender in second call or subsequent calls.

14. Tender to related firms shall not be issued simultaneously. Firms shall be termed as related if Proprietor/one or more Partners/Directors are common. Decision ofAccepting Officer on issue/deny the tender to any one of the related firms shall be final and binding.

15. The Integrity pact (IP) duly signed by officer on behalf of Accepting Officer has been uploaded alongwith NIT as per ‘Annexure “I” to Appendix ‘A’. Scanned copy of IP (Integrity Pact) duly signed on each page by the bidder(s) shall be uploaded as part of technical bid (Cover-I) and original IP duly signed on each page shall be forwared by post alongwith demand draft. **Bid of all bidder(s) who do not upload signed copy of IP along with technical bid, (cover-I) shall be rejected.**

_____ _____

(Signature of Contractor)

Dy Dir (C)/ Asst Dir (Contracts)
For Accepting Officer

752955/ A/07/E8 dated 28 Jan 2024

Military Engineer Serivices
Headquarters
Chief Engineer Jabalpur Zone
Ridge Road Jabalpur -482001

To be signed by the bidder and same signatory competent / authorized to sign the relevant contract on behalf of MES.

INTEGRITY AGREEMENT

This Integrity Agreement is made at ……on this ……. day of ……..20…….

BETWEEN

President of India represented through Chief Engineer Jabalpur Zone, Military Engineer Services, HQ Chief Engineer Jabalpur Zone Ridge Road Jabalpur Pin- 482001.

(Here-in-After referred as **'Principal/Owner',** which expression shall unless repugnant to the meaning or context hereof include its successors and permitted assigns)

AND

M/s……

(Name and Address of the Bidder / Contractor)through …….....

…….(Details of duly authorized signatory) (Here in after referred to as the **"Bidder/Contractor"** and which expression shall unless repugnant to the meaning or context hereof include its successors and permitted assigns)

Preamble

WHEREAS the Principal / Owner has floated the Tender (NIT No. 752955/ A/07/E8 Dt. 28 Jan 2025) for **PROVN OF HRDC BLDG AT INF SCHOOL, MHOW.**

(hereinafter referred to as "Tender/Bid") and intends to award, under laid down organizational procedure, contract for **PROVN OF HRDC BLDG AT INF SCHOOL, MHOW**, Herein after referred to as the "Contract"

AND WHEREAS the Principal/Owner values full compliance with all relevant laws of the land, rules, regulations, economic use of resources and of fairness/transparency in its relation with its Bidder(s) and contractor(s).

AND WHEREAS to meet the purpose aforesaid both the parties have agreed to enter into this Integrity Agreement (hereinafter referred to as "Integrity Pact" or "Pact"), the terms and conditions of which shall also be read as integral part and parcel of the Tender/Bid documents and Contract between the parties.

NOW, THEREFORE, in consideration of mutual covenants contained in this Pact, the parties hereby agree as follows and this Pact witnesses as under

1. **Commitment of the Principal/Owner**

(a) The Principal/ Owner commits itself to take all measures necessary to prevent corruption and to observe the following principles;

(i) No employee of the Principal/Owner, personally or through any of his/her family members, will in connection with the Tender, or the execution of the Contract, demand, take a promise for or accept, for self or third person, any material or immaterial benefit which the person is not legally entitled to.

(ii) The Principal/Owner will, during the Tender process, treat all Bidder(s) with equity and reason. The Principal/Owner will, in particular, before and during the Tender process, provide to all Bidder(s) the same information and will not provide to any Bidder(s) confidential / additional information through which the Bidder(s) could obtain and advantage in relation to the Tender process or the Contract execution.

(iii) The Principal/Owner shall endeavor to exclude from the Tender process any person, whose conduct in the past has been of biased nature.

(b) If the Principal/Owner obtains information on the conduct of any of its employees which is a criminal offence under the Indian penal code (IPC)/Prevention of Corruption Act, 1988 (PC Act) or is in violation of the principles herein mentioned or if there be a substantive suspicion in this regard, the Principal/Owner will inform the Chief Vigilance Officer and in addition can also initiate disciplinary actions as per its internal laid down policies and procedures.

2. Commitment of the Bidder(s)/Contractor(s)

(a) It is required that each Bidder/Contractor (including their respective officers, employees and agents) adhere to the highest ethical standards, and report to the Government / Department all suspected acts of fraud or corruption or Coercion or Collusion of which it has knowledge or becomes aware, during the tendering process and throughout the negotiation or award of a contract.

(b) The Bidder(s)/Contractor(s) commits himself to take all measures necessary to prevent corruption. He commits himself to observe the following principles during his participation in the Tender process and during the Contract execution;

(i) The Bidder(s)/Contractor(s) will not, directly or through any other person or firm, offer, promise or give to any of the Principal/Owner's employees involved in the Tender process or execution of the Contract or to any third person any material or other benefit which he/she is not legally entitled to, in order to obtain in exchange any advantage of any kind whatsoever during the Tender process or during the execution of the Contract.

(ii) The Bidder(s)/Contractor(s) will not enter with other Bidder(s) into any undisclosed agreement or understanding, whether formal or informal. This applies in particular to prices, specifications, certifications, subsidiary contracts, submission or non-submission of bids or any other actions to restrict competitiveness or to cartelize in the bidding process.

(iii) The Bidder(s)/Contractor(s) will not commit any offence under the relevant IPC/PC Act. Further the Bidder(s)/Contract(s) will not use improperly, (for the purpose of competition or personal gain), or pass on to others, any information or documents provided by the Principal/Owner as part of the business relationship, regarding plans, technical proposals and business details, including information contained or transmitted electronically.

(iv) The Bidder(s)/Contractor(s) of foreign origin shall disclose the names and addresses of agents/ representatives in India, if any. Similarly Bidder(s)/Contractor(s) of Indian Nationality shall disclose names and addresses of foreign agents/representatives, if any. Either the Indian agent or behalf of the foreign principal or the foreign principal directly could bid in a tender but not both. Further, in cases where an agent participate in a tender on behalf of one manufacturer, he shall not be allowed to quote on behalf of another manufacturer along with the first manufacturer in a subsequent/parallel tender for the same item.

(v) The Bidder(s)/Contractor(s) will, when presenting his bid, disclose any and all payments he has made, is committed to or intends to make to agents, brokers or any other intermediaries in connection with the award of the Contract.

(c) The Bidder(s)/Contractor(s) will not instigate third persons to commit offences outlined above or be an accessory to such offences.

(d) The Bidder(s)/Contractor(s) will not, directly or through any other person or firm indulge in fraudulent practice means a willful misrepresentation or omission of facts or submission of fake/forged documents in order to induce public official to act in reliance thereof, with the purpose of obtaining unjust advantage by or causing damage to justified interest of others and/or influence the procurement process to the detriment of the Government interests.

(e) The Bidder(s)/Contractor(s) will not, directly or through any other person or firm use Coercive Practices (means the act of obtaining something, compelling an action or influencing a decision through intimidation, threat or the use of force directly or indirectly, where potential or actual injury may befall upon a person, his/her reputation or property to influence their participation in the tendering process).

3. Consequences of Breach

Without prejudice to any rights that may be available to the Principal/Owner under law or the Contractor its established policies and laid down procedures, the Principal/Owner shall have the following rights in case of breach of this Integrity Pact by the Bidders(s)/Contractor(s) and the Bidder/Contractor accepts and undertakes to respect and uphold the Principal/Owner's absolute right.

(a) If the Bidders(s)/Contractor(s), either before award or during execution of Contract has committed a transgression through a violation of Para2 above or in any other form, such as to put his reliability or credibility in question, the Principal/Owner after giving 14 days notice to the contractor shall have powers to disqualify the Bidder(s)/Contractor(s) from the Tender process or terminate/determine the Contract, if already executed or exclude the Bidder/Contractor from future contract award processes. The imposition and duration of the exclusion will be determined by the seventy of transgression and determined by the Principal/Owner. Such exclusion may be forever or for a limited period as decided by the Principal/Owner.

(b) Forfeiture of EMD/Performance Guarantee/Security Deposit. If the Principal/Owner has disqualified the Bidder(s) from the Tender process prior to the award of the Contract or terminated/determined the Contract or has accrued the right to terminate/determine the Contract according to para 3(a), the Principal/Owner apart from exercising any legal rights that may have accrued to the Principal/Owner, may in its considered opinion forfeit the entire amount of Earnest Money Deposit/Performance Guarantee/Security Deposit of the Bidder/Contractor.

(c) Criminal Liability if the Principal/Owner obtains knowledge of conduct of a Bidder of Contractor, or of an employee or a representative or an associate of a Bidder or Contractor which constitutes corruption within the meaning of IPC Act, or if the Principal/Owner has substantive suspicion in this regard, to Principal/Owner will inform the same to law enforcing agencies for further investigation.

4. Previous Transgression

(a) The Bidder declares that no previous transgressions occurred in the last 5 years with any other Company in any country confirming to the anticorruption approach or with Central Government or State Government or any other Central/State Public Sector Enterprises in India that could justify his exclusion from the Tender process.

(b) If the Bidder makes incorrect statement on this subject, he can be disqualified from the Tender process or action can be taken for banning of business dealings/ holiday listing of the Bidder/Contractor as deemed fit by the Principal/Owner.

(c) If the Bidder/Contractor can prove that he has resorted/recouped the damage caused by him and has installed a suitable corruption prevention system, the Principal/Owner may, at its own discretion, revoke the exclusion prematurely.

5. Equal Treatment of all Bidders/Contractors/Subcontractors

(a) The Bidder(s)/Contractor(s) undertake(s) to demand from all subcontractors a commitment in conformity with this Integrity Pact. The Bidder/Contractor shall be responsible for any violation(s) of the principles laid down in this agreement/Pact by any of its Subcontractors/Sub-vendors.

(b) The Principal/Owner will enter into Pacts on identical terms as this one with all Bidders and Contractors.
(c) The Principal/Owner will disqualify Bidders, who do not submit, the duly signed Pact between the Principal/Owner and the bidder, along with the Tender or violate its provisions at any stage of the Tender process, from the Tender process.

6. Duration of the Pact

This Pact begins when both the parties have legally signed it. It expires for the Contractor/Vendor after the final payment under the contract has been made or till the continuation of defect liability period, whichever is more and for all other bidders, till the Contract has been awarded.

If any claim is made/lodged during the time, the same shall be binding and continue to be valid despite the lapse of this Pacts as specified above, unless it is discharged/determined by the competent Authority, MES.

7. Other Provisions

(a) This Pact is subject to Indian Law, place of performance and jurisdiction is the Head quarters of the formation of the Principal/Owner, who has floated the Tender.

(b) Changes and Supplements need to be made in writing. Side agreements have not been made.

(c) If the Contractor is a partnership or a consortium, this Pact must be signed by all the partners or by one or more partner holding power of attorney signed by all partners and consortium members. In case of a Company, the Pact must be signed by a representative duly authorized by board resolution.

(d) Should one or several provisions of this Pact turn out to be invalid; the remainder of this Pact remains valid. In this case, the parties will strive to come to an agreement to their original intensions.

(e) It is agreed term and condition that any dispute or difference arising between the parties with regard to the terms of this Integrity Agreement/Pact, any action taken by the Owner/Principal in accordance with this Integrity Agreement/Pact or interpretation there of shall not be subject to arbitration.

8. Legal and Prior Rights

All rights and remedies of the parties here to shall be in addition to all the other legal rights and remedies belonging to such parties under the Contract and/or law and the same shall be deemed to be cumulative and not alternative to such legal rights and remedies aforesaid. For the sake of brevity, both the Parties agree that this Integrity Pact will have precedence over the Tender/Contact documents with regard any of the provisions covered under this Integrity Pact.

9. Independent Monitors

(a) That the MOD has appointed following Independent Monitors for this Pact:-

Sl No	Name of IEM	E-mail id
1	Shri Narayan Murthy Ganapathy, IFoS (Retd) 3/268, Vipul Khand, Gomtinagar, Lucknow – 226010	gana_narayan@yahoo.com Mobile No. 9415147429
2	Shri Lalatendu Mohanti, IPS (Retd) A-1 Tower-5, New Motibagh, New Delhi - 110023	l.mohanti@gmail.com Mobile No. 9717095659

9.1 Details of Nodal officer nominated by E-in-C's Branch are as follows:-

Name : Shri P K S Sengar, Dir (Contracts)
E-mail id : dircont1winc-mes@nic.in
Postal Address: Room No 158
Dte of Contract Management
Engineer-in-Chief's Branch
Kashmir House, Rajaji Marg
New Delhi-110011,
Tele No – 011-23019154

Any complaint with regard to violation of IP, whenever received, will be referred to the Independent Monitors for their comments/enquiry.

(b) The task of the Monitor, is to review independently and objectively any complaint received with regard to violation of Integrity Pact and offer comments or carry out enquiry as deemed fit.

(c) That the Monitor is not subject to any instructions by the representatives of the parties and would perform his functions neutrally and independently.

(d) That the Bidder/Contractor accepts that if the Monitor needs to access any records then the Monitor shall have the right to access without restriction to all project documentation of the Principal/Owner including that provided by the Bidder/Contractor. The Bidder/Contractor will also grant the Monitor, upon his project documentation including minutes of meeting. The same is applicable to Sub-Contractors and Associates. The Monitor is under obligation to treat the information and documents of the Principal/Owner and Bidder/Contractor/ Sub-Contractors/Associates with confidentiality.

(e) That if the Monitor has reported to the Principal/Owner a substantiated suspicion of an offence under relevant Anti-Corruption Laws of India and the Principal/Owner has not, within reasonable time, taken visible action to proceed against such offence or reported it to the Vigilance Department, the Monitor may also transmit this information directly to the Central Vigilance Commissioner, Government of India.

(f) The word 'Monitor' would include singular and plural.

……

(For and on behalf of Principal/Owner)

…….

(For and on behalf of Bidder/Contractor)

WITNESS:

1……

(Signature, name and address)

2…….

(Signature, name and address)

Place:

Dated:

INTEGRITY PACT

To,

…….

…….

…….

Sub: Tender ID No 2024_MES_688696_1 CA No: CEJZ/MHOW/35(T) OF 2024-25 FOR PROVN OF HRDC BLDG AT INF SCHOOL, MHOW

Dear Sir,

It is hereby declared that MES is committed to follow the principle of transparency, equity and competitiveness in public procurement..

The subject Notice Inviting Tender (NIT) is an invitation to offer made on the condition that the bidder will sign the integrity agreement, which is an integral part of tender/bid documents, failing which the tenderer / bidder will stand disqualified from the tendering process and the bid of the bidder would be summarily rejected.

This declaration shall form part and parcel of the Integrity Agreement and signing of the same shall be deemed as acceptance and signing of the Integrity Agreement on behalf of the MES.

Yours faithfully,

Chief Engineer Jabalpur Zone

INTEGRITY PACT

To,

HQ Chief Engineer
Jabalpur Zone,
Jabalpur.

SUB: Tender ID No 2024_MES_688696_1 CA No: CEJZ/MHOW/35(T) OF 2024-25 FOR PROVN OF HRDC BLDG AT INF SCHOOL, MHOW

Dear Sir,

I/We acknowledge that MES is committed to follow the principles there of as enumerated in the integrity Agreement enclosed with the tender/bid document.

I/We agree that the Notice Inviting Tender (NIT) is an invitation to offer made on the condition that I/We will sign the enclosed integrity Agreement, which is an integral part of tender documents, falling which I/We will stand disqualified from the tendering process. I/We acknowledge that THE MAKING OF THE BID SHALL BE REGARDED AS AN UNCONDITIONAL AND ABSOLUTE ACCEPTANCE of this condition of NIT.

I/We confirm acceptance and compliance with the Integrity Agreement in letter and spirit and further agree that execution of the said Integrity Agreement shall be separate and distinct from the main contract, which will come into existence when tender/bid is finally accepted by MES. I/We acknowledge and accept the duration of the Integrity Agreement, which shall be in the line with Para 1 of the enclosed Integrity Agreement.

I/We acknowledge that in the event of my/our failure to sign and accept the Integrity Agreement while submitting the tender/bid, MES shall have unqualified, absolute and unfettered right to disqualify the tenderer/bidder and reject the tender/bid is accordance with terms and conditions of the tender/bid.

Yours faithfully,

(To be signed mentioning name by the Authorised Signatory of the Bidder)

IN LIEU OF IAFW 2159 (REVISED 1947)
(To be read in conjunction with General Conditions of contracts IAFW 2249,1989 Print)

Tele No: 0761-2603027
Fax No: 0761-2601027

CHIEF ENGINEER JABALPUR ZONE
MILITARY ENGINEER SERVICES
RIDGE ROAD
JABALPUR CANTT – 482001

752955/A/ /E8

Jan 2025

CA No: CEJZ/MHOW/ 35(T) OF 2024-25 FOR
PROVN OF HRDC BLDG AT INF SCHOOL, MHOW

1. M/s_____of_____is/arehere by authorized to tender for the above work. The tender is to be uploaded in the "www.defproc.gov.in" site before 1800 hours on _____ 2025 .

2. Any correspondence concerning this tender should be addressed quoting the reference as given to HQ Chief Engineer Jabalpur Zone, 482001 and marked as **"CEJZ/MHOW/ 35(T) OF 2024-25"**

3. "THE PRESIDENT OF INDIA DOES NOT BIND HIMSELF TO ACCEPT THE LOWEST OR ANY TENDER".

Signature of Officer

Issuing the documents

_____
(Signature of Contractor)

_____
Dy Dir/ Asst Dir (Contracts)
For Accepting Officer

NOTES TO SCHEDULE 'A'
PROVN OF HRDC BLDG AT INF SCHOOL, MHOW

Notes : GENERAL (APPLICABLE TO ALL PARTS OF SCH 'A')

1. Schedule 'A' is divided into following parts: -

1.	Schedule 'A' Part -I	Building Works
2.	Schedule 'A' Part -II	B/R works including internal services
3.	Schedule 'A' Part -III	External Water Supply
4.	Schedule 'A' Part -IV	Internal Electrification
5.	Schedule 'A' Part -V	External Electrical Supply
6.	Schedule 'A' Part -VI	Fire detector system/ Alarm System
7.	Schedule 'A' Part -VII	Water Cooler
8.	Schedule 'A' Part -VIII	Lan Conduit
9.	Schedule 'A' Part -IX	Refrigeration and air conditioning
10.	Schedule 'A' Part -X	Rain Water Harvesting
11.	Schedule 'A' Part -XI	Road, Path & Hard standing
12.	Schedule 'A' Part -XII	Sewage Disposal
13.	Schedule 'A' Part -XIII	Area Drainage
14.	Schedule 'A' Part -XIV	Site Clearance
15.	Schedule 'A' Part -XV	Miscellaneous Items of B & R and E & M (to be quoted by the tenderer in BOQ)

2. **PERIOD OF COMPLETION**

2.1 The entire work under this contract shall be completed within **84 weeks** from the date of commencement of work mentioned in the work order No.1.

3. Description of buildings works and services enumerated in various parts of Schedule 'A' is in brief. It is deemed to be amplified and read in conjunction with special conditions, particular specifications, specification for materials and workmanship and conditions in relevant trade sections of MES Standard Schedule of Rates 2009 Part-I and MES Standard Schedule of Rates 2020 Part II and contract drawings.

4. (a) For drgs related to items of schedules refer list of drgs.

(b) In case, the details in respect of items shown on main drawings are not given in the drawings referred to in main drawings, the same shall be followed from any other drawing included in the list of drawings.

5. Any Drawing mentioned in reference on various drawings or anywhere on any drawing forming part of the tender document or anywhere in the tender documents but inadvertently not included in the list of drawings, shall be deemed to form part of the tender.

6. Probable distribution of various items of internal/external services are indicated on drawings. These are tentative and may be varied where necessary at the discretion of GE. The contractor shall not be entitled for any claim on account of such varied alignment or change in quantities.

7. Layout of building work and services indicated in the site plan is tentative. No adjustment in price shall be done on account of final approved layout within the site plan area.

8. In case any variation in out to out, centre to centre and internal dimension of the rooms as mentioned on drawings is observed during execution, centre to centre dimensions shall be followed without any financial effect.

NOTES TO SCHEDULE 'A' (Contd…)

9. The lump sum quoted by the contractor shall be deemed to include for all minor extra and constructional details which may not have been specifically shown on drawings or given in particular specification but are essential for the execution of work and services in the workman like manner, sound construction and established engineering practice. The details of such items not specifically shown on drawings/specified shall be furnished by GE during execution of the work. In case of difference of opinion between the contractor and the GE as to whether or not certain items of work constitute minor extras/minor constructional details which are deemed to have been included in the contractor's lump sum, the decision of the Accepting Officer shall be final, conclusive and binding.

10. The lump sum quoted by the contractor shall also be deemed to include for all items of work specifically shown on drawings but inadvertently not specified in the specifications. The specifications for such items shall be followed as per codal provisions, technical & functional requirement and as decided by CWE. In case of any dispute, the decision of Accepting Officer shall be final, conclusive and binding.

11. For structural details like location, size of members structural drawings shall be referred. For all structural members like RCC and steel viz beams, columns, bands, lintels etc including their dimension, details and locations shall be provided as per structural drawings including TD drawings showing structural details, notes etc irrespective of whether they have been indicated in Arch Drawings or erroneously not shown, similarly in Architectural details, including provisioning of built in furniture and all other details (except structural details) shall be followed from Architectural Drawings including TD drawings. The contractor should keep above stipulations in mind while quoting his rates and nothing extra shall be admissible on this account.

12. **APPLICABLE FOR SCHEDULE 'A' PART –I (BUILDING WORKS)**

12.1 The unit lump sum rates quoted by the tenderer against Schedule 'A' Part-I shall be deemed to include for all items of work shown on drawings and as specified in particular specifications including necessary earth work, returning filling, removal, excavation required for building work complete except items of work covered under Schedule 'A' Part-II to Part XII.

12.2 BLANK

12.3 Earth obtained from Schedules Part-I to Part-XV, If found suitable as decided by the GE, shall be utilized for forming embankment/traverse. The remaining approved soil required for forming embankment/traverse shall be obtained from nearest possible MD land. The rates quoted by the contractor shall be inclusive of this.

12.4 BLANK

NOTES TO SCHEDULE 'A' (Contd…)

12.4 The cost of the following items, where ever required are also included in the rates inserted against each item of Schedule 'A' Part I.

(i) Fan hooks fixtures and fittings, niches for boxes to house MCB's /DB's /main switch boxes etc, cutting hole for exhaust fans.

(ii) Any other work specifically mentioned in notes to other parts of Schedule 'A' and particular specification.

(iii) Concealing of conduit and conduit fittings for wiring in roof slab up to the junction of roof and wall.

(iv) Anti termite treatment to the buildings listed at Sr No 01 of Schedule 'A' Part-I.

(v) Numbering of building all as directed by GE/Engineer-in-Charge.

(vi) Format made in plaster of CM (1:4) on buildings all as specified.

(vii) Waterproofing treatment to the building listed at Sch 'A' Part-I all as specified and shown on drgs.

(viii) All other items, which are not covered in notes but shown on drgs and specifications shall be deemed to be included in unit rate inserted against Schedule 'A' Part I.

13. In case of any variation provision shown in size of steel members for similar location/item in different details, higher size shall be provided.

14. The plinth height of the buildings shall be considered as shown in drawing. In case, the plinth height for any building is not shown in drawing, the same shall be considered as 450 mm above existing average ground level. Any increase/decrease in plinth height during execution of work shall be regularized through a DO.

15. **APPLICABLE FOR PROVISIONAL SCHEDULE 'A' PART II TO XV**

15.1 All items/quantities under Schedule 'A' Part-II to Part-XV are 'Provisional'. Tentative distribution of layouts of various items of internal/external services is indicated on drawings and is included in Schedule 'A' Part-II to Part-XV. These may be varied where necessary at the discretion of the GE. The contractor shall not be entitled for any claim on account of such varied alignments.

16. Specifications in MES Schedule Part-I (2009) and preamble to items given in MES Schedule Part II rates (2020) under respective trade section shall also be applicable. If any provision in the description of items of Schedule 'A' and Particular Specifications is at variance with the provisions given in specification in MES Schedule Part I (2009) and preambles to MES Schedule Part II rates (2020), provision in Schedule "A" and particular specifications shall take precedence thereof.

17. Unless specifically specified otherwise the unit rate of each items of work shall be deemed to be inclusive of supply of all new materials and labour for supply, fixing, installation, commissioning, designing and testing etc complete as applicable.

18. If, there is any discrepancy regarding General Notes on RCC works, TD (Typical Detail) Drawing and Structural Drawing, the detail shown in main structural drawing shall be followed. Similarly details shown in main architectural drawings shall always be followed in case of discrepancy between main architectural drawing and TD (Typical detail) drawings.

19. Only TMT bars shall be used in this project all as specified in PS. Diameter and spacing of TMT bars shall be same as shown in structural drgs. Where 6 mm dia CTD/MS bars are shown in drgs, 8 mm dia TMT bars shall be used in lieu thereof without changing the spacing and without any extra cost to the Govt.

NOTES TO SCHEDULE 'A' (Contd…)

20. (a) Mode of payment of RCC reinforcement bars under measurable/provisional schedules shall be net length of bars provided on site multiplied by the unit weight of respective dia of reinforcement bar(s).

(b) Offcuts of reinforcement bars less than 1.5 m in length shall not be allowed to use with lap and are to be discarded / removed from the site.

(c) Lap length provided using offcuts more than 1.5m length shall be measured for payment but only net length of bar (without lap) shall be measured to arrive weight of steel utilized in the item.

21. Makes of various materials/items have been specified hereinafter in particular specifications. However makes with model/Cat-Part No if specified in Schedule 'A' then same shall only be provided regardless what is shown in list of makes in PS. **However in case no make is specified for a particular item in the tender document, the same shall be ISI marked, Grade-I and approved by GE**.

22. The abbreviations wherever occur in tender documents shall have the meaning as indicated in Para 1.4 on srl page no. 2 of SSR-Part-I 2009.

23. **DEFECT LIABILITY PERIOD** for the subject work shall be 24 (Twenty Four) calendar months from the date of certified completion of works (Refer condition 46 of IAFW-2249)

_____ _____

(Signature of Contractor) Dy Dir/ Asst Dir (Contracts)
For Accepting Officer

SCHEDULE 'B'

ISSUE OF MATERIALS TO THE CONTRACTOR (SEE CONDITION 10 OF IAFW 2249)

Ser No	Particulars	Rates at which materials will be issued to the contractor		Place of issue by name	Remarks
		Unit	Rate in Rs		
1.	2	3	4	5	6

----------NIL----------

SCHEDULE 'C'

ISSUE OF TOOLS AND PLANTS (OTHER THAN TRANSPORT) WHICH WILL BE HIRED TO THE CONTRACTOR (SEE CONDITION 15, 34 AND 35 OF IAFW-2249)

Srl No	**Quantity**	Particulars	Details of MES crew supplied	Hire charges per unit per working day (Rs)	Stand by charges per unit per off day (Rs)	Place of issue by name	Remarks
1	2	3	4	5	6	7	8

----------NIL----------

<u>SCHEDULE 'D'</u>

TRANSPORT TO BE HIRED TO THE CONTRACTOR

(SEE CONDITION 16 & 35 OF IAFW-2249)

Srl No	Quantity	Particulars	Rates per unit per working day	Place of issue by name	Remarks

----------NIL----------

_____
(Signature of Contractor)

_____
Dy Dir (Contracts)
For Accepting Officer

TENDER

To

The President of India.

Having examined and perused the following documents: -

1. Particular specifications signed by AAD(C)/AD(C)/Dy Dir (Contracts)/SSW.

2. Drawings detailed in the list of drawings.

3. Schedule ‘A’, ‘B’, ‘C’ and ‘D’ attached herewith.

4. MES Standard Schedule of rates 2009 (Part-I) specifications together with amendment No. 1 to 3 and Part-II 2020 (rates).

5. General Conditions of Contracts (IAFW-2249 - (1989 Print) together with Errata Nos 1 to 20 and amendments Nos 1 to 49.

6. Water will **NOT** be supplied by the MES. Contractor has to make suitable arrangement.

7. Should this tender be accepted I/We agree:-

(a) ***That the sum of **Rs 5,80,300 (Rupees Five Lakh Eighty Thousand and Three Hundred only)** forwarded as Earnest Money shall be refunded by the Govt on receipt of the required amount of Performance security deposit all as per condition 19 of the IAFW-2249.

(b) To execute all the works referred to in the said documents upon the terms and conditionscontained or referred to therein and as detailed in the General Summary hereinafter and to carryout such deviations as may be ordered vide condition 7 of IAFW-2249 up to maximum of 10%(Ten percent).

(c) Further agree to refer all the disputes in terms of condition 71 of GCC during execution ofthe work or after determination/cancellation/termination of the contract arising out of the contractexcept/excluding those for which decision of the Accepting Officer/CWE/GE is final & binding asstipulated in the subject contract Agreement & in the GCC however including any disagreement by either party with any action/ inaction/opinion/ instruction certificate or valuation by the Accepting Officer or his Nominees hall in the first place be referred to the DISPUTE DISPUTERESOLUTION BOARD (DRB) (Applicable in case of contracts of value more than 10 Cr)/ CONCILIATION (Applicable in case of contracts of value upto 10 Cr) all as mentioned in the special condition of the Tender Documents.

(d) Further agree to refer all disputes, as required by condition 70 to the sole arbitrator from MoD panel of Arbitrator to be appointed by the Engineer-in-Chief or officiating E-in-C or Director General of Works if specifically delegated in writing by Engineer-in-Chief, Army Headquarters, New Delhi whose decision shall be final, conclusive and binding.

*** To be deleted where not applicable

Contd…./-

Brought forward from Sl Page No. the lump sum Rs. _____ _____ Rupees _____ _____

Signature _____Name (in block/capital letters) _____ in the capacity of _____ duly authorized to sign the tender for and behalf of _____ (in block capital).

Witness_____ Date _____

Address _____ Address :_____

_____ _____

_____ _____

_____ Telephone No.:_____

ACCEPTANCE

_____alterations have been made in these documents and as evidence that these alterations were made before the execution of the contract agreement; they have been initiated by the contractor and Shri _____ _____. The said officer is hereby authorized to sign and initial on my behalf the documents forming part of this contract.

The above tender was accepted by me on behalf of the President of India for the Lump sum of Rs. _____(Rupees_____ _____) on the _____day of _____.

Signed Dated this _____day of_____

Chief Engineer
Jabalpur Zone
Accepting Officer
(For and Behalf of the President of India)

Contd…./-

General Conditions of contracts
IAFW-2249 (1989 print)
Lump Sum Contracts (IAFW-2159)

A copy of the General Conditions of Contracts (IAFW-2249) 1989 PRINT with errata 1 to 20 and amendment No 1 to 49 has been supplied to me/us and is in my/our possession. I/We have read and understood the provisions contained in the aforesaid General Conditions Of Contracts before submission of this tender and I/We shall abide by the terms and conditions there of, as modified, if any, elsewhere in these tender documents.

It is hereby further agreed and declared by me/us that the General Conditions Of Contracts (IAFW-2249-1989 Print) including condition 70 thereof pertaining to settlement of disputes by Arbitration and Errata 1 to 20 and amendment No 1 to 49 form part of these tender documents.

Dy Dir (Contracts)
for Accepting Officer

Signature of Contractor
Date

Contd…./-

SCHEDULE OF MINIMUM WAGES

It is hereby agreed that the “Schedule of Minimum Wages” as published by Government of India or provincial Government’s Notification whichever is higher issued upto date of submission of tender shall form part of these tender documents.

My/our signature here under amounts to my/our having signed the aforesaid documents forming part of the tender.

Note :- “Schedule for minimum wages” referred to above is available for reference, in the office of HQ Chief Engineer Jabalpur Zone, 482001.

_____ _____

(Signature of Contractor) Dy Dir / Asst Dir (Contracts)
For Accepting Officer

Contd…./-

SPECIAL CONDITION

GENERAL

1. (a) General specifications mentioned any where in the contract including those in General Conditions of Contracts, viz IAFW-2249, shall mean the specifications given in the MES Standard Schedule of Rates 2009 (Part-I). Particulars specifications given here-in - after are in brief and only to particularize/amend and emphasis the specification given in the MES Standard Schedule of Rates. Specifications general rules, Preambles and special conditions contained in MES Standard Schedule of Rates 2009 Part-I & Part-II 2020 shall, unless specifically described otherwise, be applicable to this contract.

(b) Any work shown on drawing(s) but not described in Particular specifications shall, unless specifically described to be excluded from the contract, be deemed to be included in the lumpsum quoted by contractor for Schedule 'A'. In case of doubt with regard to the manner of detail for executing the work, the decision of the Accepting Officer shall be final, conclusive and binding. This shall not be applicable to the items of work pertaining to services for which separate parts of Schedule 'A' setting out the requirements have been given. Should there be item(s) or parts as shown on drawings but not included in the relevant sections of Sch 'A' these shall, unless otherwise described or stated in the contract, constitute an additional work.

(c) All heading and marginal notes to these special conditions or to any other documents of this contract are solely for the purpose of giving a concise indication and not a summary of the contents thereof, and they shall never be deemed to supersede the contents of the clauses.

(d) The tender is advised to point out in writing discrepancy(s) if any of the tender documents, at least 10 days before the due date of receipt back of tender. In case discrepancy(s), in contract documents is/are noticed by the contractor during execution of the works, it shall be brought out to the notice of the Accepting Officer of contract and others concerned in writing before commencement of the particular work where discrepancy(s) is/are noticed failing which the contractor shall forfeit his right to claim on account of such discrepancy(s).

(e) Also where there are obvious mistakes in any of the contract provisions, the Accepting Officer shall be sole deciding authority with regard to the intention of the contract documents and his decision in this respect shall be final, binding and conclusive.

(f) Where the drawings show finishes to wooden members, such finishes shall be applied to wooden members as well as wood based members. However, surfaces covered with laminates shall not be given any finish such as painting, polishing etc.

(g) Where reinforcement details/thickness of RCC members such as slab, beams, lintels etc are not shown on the drawings, the same shall be supplied by the GE on the basis of provisions in IS-456- 2000 and the same shall be provided and incorporated in the works by the contractor and no extra payment will be admissible on this account.

2. **CONDITIONS OF CONTRACTS AND EXECUTION OF WORKS**: - The various sections /sub-divisions of the tender documents shall be deemed to be supplementary/ complementary to each other.

3. **INSPECTION OF SITE BY THE TENDERER** :-

(i) The tenderer is advised to contact the GE for the purpose of inspection of site(s) and relevant document other than those sent herewith, who will give reasonable facilities for the purpose. The tenderer shall also make themselves familiar with the working conditions accessibility of site, availability of materials and other cogent conditions which may effect the entire completion of work under this contract.

(ii) The tenderer shall be deemed to have inspected the site(s) and made themselves familiar with the working conditions, whether they have actually inspected the site(s) or not.

SPECIAL CONDITION (CONTD…)

4. **CONDITIONS OF WORKING**

4.1 CONDITION FOR WORKING IN UNRESTRICTED AREA. All works lie in Unrestricted Area. The restrictions for entry to work site and conditions of working in unrestricted area shall be as under:-

(a) The work lies in an Unrestricted Area. However, the contractor, his agents, servants, workmen and vehicles may pass through the unit lines, in which case, the Engineer-in-Charge at his discretion has the right to issue the passes, control their admission to the site of work or any part thereof. The contractor shall, on demand by the Engineer-in-Charge, shall submit a list of personnel etc concerned and any other information called for by the Engineer-in-Charge and shall satisfy the Engineer-in-Charge as regards the bonafide of such people. Passes shall be returned at any time on demand by the Engineer-in-Charge and in any case on completion of work.

(b) The contractor and his work people shall observe all the rules promulgated from time to time by authority controlling the area where the work is to be carried out eg prohibition of smoking etc. Any person found violating the security rules laid down by the authority shall be immediately expelled from the area without assigning any reasons whatsoever and the contractor shall have no claim on this account. Nothing shall be admissible for any man hours lost on this account.

5. **SECURITY OF CLASSIFIED DOCUMENTS** :- Contractor's special attention is drawn to condition 2-A & 3 of General Conditions of Contracts (IAFW-2249). The contractor shall not communicate any classified information regarding the works either to Sub-contractors or others without prior approval of the Engineer-In-Charge. The contractor shall also not make copies of the design/drawings and other documents furnished to him in respect of the work and he shall return all documents on completion of the work or earlier or termination of contract. The contractor shall alongwith final bill attach a receipt from the Engineer-in-Charge in respect of his having returned the classified documents as per condition 3 of General Conditions of contracts (IAFW-2249).

6. **OFFICIAL SECRET ACT** :- The contractor shall be bound by the Indian Official secret Act 1923, particularly section 5 thereof, all as per condition 2A of IAFW-2249

7. **EMPLOYMENT OF PERSONNEL** :-

(a) Contractor shall employ only Indian Nationals as his representatives, servants and workmen after verifying their antecedents and loyalty. He shall ensure that no person of doubtful antecedents and nationality is, in any way, associated with the work. If for the reasons of technical collaboration or other considerations the employment of foreign national(s) is unavoidable, the contractor shall furnish full particulars to this effect to the Accepting Officer at the time of submission of the tender. As a proof that the contractor has employed only Indian Nationals, he shall render a certificate to GE within one month from the date of acceptance of tender to this effect. In case the GE desires, contractor will have the police verification done of personnel employed by him.

(b) The GE shall have full powers, and without giving any reason to order the contractor immediately to cease to employ in connection with this contract, any agent, servant or employee whose continued employment is, in his opinion, undesirable, The contractor shall not be allowed any compensation on this account.

(c) The contractor's attention is also drawn to condition 25 of IAFW-2249 in this connection.

SPECIAL CONDITION (CONTD…)

8 **AVAILABILITY OF LAND FOR STORAGE OF MATERIALS, TEMPORARY WORKSHOP AND OFFICES ETC**: -Refer condition 24 of IAFW-2249. The contractor shall be permitted to store his materials including erecting temporary sheds at the area of land marked on site plan and he shall pay a license fee of Rs 1/- per year or part thereof in respect of each and every separate area of land allotted to him.

9. **DAMAGE TO EXISTING STRUCTURES**:- Any damage to the existing structures, any existing road, etc during the execution of work shall be made good by the contractor at his own expense. Rectification, replacement, making good and touching up etc, shall be carried out to conform to the materials and workmanship originally provided and to the satisfaction of the Engineer-in-Charge. In case of any dispute on this account, the decision of the GE shall be final conclusive and binding.

10. **QUARRIES AND ROYALTIES** :- Quarries are not available on land which is in the charge of MES authorities, Condition 14 of IAFW-2249 shall be treated as not applicable.

11. **APPROACHES**: - The contractors shall make arrangements for and provide at his own cost all temporary approaches, if required to the site(s), after obtaining approval in writing of the GE to the layout of such approaches.

12. **ACTION WHERE NO SPECIFICATIONS EXIST** :-

(a) In case any class of work for which there is no such specifications in the tender documents including SSR, such work shall be carried out in accordance with Indian Standard Specifications and where Indian Standard Specifications do not cover the same, the work shall be carried out as per standard engineering practice subject to the approval of the GE.

(b) No claim what-so-ever will be tenable for any misunderstanding or for not having ascertaining requisite information /knowledge to muster the resources required in satisfactory completion of the entire work to requirements as per contract agreement.

13. **METHOD OF MEASUREMENT**: - Measurement of all measurable work executed under this contract shall be taken in accordance with the rules of measurements as laid down in SSR 2009 Part-I and SSR 2020 Part-II pertaining to various items of work as applicable, unless otherwise indicated in the respective sections of schedule “A”. In case the method of measurements for a particular item of work is not given in the SSR, a reference will be made to the relevant provision contained in the latest edition of IS-1200.

14. CRITICAL PATH METHOD/NET WORK ANALYSIS AND THE TIME SCHEDULE:

(a) The time and progress chart, to be prepared as per condition 11 of IAFW-2249 (General Conditions of contracts), shall consist of detailed net work analysis and a time schedule based on latest project management tools / techniques & software. The critical path net work will be drawn by the contractor and approved by GE soon after acceptance of the tender.

(b) (i) The time scheduling of the activities will be done by the contractor so as to finish the work within the stipulated time. The contractor is advised to use the facilities of computerization available with any Organization which affords such facility in doing the same. On completion of the time-schedule, a firm calendar date schedule will be prepared and submitted by the contractor to GE, who will approve it after scrutiny. No financial liability shall accrue to the Govt in connection with computerization

(ii) The schedule will be submitted in quadruplicate within two weeks from the date of handing over of site. In case the contractor fails to submit the CPM net work diagram, the net work prepared by the GE will be binding on him and he will be bound to employ resources of men and materials in the manner directed by the GE.

SPECIAL CONDITION (CONTD…)

(c) During currency of the work, the contractor is expected to adhere to the time schedule and this adherence will be part of the contractor's performance under the contract. During the execution of the work, the contractor is expected to participate in the reviews and updating of the net work undertaken by GE. These reviews may be undertaken at the discretion of the GE either as a periodic appraisal measure or when the quantum of work ordered on the contractor is substantially changed through deviation order (s)/or amendment. Any revision of the schedule as a result of the review will be submitted by the contractor to the GE within a week, who will approve it after due scrutiny. The contractor will adhere to the revised schedule thereafter.

(d) In case of Contractor not agreeing to the revised Schedule, the same will be referred to the Accepting Officer whose decisions will be the final, conclusive and binding. GE's approval to the revised Schedule resulting in completion date beyond the stipulated date of completion shall not automatically amount to grant of extension of time, Extension of time shall be considered and decided by the appropriate authority in terms of condition 11 of IAFW-2249 and separately regulated.

(e) Contractor is expected to mobilise and employ sufficient resources to achieve the detailed schedule within the broad frame work of the accepted methods of working and safety.

(f) No additional payment will be made to the Contractor for any multiple shift work or other incentive methods contemplated by him in this work schedule even though the time schedule is approved by the Department.

15. **CO-ORDINATION WITH OTHER AGENCIES**: - The contractor shall permit free access and afford normal facilities and usual convenience to other agencies or departmental workmen to carryout connected works or service under separate arrangements. The contractor will not be allowed any extra payment on this account.

16 **SECURITY AGAINST LOSS OR DAMAGE**

(a) Contractor's plants/equipments at site. The contractor shall furnish to the Engineer-in-Charge every morning a distribution return of his plants/equipments on the site of the work, stating the following particulars:-

(i) Particulars of plants/equipments, its make, manufacturer No, Model No if any, Registration No if any, capacity year of manufacture and year of purchase etc.

(ii) Total No (Qty) on site of work.

(iii) Location and indication of No (Qty) at each location of the site of work.

(iv) Purchase value on the date of purchase .For the purpose of this condition, plant/equipment shall include vehicles i.e. trucks and lorries but neither the workman's tools nor any manually operated tools/equipment. The Engineer-in-Charge shall record the particulars supplied by the contractor in the works dairy and send monthly return to the GE for record in his office.

(b) Loss on account of Enemy Action: - If as a result of enemy action the contractor suffers any loss or damage, the Govt shall reimburse to the contractor such loss or damage to the extent and in the manner provided here-in-after.

(i) The loss suffered by him on account of any damage or destruction of his plant / equipment (as defined in special condition (a) above), material or any part or parts thereof the amount of loss assessed by the Accepting Officer of the contract on this account shall be final, binding and conclusive.

(ii) The compensation paid by him under any law for the time being in force, to any workmen employed by him for any injury caused to workmen or workman's legal successor for loss of the workmen's life.

SPECIAL CONDITION (CONTD…)

(iii) Payment of compensation for loss or damage to any work of part of work carried out. The amount of compensation shall be determined in accordance with condition 48 of General Conditions of Contracts (IAFW-2249).

(iv) No reimbursement shall be made nor shall any compensation be payable under the above provision unless the contractor had taken Air Defence precautions ordered by the GE/or concerned, or in the absence of such order, reasonable precautions taken by the contractor. No reimbursement shall be made nor shall any compensation be payable for any plants/equipment or materials not lying at site of work at the time enemy action.

17. **FORCE MAJEURE**

Should any force majeure circumstances arise, each of the contracting party shall be executed for the no fulfillment or for the delayed fulfillment of any of its contractual obligations, if the affected part within 15 days of its occurrences informs the other party in writing**.**

Force majeure shall mean fires, floods, natural calamities or other acts such as war, turmoil, strikes (as no limited to be establishment of the selled), sabotage, explosions, quarantine restrictions beyond the control of either party.

It is understood and agreed between the parties hereto that the rights and obligations of the parties shall be deemed to be in suspension during the continuance of the force majeure event as aforesaid and the said rights and obligations shall automatically revive upon the cessation of intervening force majeure event. The period within which the rights & obligations of the parties shall be in suspension due to the force majeure event shall not be considered as a delay with respect to the period of delivery and /or acceptance of delivery under the contract or otherwise the detriment of either party.

Not withstanding the provisions of the immediately foregoing clauses it is further understood and agreed between the parties hereto that in the event of any force majeure persisting for an uninterrupted period exceeding 6 (Six) months, either party hereto reserves the right to terminate this contact upon giving prior written notice of 30 (Thirty) days to the other party of the intention to terminate without any liability other than reimbursement on the terms provided in this agreement of the goods received.

18 WATER (Refer to condition 31 of IAFW-2249)

18.1 (a) **Refer condition 31 of General Condition of Contract IAFW-2249, clause 1.13 of MES Schedule, Part-I & SI No 06 on Tender page**

18.2 Refer to condition 31 of IAFW-2249.**Water will not be supplied by the MES**. The tenderers are advised to visit the site of works to ascertain availability of water from civil sources or from nearby natural sources outside ministry of Defence land. The contractor shall be allowed, if he so desires, to install hand pumps, tube wells at site of work at places as approved by Engineer-in-Charge and nothing shall be charged from the contractor. Thecontractor shall remove the hand pumps, tube wells as and when asked to do so by Engineer–in-Charge/GE and in any case on completion of the work and before issue of completion certificate, unless GE desires that these hand pumps, tubes wells be left in position and the contractor agrees to do so without claiming cost thereof from department. No compensation whatsoever shall be admissible to the contractor, if he is required to removethe pump (s) tube wells before completion of work. Use of water from such sources shall only be permitted if, found after testing, potable and fit for use in the work. The water from such sources shall be got tested by contractor initially once for all individual source of supply and every time if there is a change of bore well within MOD land and once in every month (monthly) or whenever there is a change of source when water is fetched from the source outside MOD land. The water from such sources shall be got tested regularly by the contractor from laboratory approves by the GE, who shall after satisfying himself permit the contractor to use the water from such sources. Testing charges shall be borne by the contractor.

SPECIAL CONDITIONS (CONTD...)

18.3 The contractor, if he so desires, will be allowed to augment the departmental water supply by boring wells at his own cost at site(s) approved by the GE. On completion of the work the contractor shall remove pipes etc. installed and make good the site as directed by the Engineer-in-Charge without any extra cost to Govt. The contractor shall be given no rebate for augmentation of water supply and charges for water shall be recovered at the rate mentioned hereinbefore.

19. **ELECTRIC SUPPLY** :-

(a) In case the contractor desires to buy electricity from the MES, he shall be charged for electric energy consumed in the following rates:-

(i) At Rs 13.69 per unit for the light.

(ii) At Rs 13.69 per unit for power.

(b) Electric supply required for the work shall be made available by the MES at the point(s) marked on the layout plan. The main switch and energy meter to register the electric energy supplied shall be provided and installed by the MES. The contractor shall provide all necessary cable, fittings etc from the main switches in order to ensure a proper and suitable supply of electricity for the execution of work. The contractor shall also provide necessary earthing keeping in view the security requirements. The MES does not guarantee for the continuity of supply and no compensation what so ever shall be allowed for supply becoming intermittent or for breakdown in the system.

(c) The actual position of supply point(s) for electric energy will be ascertained by the contractor form the GE before submitting his tender. Irrespective of whether he actually inspects or not, he shall be deemed to have inspected and seen the supply point (s) in consultation with GE before submitting his tender. No claim, what so ever, shall be entertained on this account.

20. **VALIDITY OF TENDER**: - Tender shall remain open for acceptance for a period of **90 days** from the next date subsequent to last date of bid submission.

SPECIAL CONDITION (CONTD…)

21. **SAMPLE OF MATE RIALS/ARTICLES**.

(a) Materials/Articles provided by the contractor for in-corporation in the works shall unless otherwise specified in the Particular specifications, comply with the requirements of the relevant Indian standards (IS) of the year of publication/ edition specified in the MES Schedule Part-I and shall have IS certification marking.

(b) Sample of all materials before incorporating in the work, ordering in bulk shall be got approved by the contractor from GE in writing .

(c) In case of material/ articles for which Indian standards are not finalised/ issued and therefore not include in the MES schedule contractor shall produce the samples and get them approved by the GE in writing before incorporation in the works.

(d) The contractor is advised to inspect the samples of local building materials, e.g bricks, stone aggregate, sand etc and other materials maintained by the CWE/GE before submitting his tender. The contractor will be deemed to have inspected those samples and have full knowledge thereof, whether he inspects them or not. The materials to be provided by the contractor shall conform to or be superior to the sample.

(e) The selection and use of sources for materials shall be subject to the compliance of these specifications read in conjunction with MES schedule and relevant Indian Standard Specifications.

22. **PROPRIETORY MATERIAL/ ARTICLES**.

(a) The proprietary articles such as cement , electrical cables , electrical fittings and sanitary fittings etc when brought to site shall be inspected and approved by the GE.

(b) The quantity of proprietary articles brought to site shall be recorded in measurement book (IAFW-2261) and signed by the contractor and the Engineer-in-Charge as a check to ensure that required quantity has been brought to site for incorporation in the works.

(c) The proprietary materials shall be stored as directed. Those already recorded in the measurement book shall be marked suitably for identification.

(d) The contractor, shall on demand, produce to the GE, original receipt vouchers/ invoices in respect of the supplies, to ensure that the contractor has actually brought the articles of required quality and quantity from the authorized agent/manufacturer/ supplier and also to find out the rates thereof.

(f) **Purchase voucher** of following materials from manufacturer/dealer shall be produced by the contractor to Engineer-in-Charge. Defaced copies of voucher shall be held on record by the Engineer-in-Charge. Details of vouchers of proprietary material shall be entered in MB.

(i)	HDPE Pipe	(ix)	Cement
(ii)	Gunmetal gate valve or globe valve	(x)	LT and HT XLPE Cables
(iii)	Road marking paint	(xi)	Cable Jointing Kit
(iv)	GI Pipe	(xii)	Reinforcement
(v)	Steel tubular swaged pole.	(xiii)	Cable cover
(vi)	DG Set	(xiv)	Transformer
(vii)	Water pumps/ Mono block pump	(xv)	APFC Panel
(viii)	LED , Solar light and fittings thereof		

SPECIAL CONDITION (CONTD…)

23. **RECORD OF CONSUMPTION OF CEMENT**.

(a) For the purpose of keeping record of cement issued and consumed in works one properly bound register serially numbered and all pages initialed against the numbering by the Engineer-in-Charge and in the form approved by Engineer-in-Charge showing daily receipt quantity used in the work, balance in hand, shall be maintained.

(b) The register shall be kept at site in the safe custody of the contractor's representative during the progress of the work and shall be produced for verification of the Inspecting officer on demand.

(c) The entries in the aforesaid register will be signed daily by the contractor or his authorised representative and the JE for supervision and once in a week by Engineer-in-charge.

(d) On completion of the work, the contractor shall deposit the cement register with Engineer-in-Charge for record.

24. **WATCH/LIGHTING** ; - The contractor shall at his own cost take all possible precautions to ensure safety of life and property by providing necessary fencing, barrier, light, watchmen etc, during the progress of work and as directed by Engineer-in-Charge .

25. **MAKING GOOD** ; -The contractor shall leave or form holes of concrete (Plain or RCC), brick work , stone masonry and in any other situation as required for the work or as directed by Engineer-in-charge and make good all the holes in the same mortar and mix as specified for that portion of work.

26. **CLEANING DOWN**:-Refer condition 49 of IAFW-2249. The contractor shall clear work site & area allotted for storage of material & erection of plants so as to leave the whole premises clean and tidy to the entire satisfaction of Engineer-in-Charge before handing over the items/works. No extra payment shall be admissible to the contractor for this operation.

27. **CONSTRUCTION LABOUR WELFARE CESS/TAX**:- Consequent upon promulgation of ordinance by the President of India, the contractor is liable to pay element for construction labour welfare CESS/Tax. The rates quoted by the contractor shall be deemed to be inclusive of this CESS/TAX.

28 **PRICING OF DEVIATIONS**:-The principle enumerated in condition 62 of IAFW-2249 shall apply to this contract.

29. **MINIMUM WAGES PAYABLE**:-

(a) Refer condition 58 of IAFW-2249. Govt. of India Min of Labour (Deptt of labour and Employment) vide notification dated have fixed minimum rates of wages as up to dated from time to time for various categories of employees under the Minimum Wages Act 1948. The contractor shall not pay wages lower than minimum wages including allowances as fixed under said notification as updated till the date of receipt of tender. Where in any area the minimum rates of wages are also fixed by state Government for employees in employment in the construction/maintenance of buildings/roads/runways, the labour wages as fixed vide above said updated notification of Ministry of labour or respective state Government whichever is higher in respect of these areas shall be deemed minimum rates of wages payable under this contract and the contractor shall not pay lower than this minimum rates of wages.

(b) Contractor's attention is also drawn, amongst other things to the "explanations" to the schedule of minimum wages referred to above.

(c) The fair wages referred to in condition 58 of IAFW-2249 will be deemed to be the same as the minimum wages, referred to above as up to dated from time to time.

SPECIAL CONDITION (CONTD…)

(d) Sch. of minimum wages has not been enclosed along with tender documents. However, contractor shall be deemed to have verified the minimum wages payable as on the last due date of receipt of tender.

(e) The contractor shall have no claim whatsoever, if on account of local factor and/or regulations, he is required to pay the wages in excess of minimum wages as described above during the execution of work.

(f) Wages/salary of employed labourers and supervisor/deployed staff shall be paid by the contractor through Aadhar Linked bank account of individual labour and deployed staff. Relevant details of payment made by the contractor through bank shall be produced for verification of principal employer (Garrison Engineer).

30. TAXES

Tendered rates are inclusive of :-
(i) Goods & Services Tax (GST)
(ii) Laobur welfare cess & (iii) Other levies, royalties & duties.

31. RE-IMBURSEMENT/REFUND ON VARIATION IN TAXES DIRECTLY RELATED TO CONTRACT VALUE

(a) The rates quoted by the contractor shall be deemed to be inclusive of all taxes including GST, Labour welfare cess/Tax, duties, royalties and other levies.

(b) (i) The taxes which are levied by Govt at certain percentage rates of contract Sum/Amount shall be termed as “taxes directly related to contract Value” such as GST on works contracts, Labour Welfare cess/tax and like but excluding Income Tax. The tendered rates shall be deemed to be inclusive of all “ taxes directly related to contract value” with existing percentage rates as prevailing on last due date for receipt of tenders. Any increase in percentage rate of “taxes directly related to contract value” with reference to prevailing rates on last due date for receipt of tenders shall be reimbursed to the contractor and any decrease in percentage rate of “taxes directly related to contract value” with reference to prevailing rates on last due date of receipt of tenders shall be refunded by the contractor to the Govt/deducted by the Govt from any payments due to the contractor. Similarly imposition of any new “taxes directly related to contract value” after the last due date for receipt of tenders shall be reimbursed to the contractor and abolition of any taxes directly related to contract value” prevailing on last due date for receipt of tenders shall be refunded by the contractor to the Govt/deducted by the Govt from the payments due to the contractor.

(ii) The contractor shall, within a reasonable time of this becoming aware of variation in percentage rates and/or imposition of any further “taxes directly related to contract value”, give written notice thereof to the GE stating that the same is given pursuant to this special condition, together with all information relating there to which he may be in a position to supply. The contractors shall submit the other documentary proof/informations as the GE may require.

(iii) The contractor shall, for the purpose of this condition keep such books of account and other documents as are necessary and shall allow inspection of the same by a duly authorized representative of Govt, and shall further, at the request of the GE furnish, verified in such a manner as the GE may require, any documents so kept and such other informations as the GE may require.

(iv) Re-imbursement for increase in percentage rates/imposition of “taxes directly related to contract value ”shall be made only if the contractor necessarily and properly pays additional “taxes directly related to contract value” to the Govt, without getting the same adjusted against any other tax liability or without getting the same refunded from the concerned Govt authority and submits documentary proof for the same as the GE may require.

SPECIAL CONDITION (CONTD…)

32 GST AND LABOUR WELFARE CESS :- It shall be deducted by the disbursing officer at the prevailing rates from gross amount payable to the contractor. Contractors quoted rates shall include taxes to be paid/ deducted at source by the disbursing officer. Rate of recovery of taxes shall be all as per Govt notification /SRO as applicable on last date of receipt of tender. Nothing extra is payable on this account.

32.1 Rate of Tax to be paid/recovered at the source under Goods & Services Tax (GST) shall be the percentage on gross value of contract as prescribed by the Central Board of Excise & Custom authorities.

32.2 Contractor shall get registered with the Central Board of Excise & Custom for allotment of ARIN / ID / GSTIN. Contractor shall file monthly/ quarterly returns for GST as per instructions in the matter from the concerned authorities.

33. **WORK ON HOLIDAYS** :- The contractor shall not carry out any work on gazetted holidays, weekly holidays and other non-working days except when he is specially authorised in writing to do so by the GE. The GE may at his sole discretion declare any day as holiday or non-working day without assigning any reason for such declaration.

34. TESTING OF MATERIAL

(a) The contractor shall be responsible for various tests to be carried out as listed in Appx ‘A’ in the site laboratory which will be set up by the contractor for testing of material at his own cost and arrangement and for that, they shall employ a competent technical representative & staff as approved by GE. All such tests shall be carried out in presence of Engineer-in-Charge. The cost of labour & materials for testing shall be borne by contractor.

(b) The rates of testing given in Col 7 of Appx ‘A’ shall be per lot of samples. No charges for tests carried out in site laboratory shall be recovered from the contractor. However, in case, a test which was specified to be carried out in site lab, and which could not be carried out in the site lab due to some reason, such test will be carried out in Zonal lab/ Central test lab (CTL) & any other approved lab . Recovery for the same shall be effected from the contractor at the rates indicated in Appx ‘A’.

(c) The tests which are specified to be carried out in Zonal lab and any other approved lab, charges for the same shall be borne by contractor at the rates indicated in Appx ‘A’, only if the test results disclose that materials are not in conformity with provisions of the contract. However, if the tests disclose that materials are in conformity with the provisions of the contract no recovery shall be effected from the contractor.

(d) For the test to be carried out in laboratories other than site laboratory, the facility & cost for conveyance of samples shall be provided by the contractor. The charges for testing shall however be borne by the department if test results are in conformity with contract provisions, otherwise the recovery for such test shall be made from the contractor as per the rates indicated in Appx ‘A’ or the actual charges whichever is higher.

35. ENFORCEMENT OF EMPLOYEES PROVIDENT FUND AND MISCELLANEOUS PROVISIONS (EPF & MP) ACT 1952

(a) Contractor shall get registered with employees provident fund organization. PF organization will allot **Provident Fund Code Number** to the contracting firm. Scanned copy of letter of allotment of Provident Fund Code number, (showing allotted code number) issued by the EPF organization shall be uploaded by the contractor in cover 1 (T bid) along with scanned copy of DD towards tender cost and application letter. Those bidder not registered with PF organization and not upload above scanned copy of letter of allotment of PF code number in cover 1 (T bid), they shall be disqualified during evaluation of T bid and consequently their financial bid (cover2) shall not be opened.

SPECIAL CONDITION (CONTD…)

(b) All workers/tradesmen deployed by the contractor under subject contract shall be enrolled as members of Provident Fund and shall be given Universal Account Number (UAN).

(c) While submitting payment under RAR and final bill contractor shall render relevant certificate duly supported with challans to Engineer-in-Charge/Garrison Engineer giving confirmation that all workers employed directly/indirectly by him under subject contract are enrolled for EPF and contribution toward EPF credited till date into their respective accounts (UAN).

(d) Failure by the contractor of compliance of EPF and MP Act shall be an offence and attract penalty from the competent authorities in the matter.

37. QUALIFIED TRADEMEN : (APPLICABLE FOR WORKS COSTING RUPEES ONE CRORE OR MORE)

In compliance with the condition 26 of IAFW-2249 (General Conditions of Contracts), the contractor shall employ skilled/semi skilled tradesmen who are qualified and possessing certificate in particular trade from Industrial Training Institute/(ITI)/ National Institute of Construction Management and Research (NICMAR)/ similar reputed and recognized Institutes by State/Central Government, to execute the works of their respective trade. The number of such qualified tradesmen shall not be less than 25% of total skilled/semi skilled tradesmen required in each trade. The contractor shall submit the list of such tradesmen along with requisite certificates to Garrison Engineer for verification and approval. Notwithstanding the approval of such tradesmen by GE, if the tradesmen are found to have inadequate skill to execute the work of their trades, leading to un-satisfactory workmanship, the contractor shall remove such tradesmen within a week after written notice to this effect by the GE and shall engage other qualified tradesmen after prior approval of GE. GE's decision whether a particular tradesmen processes requisite qualification, skill and expertise commensurate with nature of work, shall be final and binding. No compensation whatsoever on this account shall be admissible.

38. BLASTING : Blasting in any form is prohibited.

39. SITE DOCUMENTS

In addition to the site documents which are normally maintained for works, the following additional documents (as applicable for this work) shall be maintained at site and signed by GE / his authorized representative and the Contractor :-

a. Sieve analysis for coarse / fine aggregate.

b. Water absorption test for aggregates of different sizes/ grading.

c. Stage passing register.

d. Daily works diary register.

e. Register of materials daily arrived at site.

f. Material approval register.

g. Cement consumption register.

h. Register for clay, silt and impurities in aggregate test.

i. Impact value of stone aggregates of different sizes.
j. Crushing value of stone aggregates of different sizes.
k. Flakiness index of stone aggregates of different sizes.
l. Rolling hours Register.
m. Aggregate abrasion value register.
n. Register for soundness test.
o. Register for temperature control at boiler out at paver.
p. Cube testing register.
q. Beam testing register.

SPECIAL CONDITION (CONTD…)

40. DISPUTE RESOLUTION BOARD (DRB) (APPLICABLE IN CONTRACTS CONCLUDED BY ZONAL CE/CCE OF VALUE ABOVE RS 10 CRORES).

i. During execution of the works or after completion or after determination/cancellation/termination of the contract, all disputes between the parties to contract arising out of the contract (except those for which decision of Accepting Officer or any other officer (CWE and/or GE) is expressed to be final and binding), including any disagreement by either party with any action, inaction, opinion, instruction, certificate or valuation by the Accepting Officer or his nominee, the matter in dispute shall, in the firstplace be referred to the Dispute Resolution Board (DRB). In case of disagreement with the decision of such DRB, any party may invoke arbitration as per contract conditions.

ii. The constitution of the DRB shall be a three member body as under :-

a. **Chairman:** Jt DG (C) of the concerned Command Chief Engineer/ADG. WhereJt DG (C) is not posted in the Command/ADG office, any other Chief Engineer/ Brig level Officer posted in CE Command/ADG shall be nominated by Command CE athis sole discretion.

b. **Member 1 :** Col/Director level officers of command/ADG office or any otherZonal CE to be nominated by Command CE/ADG.

c. **Member 2 :** Col/Director level officers of command/ADG office or any otherZonal CE to be nominated by Command CE/ADG.

iii. The name of chairman and members shall be notified by the Accepting Officer within one month of the date of acceptance of contract.

iv. Once the DRB is constituted the members and chairman shall disclose in writing their neutrality and impartiality about any personal interest in the work.

v. The dispute shall be referred to the chairman of the DRB by the concerned party after giving notice to the other party for invoking of this clause.

vi. The DRB shall decide the dispute in accordance with the terms of the contract, principle of natural justice, equity and fair play.

vii. The DRB may fix oral hearing at a place, date and time as decided by the Chairman.

viii. The requisite administrative support to the DRB shall be provided by the Accepting Officer.

ix. All the contract documents pertaining to the case shall be provided by the Accepting Officer for reference by the DRB.

x. DRB shall give its decision on the disputes within three months of notice from anyparty invoking the DRB clause. This period can be extended by one month with the consent of the parties.

xi. All the decisions given by the DRB shall be by majority and such decisions shall be communicated in writing by the Chairman to the parties.

xii. If the decision of the DRB is not to the satisfaction of either party or if the DRB failsto give decision within the laid down time, either party shall indicate his reservations to the other party within 30 days of such decision and to refer that dispute for arbitration withinthe provisions of Condition 70 of IAFW-2249 General Conditions of Contract.

xiii. It shall be mandatory for the party invoking arbitration on any particular dispute tohave first exhausted the remedy provided under the DRB clause for that particular dispute.

xiv. The mandate of the DRB shall terminate on completion of one year from the date of completion/ determination/cancellation/termination of the contract.

xv. If any member or Chairman of the DRB is unable to function due to any reason whatsoever, or he resigns his appointment, Chief Engineer Command/ADG as the casemay be, shall fill the vacancy so caused within 15 days of the occurrence of such vacancy.

xvi. Any dispute referred to the DRB and having been decided by the DRB and not objected to by either party within 30 days shall attain finality and shall not be referable to arbitration.

xvii. Accepting Officer shall ensure implementation of the decisions of the DRB which attain finality, ie except those which are objected to by him or by contractor within 30 days as per Para 12 above. The amount so decided shall be processed as per AIP and the authority be mentioned as 'decision of DRB'.

xviii. Findings and decision of DRB shall be admissible as evidence, to the extent permissible as per law, in the subsequent arbitration and/or litigation.

xix. DRB Chairman/members shall not, in any case, be liable to be called as witness or to produce any evidence in any Arbitration or departmental proceedings of any kind.

xx. During execution of work the disputes may be referred to the DRB as per the requirement of each party after having exhausted the decision making process provided in the contract. In case of completion of work or after determination/cancellation/termination of the contract all the disputes including payment/ nonpayment/delay in final bill shall be simultaneously referred to the DRB within six months of completion/determination/ cancellation/termination of the contract.

41. **ROLLERS (**Reference condition 15 of IAFW-2249)

41.1 The contractor shall deploy Pneumatic type Roller, Tandem vibratory Roller and smooth wheel road roller as per requirement basis and a log book for each road roller shall be maintained by him for recording working hours. Entries in the log book shall be signed by the contactor or his authorized representative and by the Engineer-in-Charge. To ensure proper consolidation, roller must work for at least the number of days assessed on the basis of output given here under. If the roller has not worked for the number of days so assessed, recovery shall be affected from the contractor for the number of days falling short of the days assessed on the basis of output stipulated. The recovery shall be affected as under:-

(a) Roller deployed/ hired by the contractor then recovery at the rate of Rs 5000/- per working day of 8 hours. The above provision shall not, however absolve the contractor of his responsibility of properly consolidating surfaces as required under the provisions of the contract.

41.2 **OUTPUT PER DAY OF 8 HOURS WORK**

(i)	Consolidation of formation surface/sub grade	1850 Sqm
(ii)	Consolidation of stone soiling 20cm thick (spread thickness) with 8 to 12 tonne power road roller	590 Sqm
(iii)	do- 15 cm thick –do-	800 Sqm
(iv)	Consolidation of water bound macadam (stone metal) 10 cm compacted thickness including spreading and consolidation with blinding material	185 Sqm
(v)	Consolidation of water bound macadam (stone metal) 7.5 cm compacted thickness including spreading and consolidation with blinding material	248 Sqm
(vi)	Consolidation of single coat surface dressing Consolidation of single coat surface ressing with blinding material	774 Sqm
(vii)	-do- two coat surface dressing	558 Sqm
(viii)	Consolidation of 2.5cm thick premixed carpet including seal coat	600 Sqm
(ix)	Consolidation of 2 cm thick premixed carpet including seal coat	744 Sqm
(x)	Consolidation of 4 cm thick premixed carpet including seal coat	372 Sqm

Note: -

(i) Wherever tandom vibratory roller and pneumatic tyred rollers & vibratiory earth compactor is specified, the output of the same shall be as per manufacturer's instructions for the respective items so as to achieve proper compaction.

(ii) Any other Sch 'A' items viz WMM, BM & SDAC for which output not covered above shall be determined on actual site basis through BOO ordered by the CWE well before Commencement of road work.

SPECIAL CONDITION (CONTD…)

42 **EMPLOYMENT OF PERSONNEL**

42.1 The contractor shall employ only Indian National as his representative, servants and workmen and verify their antecedents and loyalty before employing them for work. He shall ensure that no person of doubtful antecedents and nationality is in any way associated with the works. If for reasons of technical collaboration or other consideration the employment of foreign national is unavoidable, the contractor shall furnish the particulars to this effect to the Accepting Officer at the time of submission of the tender. As proof that the contractor has employed only Indian Nationals he shall render a certificate to the GE within one month from the date of acceptance of the tender to this effect. In case the GE desires, contactor will have the police verification done of the personnel employed by him. Contractor's attention is also drawn to condition 25 of IAFW-2249 in this connection.

42.2 The GE shall have full powers and without giving any reason, to require the contractor immediately to cease to employ in connection with this contract, any Agent or servant or employee whose continued employment is, in his opinion, undesirable. The contactor shall not be allowed any compensation on this account.

42.3 Contractor's Supervision : Refer Condition 25 of IAFW-2249 along with latest amendments.

43 GUARANTEE IN RESPECT OF ANTI TERMITE TREATMENT

43.1 The contractor shall stand a guarantee of 10 (TEN) years to the Government for the work against recurrence of termite infestation (viz. effectiveness of the treatment carried out). For giving guarantee, if any additional treatment over and above that is specified is required, the same shall be carried out by the contractor without any extra cost to the Government. The contractor shall furnish a written guarantee for the effectiveness of the treatment for a period of TEN YEARS in favour of Garrison Engineer.

43.2 The amount of security deposit to be held back from the Contractor's bill against the guarantee for anti -termite treatment shall be calculated on the amount of anti- termite treatment at contract rates as per the scales given below enhanced by 1.25 times of the amount so calculated. This shall be worked out by GE and intimated to the Contractor. This amount shall be refunded to him after the expiry of the guarantee period. Alternatively, the Contractor may give a separate interest bearing security deposit to GE valid for 10 Years for this amount.

	Cost at contract rates	Amount of Security Deposit
(i).	Upto Rs. 50 lakhs	2% of the amount subject to a minimum of Rs. 5000/-
(ii).	Over Rs. 50 lakhs and upto Rs. 100 lakhs	Rs. 1,00,000/- + 1.5% of the amount exceeding Rs. 50 Lakhs
(iii).	Over Rs. 100 lakhs and upto Rs. 500 lakhs	Rs. 1,75,000/- + 1% of the amount exceeding Rs. 100 Lakhs
(iv).	Over Rs. 500 lakhs and upto Rs. 1500 lakhs	Rs. 5,75,000/- + 0.5% of the amount exceeding Rs. 500 Lakhs
(v).	Over Rs. 1500 lakhs	Rs. 10,75,000/- + 0.5% of the amount exceeding Rs. 1500 Lakhs subject to Rs. 18.75 Lakh

43.3 Should the GE at any time during construction or reconstruction or prior to the expiry of the guarantee period, finds that the buildings have been infested with termites, the contractor shall on demand in writing from the GE specifying the building(s) complained of, not-with-standing that the same may have been inadvertently passed, certified and paid for, undertake to carryout such treatment as may be necessary forthwith to render the building(s) free from termite infestation at his own expense till expiry of the guarantee period. In the event of his failure to do so within the specified period to be specified by the GE in his demand aforesaid, the GE may undertake such treatment at the risk and expense in all respects of the contractor. The liability of the contractor under this condition shall not extend beyond the period of ten years from the date of completion, unless the GE had previously given notice to the contractor to rectify the defects.

SPECIAL CONDITION (CONTD…)

43.4 The contractor shall provide a plate/board of adequate dimension but not less than 90cm x 45 cm x 2 cm thick in cement mortar (1:4) on one side of each building and indicated thereon the Contract No, Name of contractor and Name of the agency executing the anti termite and water proofing treatment and the date of expiry of the guarantee period.

44. **Conciliation (APPLICABLE IN CONTRACTS CONCLUDED BY ZONAL CE/CCE OF VALUE UPTO RS 10 CRORES).**

(a) Refer condition 71 IAFW-2249. Consequent to enactment of the Arbitration and Conciliation Act 1996 and with an objective to encourage settlement of disputes through Conciliation.

(b) On finalization of conciliation proceedings, the settlement agreement shall be signed by the Accepting Officer on behalf of Union of India after ascertaining the availability of funds.

(c) In case the net total amount of payment involved in the settlement agreement exceeds Rs 2.00 Lakhs, the Accepting Officer will consult the next higher authority and obtain clearance before conclusion of conciliation proceedings.

(d) The amount of settlement agreement shall be treated as an agreed amount payable under the contract and such payment shall be processed through a hand receipt.

(e) This provision is applicable to contracts of value less than Rs 10 Crores concluded by Zonal CEs/CCEs.

(f) The following special condition shall be inserted in the tender documents

1. Scope of Conciliation. The scope of conciliation shall be restricted to the following types of disputes with financial limits as indicated there in:-

(a) Disputes relating to levy of compensation for delay in completion-actual amount of compensation.

(b) Disputes relating to technical examination of works.

(c) Disputes relating to interpretation of the provisions of the contract with reference to their application to parties.

(d) Disputes relating to non return of Schedule ‘B’ stores over-issued to the contractor.

(e) Any other dispute having fair chances of being resolved by conciliation and considered fit to be referred to conciliation by the parties.

For item (b), (c), (d) and (e) as stated above, the financial limit shall be rupees two Lakhs or one percent of the contract amount whichever is less for each item.

2. Commencement of Conciliation Proceedings .

(a) The party initiating conciliation shall send to the other party a written invitation to conciliate, briefly identifying the subject of the dispute.

(b) Conciliation proceedings shall commence when the other party accepts in writing the invitation to conciliate.

(c) If the other party rejects the invitation, there will be no conciliation proceedings. If the party initiating conciliation does not receive a reply within 30 days from the date on which he sends or within such other periods of time as specified in the invitation, he may elect to treat this as a rejection of the invitation to conciliate and if he so elects, he shall inform in writing the other party accordingly.

SPECIAL CONDITION (CONTD…)

3. Number of Conciliators. There shall be a Sole Conciliator.

4. Appointment of Conciliator. All disputes brought out in Para 1 (a) to (e) above shall be referred to the Sole Conciliator viz. Serving officer not below the rank of Superintending Engineer/Superintending Engineer (QS&C) having degree in engineering or equivalent or having passed final/direct final examination of Sub Division II of Institution of Surveyors (India) to be appointed by the Engineer-in-Chief, Army Headquarters, New Delhi or in his absence the officer officiating as Engineer-in-Chief or Director General of Works if specifically delegated by the Engineer-in-Chief in writing.

5. Status of Effect of Settlement Agreement. The settlement agreement signed by the parties as a result of conciliation proceedings shall have the same status and effect as an arbitral award on agreed terms.

_____ _____

(Signature of Contractor) Dy Dir (C)/ Asst Dir (C)
For Accepting Officer

SPECIAL CONDITIONS [CONTD. /.] ANNEXURE A TO SPECIAL CONDITION

MATERIALS AND THEIR TESTS (OTHER THAN STEEL / CEMENT)

Srl No	Material	Tests	Method of testing	Frequency of tests			Level of test	Rate per test Rs.	Remarks
1	2	3	4	5			6	7	8
1.	Fly ash Brick	1) Compressive strength	IS-3495 (Part-II)	As per IS-5454 as given under:-			A	330/-	Checks for visual and Dimensional characteristics shall also be carried out
		2) Water Absorption 3) Efflorescence	--Do– ---Do… (Part-I)	Lot Size. 1001 to 10000 10001 to 35000 35001 to 50000	sample Size 5 10 15	permissible Nos of defective bricks 0 0 1	A A	330/- 330/-	as per IS:5454 Legend A-Site Lab B-Zonal Lab/Govt Engg college. C-National test house/SEMTWING/Engg. College.
2.	Coarse Aggregate	1) Sieve Analysis	IS:2386(Part-I)	One test for every 100 cum of aggregates or part there of brought to site.			A	660/-	
		2) Flakiness Index	---do--				A	250/-	
		3) Estimation of deleterious materials	---do–	One test for every 100 cum of aggregate or part thereof.			A	600/-	
		4) Organic impurities	---do--	One test per source of supply			C	275/-	
		5) Moisture content	IS:2386(Part-II)	Regularly as Reqd.			A	330/-	
		6) Specific gravity	--do--	One test for each source of supply.			B	330/-	

CONTD. /.

SPECIAL CONDITIONS [CONTD. /.] ANNEXURE 'A' TO SPECIAL CONDITION

MATERIALS AND THEIR TESTS (OTHER THAN STEEL / CEMENT)

Srl No	Material	Tests	Method of testing	Frequency of tests		Level of test	Rate per test Rs.	Remarks
1	2	3	4	5		6	7	8
3.	Fine aggregate	1)Sieve Analysis	(IS:2386Pt-I)	One test for every 100 cum of FA or part thereof when brought to site.		A	660/-	
		2) Test for clay, silt and impurities	--do–	--do--		A	500/-	
		3) Specific gravity	-do- But (Part- II)	One test for each source of supply		B	330/-	
		4) Test for organic impurities	--do--	One test for each source of supply.		C	275/-	
		5) Moisture content	-do-	Regularly as required subject to 2 tests/per day when being used.		A	330/-	
4.	Cement	1)Setting time	IS-4031-63 Reaffirmed 1980	Once for each consignment or as and when Reqd.		B	500/-	
		2) Soundness	--do--	--do--		C	550/-	
		3) Compressive strength	--do--	--do--		B	550/-	
		4) Fineness	--do--	--do--		C	275/-	
5.	Structural concrete	1) Slump test or compacting factortime or VEE-BEE test	IS-1199	The Min frequency of sampling ofconcrete of each grade shall be as under :-		A	300/-	(1) Random sample shall be carried out to cover all mixunits.
		2) Compressive strength	IS-516	Qty of Conc. IN THE work(M^3)	No of samples	A	900/- per sample	(2) Refer IS-456-2000 Clause. 15.2 for frequency of sampling.
				1-5 6 -15 16-30 31-50 51 and above	1 2 3 4 4 + 1 for each additional 50 Cum or part thereof			
6.	Water for constn purpose	1)Test for acidity 2) Test for alkalinity 3) Test for solid content	IS:456 & 3025 -do- -do-	Once at the stage of approval of source of water		B B C	500/- 500/- 500/-	Refer relevant clause of IS-456:2000

(Signature of Contractor)

Deputy Dir/ AD (Contract)
For Accepting Officer

PARTICULAR SPECIFICATION

1.0 GENERAL

1.1 Work under this contract shall be carried out in accordance with Schedule A', Special Conditions, Particular Specifications, drawings including notes thereon (unless specified otherwise) and general rules and specifications given in MES SSR Part I –(2009) as well as general rules, Special Conditions and preambles to the various rates given in MES SSR Part II– (2010) (MES SSR Part I & Part II hereinafter called MES Schedule) and these shall be read in conjunction with eachother.

1.2 The term General Specification referred to here in before as well as referredto in IAFW-2249 (General Conditions of Contracts) shall mean the specifications contained in the MES Schedule Part I & IS.

1.3 General Rules, Specifications, Special Conditions, method of measurements, preambles in the MES Schedule shall be deemed to be applicable to the work under this contract, unlessspecifically mentioned otherwise in these documents.

1.4 The term —as specified, wherever appears in tender documents and drawings, relates to relevant particular specifications and in its absence general specifications.

1.5 Particular specifications in this section given hereinafter shall be generally applicable to all works covered under Schedule —A". The particular specifications are brief and are only to particularize, amend and emphasize the specifications given in MES Schedule, which are not repeated.

1.6 Where specifications/provisions given in these particular specifications are at Variance with the provisions/specifications given in MES Schedule, specifications/provisions given in these particular specifications, hereinafter, shall be followed.

1.6 Where specifications for any item of work are not given in MES Schedule or in these particular specifications, specifications as given in relevant Indian Standard or Code of practice shallbe followed.

1.7 Referencetoanydrawingswhichismentionedonthedrawingformingpartof the tender but not specifically mentioned in the list of drawings shall be deemed to be forming part of the tender. The tenderer shall see such drawings/details in the office of Accepting Officer/concerned CWE/GE before quoting his tender.

1.8 The contractor shall not take cognizance of Note (s) appearing on drawings regarding the bearing capacity of soil taken into consideration while designing the foundation, as it is for departmental purpose only.

2.0 LAYOUT

2.1 In laying out buildings center line dimensions mentioned in the drawings or derived there from shall be strictly followed.

3. MATERIALS THAT LOOSE IDENTITY AFTER INCORPORATION:

3.1 Materials such as water proofing compound, paint, chemicals for anti termite treatment, cement, steel for reinforcement etc. the quantity of which cannot be checked after incorporation in the work shall, when collected at site be recorded in the Measurement Book and signed by both the MES representative and the contractor so as to check and ensure that the required quantity has been brought to site for incorporation in the work. Reference to Purchase Vouchers/invoice No, Name of the manufacturer/dealer from whom materials have been purchased shall be recorded in the Measurement Book (s)

3.2 Unless otherwise specified, the contractor shall obtain/procure materials direct from manufacturers or their authorised dealers/stockists where such authorised stockists have been appointed. The contractor shall produce original voucher/ invoice of supplier to the GE to ensure that the contractor has actually brought all the required quantity and the quality of the materials from the authorised dealer/manufacturers and also to find out the rate. The vouchers/invoice shall be endorsed date and verified and signed by Engineer-in-Charge and the Contractor and kept with the Engineer-in-Charge who shall forward these to the GE soon after completion of works.

3.3 Materials brought at site shall be stored as directed by the Engineer-in- Charge and those already recorded in Measurement Book shall be suitably marked for identification.

PARTICULAR SPECIFICATION

3.4 The contractor shall ensure that the materials are brought to the site in original sealed container or packing, bearing manufacturer‘s marking.

3.5 A register showing the daily consumption of the chemical for anti termite treatment shall be maintained, which shallbe signed by the Engineer-in-Charge andthe contractor in token of correctness of the entries therein. The register shall also indicate the following:-
(a) Name and address of the manufacturer and supplier.
(b) Requirement of chemical in the work as per specifications.
(c) The quantity actually used in the work.
(d) Batch number of chemical received and it‘s date of expiry.

4. **Blank**

5.0 EXCAVATION AND EARTHWORK:

5.1 SURFACE EXCAVATION:

5.1.1 Surface excavation as catered in sch A Part-XI (site clearance, excavation & earth work) for Bldg work under SI Item Nos 1.01 to 1.07 shall be carried out to a depth not exceeding 30 cm and 15 cm (average) in hard/dense soil to the extent required for the area covered all around the building & as directed by the EIC in order to remove inequalities.

5.1.2 All spoils, vegetation/shrubs & soil obtained from surface excavation shall be disposed off to a distance exceeding 500 m and not exceeding 1000 m in all as directed by the Engineer-in-Charge.

5.1.3 Surface excavation shall be carried out before the excavation for foundation is started.

5.1.4 Depth of foundation shown in drawings for the buildings is the depth after surface excavation.GL marked ondrawings shall be average GL as fixed byGE after surface excavation.

5.2 EXCAVATION

5.2.1 GENERAL:

5.2.2 The lump sum price quoted by the Contractor for Schedule —A‖ Part-I shall be deemed to include for all excavation and earthwork in hard/dense soil and soft/disintegrating rock. Deviation if any shall be priced based on actual soil met with at site at the applicable rates available in MES Schedule of Rates 2010 subject to contractor‘s percentage for both Add & omit portion.

5.2.3 If hard rock is met with during excavation, it shall be taken out and stacked as directed. Hard rock shall become Contractor‘s property for which the Contractor shall give a credit at the rate of Rs. 500/- per Cum. of solid measured of hard rock to the Govt (Stack measurement less than 50 % for voids). Soft/disintegrated rock shall be disposed off/reused in filling as and where permitted by the GE. The contractor may use the hard rock obtained from excavation in the work for building as stone, for hard core, lean concrete and WBM if the rock is approved by the GE in writing for use. The hard rock becoming contractor‘s property shall be removed off the site only after writtenpermission of the Engineer-in-Charge/GE.

5.2.4 On commencement of work as well as after carrying out the works under site clearance/surface dressing, levels will be taken jointly by the contractor and Engineer- in-Charge at suitable intervals in the grid system as directed by Engineer-in-Charge. These shall be noted down on graph sheet and noted down in level book/register and these shall be signed by both the contractor and Engineer-in-Charge and shall be used to arrive at average ground level after getting the same approved by GE in writing.

5.2.5 Decision of the Garrison Engineer shall be final, conclusive & binding as regards to classification of soils & rocks.

5.3 **DEWATERING**:
No extra payment over the lump sum amount quoted for buildings and services shall be admissible for dewatering, if water is met with or accumulated in the foundations or any other excavations due to any cause, whatsoever, and for excavation in mud, bailing and pumping of water if required, shall be done as described in Para 3.17 of MES Schedule Part-I.

PARTICULAR SPECIFICATION

5.4 **TIMBERING**:

The contractor's lump sum shall include the cost of timbering, if any, required to uphold the sides of excavation. If any timbering as specifically ordered to be provided in writing by the Garrison Engineer, consequent upon necessity arising as per site conditions, the same will be paid for under a deviation order.

In case timbering to excavation is required and specifically ordered by GE in writing this shall be paid as a deviation.

5.5 **FILLING:**

5.5.1 Earth obtained from surface excavation shall not be used for filling purposes for works under this contract and shall be removed from the site and filled in low lying area or at location directed by EIC/GE.

The approved earth free from roots and other vegetations obtained from excavation shall be used for filling in trenches, under floors and in other location as approved by Garrison Engineer.

Expansive or other unsuitable soil obtained from excavation shall not be used in filling. The decision of Garrison Engineer as to whether the soil obtained from excavation is suitable or not for filling, either partly or fully shall be final and binding. If the quantity of suitable soil obtained from excavation falls short of the filling required, the contractor shall bring the requisite quantity of approved earth from the places out side Ministry of Defence land/outside AF Station Maharajpur without any extra cost to Government.

5.5.2 **FILLING FOR NON EXPANSIVE SOIL:**

Contractor's tendered lump sum against Building works, Schedule 'A' Part-I shall include for the following: -

Contractor shall quote his lump sum taking into consideration that entire quantity of earth obtained from excavation (except that from surface excavation) shall be utilised for filling purposes like, returning, filling in trenches, filling under floors to obtain plinth height as shown on drawing etc and removal of surplus soil to a dist n exc 50 m

.

In case of earth obtained from excavation under this contract is insufficient for filling purposes, then contractor shall bring approved earth required to meet the quantity of filling from quarries situated outside MD land, outside Mhow Station without any extra cost to Government

The thickness of filling shown in drawings is after consolidation

All filling shall be spread in layers not exceeding 25 cm thickness, watered and consolidated.

5.6 **DISPOSAL OF SURPLUS EXCAVATED EARTH:**

All surplus spoil obtained from excavation for building work (Schedule 'A' Part-I) shall be disposed off to a distance n exceeding 250 m at the location (s) as directed by Engineer-in-Charge, deposited, spread and leveled all as specified in MES Schedule.

5.7 **MOORUM FILLING**

Moorum shall be as specified in clause No 3.21 and shall be spread and leveled as specified in clause No 3.21.4 of SSR Part I .

5.8 **PRE CONSTRUCTION ANTI TERMITE TREATMENT IN FOUNDATION AND GROUND FLOORS FOR THE BOQ ITEM**

5.8.1 The pre-construction anti termite treatment shall be carried out in buildings as catered under item No 1.01 of schedule A Part I of BOQ with the proportions and with specifications as speci-fied in clauses 3.26.1 of MES Schedule Part-I of 2009 except that no mound treatment shall be provided. Wherever 5 litre or 7.5 litre is mentioned in these clauses of MES Schedule Part-I it shall mean 5 litre or 7.5 litre of emulsion of concentration as indicated in MES Schedule. Antitermite treatment shall be car-ried out with emulsion of the chemical —chemical chlororpyrifos 20% EC (IS-8944) or Imidacloprid 30.5% SC (IS-16131) as per IS 6313 parts I and II- 2001. Con-centration by weight percent (Active Ingredients) shall be 1% in both cases of chemicals.

5.8.2 The work of anti termite treatment shall be got done from a specialist pest controller and exterminator agency, which shall be a member of Indian Pest Control Association, holding license as per Clause 13 of the Insecticides Act, 1968

PARTICULAR SPECIFICATION

5.8.3 The Lump sum rates quoted for the building works (Sch A Part-I) mentioned in clause 5.8 above shall include the following pre-construction Anti termite-treatment:-(a) Treatment to Masonry & column foundations, trenches/pits/ basements and treatment of RCC foundations and basements and backfill using chemical all as specified in Para 3.26.6 of MES Schedules Part I.

(b) Treatment to top surface of plinth filling and plinth protection all as specified in para 3.26.7 of MES Schedule Part I.

(c) Treatment at junction of walls and the floor all as specified in para 3.26.8 of MES Schedule part I

(d) Treatment along external perimeter of building all as specified in para 3.26.9 of MES Schedule Part I.

5.8.4 Chemicals brought to site in sealed container bearing ISI certification marks shall only be permitted to be used in the work.

5.8.5 (a)Chemicals shall be stored carefully at site. Seals of containers shall be broken only in the presence of Engineer-in-charge. Empty containers shall be removed off the site promptly. On any particular day, if the contentsofthe fullcontainercouldnotbe used in the work, the container shall be sealed at the end of the day in the presence of Engineer-in-Charge.

(b)Total quantity of chemical required for the work as per specification shall be worked out in order to ensure that full quantity is brought to site and used in the work. For this purpose, entries shall be made in measurement book (Not for payment/Not to be abstracted) indicating the brand, name, batch number, date of manufacturing, date of expiry, quantity brought etc, and signed by Engineer-in-Charge as well as the representative of the agency executing the work and contractor‘s representative.

5.8.6 **GUARANTEE**

5.8.7 The contractor shall stand a guarantee of 10 (TEN) years to the Government for the work against recurrence of termite infestation (viz. effectiveness of the treatment carried out). For giving guarantee, if any additional treatment over and above that is specified is required, the same shall be carried out by the contractor without any extra cost to the Government. The contractor shall furnish a written guarantee for the effectiveness of the treatment for a period of TEN YEARS in favour of Garrison Engineer.

The amount of security deposit to beheld back from the Contractor‘s bill against the guarantee for anti -termitetreatment shall be calculated on the amount of anti-termite treatment at contract rates as per the scales given below enhanced by 1.25 times of the amount so calculated. This shall be worked out by GE and intimated to the Contractor. This amount shall be refunded to him after the expiry of the guarantee period. Alternatively, the Contractor may give a separate interest bearing security deposit to GE valid for 10 Years for this amount.

	Cost at contract rates	Amount of Security Deposit
(i).		
	UptoRs.50 lakhs	2% of the amount subject to a minimum of Rs. 5000/-
(ii).		
	Over Rs. 50lakhsand upto Rs. 100 lakhs	Rs.1,00,000/-+1.5% of the amount exceeding Rs. 50 Lakhs
(iii).		
	Over Rs. 100 lakhsand upto Rs. 500lakhs	Rs.1,75,000/-+ 1% of the amount exceeding Rs.100 Lakhs
(iv).		
	Over Rs. 500 lakhsand upto Rs. 1500 lakhs	Rs.5,75,000/-+ 0.5% of the amount exceeding Rs.500 Lakhs
(v).		
	OverRs.1500lakhs	Rs.10,75,000/-+0.5% of the amount exceeding Rs.1500 Lakhs subject to Rs. 18.75 Lakhs

PARTICULARSPECIFICATION

5.8.8 Should the GE at any time during construction or reconstruction or prior to the expiry of the guarantee period, finds that the buildings have been infested with termites, the contractor shall on demand in writing from the GE specifying the building(s) complained of, not-with-standing that the same may have been in advertently passed, certified and paid for, undertake to carry out such treatment as may be necessary forthwith to render the building(s) free from termite infestation at his own expense till expiry of the guarantee period. In the event of his failure to do so within the specified period to be specified by the GE in his demand aforesaid, the GE may undertake such treatment at the risk and expense in all respects of the contractor. The liability of the contractor under this condition shall not extend beyond the period of ten years from the date of completion, unless the GE had previously given notice to the contractor to rectify the defects.

5.8.9 The contractor shall provide a plate/board of adequate dimension but not lessthan 90cm x 45 cm x 2 cm thick in cement mortar (1:4) on one side of each building and indicated there on the Contract No, Name of contractor and Name of the agency executing the antitermite and water proofing treatment and the date of expiry of the guarantee period.

5.9 HARDCORE

Refer Clause 3.27 of MES Schedule Part-I. Hard core shall be of broken stone of gauge not exceeding 63 mm, well graded to provide dense and compact sub base. Unless otherwise indicated or shown in drawings/specifications, the thickness of hard core shall be 100 mm consolidated. Hard core filling where indicated on drawings shall be spread, levelled in layers not exceeding 15cm thick well rammed, watered and consolidated. Moorum may be used to fill interstices in hard core. Thickness wherever shown on drawings/schedule of finishes shall be treated as consolidated thickness

6. **CONCRETE**

6.1 MATERIALS

6.1.1 **CEMENT**

Refer clause 4.3 on page 51 of MES Schedule Part-I.

6.1.2 **PROCUREMENT AND TESTING OF CEMENT:** cement required for completion of this contract shall be ordinary Portland cement 43 grade conforming to IS: 8112-1989 andeach bag shall have ISI certification mark and manufacture date.

All cement required for completion of this contract shall be procured by the contractor, under his own arrangement direct from the manufactures. The cement shall be procured directly from the following manufacturers. Contractor shall furnish particulars of the manufacturer of cement listed below along with the date of manufacturing to the GE for every lot of cement separately. The cement so brought shall be fresh & in no case older than 60days. From the date of manufacturing

Srl No	COMPANY NAME	BRANDNAME	ADDRESS	Type of Cement
1	2	3	4	5
1.	The Associated Cement Companies Ltd	ACC	414/421, Splendor Forum (4th Floor), 3, Distt Centre, Jasola, New Delhi-110044 Ph-011-46583600	All
2.	M/s Ultra Tech Cement	ULTRATECH	_B‘ Wing, 2nd Floor, Mahakali CavesRoad Andheri (East), Mumbai-400093	All
3.	The India Cement		Dhun Building, 827, Anna Salai, Chennai-600002	All
4.	M/s Dalmia Cement (Bharat)Ltd	DALMIA KONARK	Dalmiapuram, Dist tiruchirappalli, TamilNadu–621651	All
5.	Century cements	CENTURY	Industry House, 159 Chruch gate Reclamation, Mumbai-400020 Ph-022-22023936	All
6.	M/s Saurashtra Cement	SAURASHTRA	Gala NoA-1, Ground Floor, Udhyog Sadan No 3 MIDC,	All
7.	The Ramco Cements Ltd. (Formerly Madras cement)	RAMCO	Auras Corporate Centre, 98-A, Dr. Radhakrishnan Salai, Mylapore, Chennai-600004Ph-04424878666	All
8.	Mangalam CementLtd	MANGALAM	PO dityanagar, Morak, Dist- Kota, Rajasthan-326520 Mob-9351468076	All
9.	Birla Corporation Ltd	BIRLA	Birla Building (3rd & 4th Floor) 9/1 R.N Mukherjee Road Kolkata-700001 Ph-033-30573700	All
10.	Orient Cement	ORIENT	5-9-22/57/D, 2nd and 3rd Floor, GP Birla Centre, Adarsh Nagar, Hyderabad-500063Ph-04423688600	All
11.	M/s Nuvoco Vistas	NUVOCO	Equinox Business Park Tower-3, EastWing, 4th	All
12.	Shree Cement	SHREE	Bangur Nagar, Beawar, Distt-Ajmer, Rajasthan-305901 Ph-01462228101/06	All

13.	JKCement	JK	Kamla Tower Kanpur-208001	All
14.	JK Lakshmi CementLtd	JKLAKSHMI	Jaykaypuram, Dist-Sirohi, Rajashthan Ph-02971 244409/10	All
15.	JaypeeRewa Cement	JAYPEE	Japee Nagar P.O. Japee nagar, Rewa-486450 (M.P.)	All
16.	Ambuja CementLtd	AMBUJA	Kodinar, PO-Ambujanagar, Taluka-Kodinar, Distt-Junagadh, Gujrat-362715 Ph-02795237000	All
17.	M/S Shree Guru Kripa Cement (Pvt) Ltd	SARTAJ	4/4 Trikuta Nagar Jammu Ph- 0191-2472043	(a)OPC43 (b)PPC
18.	M/S Prasakti Cements Ltd.	PRASAKTI cements	123/3 RT, Plot#8-3-214/21 Srinivasa Nagar Colony (west) Hyderabad-500038 Tel- 040-44119100/200, Fax: 040-23747562	(a)OPC43 (b)PPC
19.	M/S My Home Industries Ltd	MAHA CEMENT	9th Floor, Block-3, My Home Hub, Madhapur, Hydrabad- 500081	(a)OPC43 (b)PPC (c)PSC
20.	M/S Chettinad Cement Corporation Limited	CHETTINAD Cement	4th Floor, Rani Seethai Hall, Anna Salai Hall, Chennai, Tamil Nadu 600006 Tel-044-42951800 Fax-044-28291558	(a)OPC43 (b)PPC
21.	M/S Sanghi Industries	Sanghi	10th Floor, kataria Arcade Off SG Highway, PO- Makarba, Dist- Ahmedabad, Pin-380051, Ph- 079-26838000	(a) OPC53 (b)PPC
22.	M/s Wonder Cement Ltd	WONDER Cement	17, Old Fatehpura Sevamandir Road, Udaipur-313004, Rajasthan (India) Tele +91-294-33991133 Fax-+91-294-30063333	(a) OPC43 (b) OPC53 (c) PPC
23.	M/S Kerosam Industries Ltd	Birla Shakti	Cement Division, Office No. 613 to 616, Block-3, White House, VI Floor, 6-3-1192/1/2, Kundanbagh, Begumpet Hyderabad - 500016, Telangana Phone:040-43344555 Fax-+91-40-4334-4534	(a) OPC43 (b) OPC53 (c) PPC

Note:- Contractor can procure cement even from those makes/ brandapproved by theAHQ E-in-C‘s Branchduring currency of work.)

6.1.3 The contractor shall frurnish the particulars of the manufacturer of cement along with the date of manufacture to the Garrison Engineer for every lot of cement separately. The cement so brought shall be fresh and in no case older than 60 days from the date of manufacture. The GE shall verify the document in support of the purchases of cement. Before placing order for supply of cement by the contrctor, he shall obtain written approval from the GE regarding name of manufacturer, quantity of cement etc. Cement shall be procured for minimum requirement of one month and not exceeding the requirements of the same for moe than two monts at a time. The cement shall be consumed in the work within three months after receipt. Cement shall conform to the requirement of IS Specification and each bag of cement shall bear relevant ISI mark. The weight of each consignment shall be verified by the GE and recorded. The content of cement shall be checked at random to verify the actual weight of cement per bag. However, the content of cement per bag shall be 50 kg only subject to tolerance given in Clause 9.2.1.1 and Annexure ‗B of IS-8112 and Clause 10.2.1.1 and Annexure ‗B of IS-1489 for OPC.

6.1.3 **TESTING:**

a) The manufacturer is required to carry out inspections and testing of cement in accordance with the relevant BIS provisions. The contractor shall submit the Manufacturer‘s Test Certificate in original along with the test sheet giving the result of each physical test as applicable andchemical composition of the cement or authenticated copy thereof duly signed by the manufacturer, with each consignment. The Engineer-in-Charge shall record these details in the cement Acceptance/Rejection Register as appended herein below after due verification.

b) The GE may organize independent testing as per IS:3635 (method of sampling hydraulic cement) and IS: 4001-1995 (Method of Physical test for hydraulic cement) and IS: 4030-1985 (Methods of chemical analysis of hydraulic cement) of random sample of cement drawn from each lot of the consignment from the National Test House, SEMT Wing, CME, Regional Research Laboratories, Govt Engineering college, IS approved laboratories and Zonal lab etc.

c) The contractor shall make available required quantity of cement and other facilities for testing of cement by GE as specified here-in-before and shall bear the cost of cement and testing there to irrespective of whether sample passes or fails. GE shall make paymentto thetestinghouse lab and direct the contractor to deposit the amount in Govt. treasury in favour of GE concerned and submit the treasury challan to the GE within 10 days of the letter issued by the GE for payment of cost of testing of cement. If contractor fails to do so, the testing charges shall be recovered from the dues of the contractor.

d) The required quantity of samples of the cement shall be taken in two polyethylene bags and sealed by the Engineer-in-Charge in the presence of representative of the contractor. The seal shall be authenticated both by the Engineer-in-Charge and representative of the contractor. Out of two samples, one sample shall be sent for testing and other retained by GE in his office. Record of these samples shall be kept by Engineer-in-Charge in the register and copy of the same shall be sent to GE/CWE/CE by him for record.
Cement brought by the contractor shall be allowed to be incorporated in the work after satisfactory test results of sample sent for testing. GE shall inform the contractor, about the test results and permission to incorporate in the work, in writing. GE shall ensure that word ―pass‖ is embossed on both sides of the cement bags of lot, which has been found as per specification as per test result. The cost of embossing as indicated herein above shall be borne by contractor.

e) In case the sample got tested by the GE from the institutions/ laboratories, mentioned here in before, fails, the concerned lot of cement shall stand rejected. The GE shall emboss a mark ‗X on both sides of each bag of the rejected lot. The GE will inform the contractor in writing about the test result and direct him to remove the rejected cement from the site of
works within two days of the order by the Engineer-in-charge. Contractor shall have no claim whatsoever on account of rejection and removal of cement.

f) It shall be the responsibility of the contractor to arrange cement well in advance, tofacilitate it‘s testing before use in the work. The contractor shall submit the program of procurement of cement to the GE before the date of commencement of the work indicating date of procurement and quantity.

6.1.1.4 STORING:

(a) Cement shall be stored in covered godown over dry platform at least 20 cm high in such a manner as to prevent deterioration due to moisture or intrusion of foreign matter. In case of store room the stack should be atleast 20cm away from floors and 60 cm from walls. The stacking of cement shall be done as specified in relevant IS. The storage accounting and preservation of cement supplied by the contractor shall be done as per standard Engineering practice till the same is incorporated in the work and the cost of the same shall be deemed to be included in the unit rate/amount quoted by the tenderer. The Engineer-in-charge (EIC) shall inspect once a day to verify that cement lying at site is stored, accounted, preserved and maintained as per the norms. The cement shall be stored so as to differentiate each tested and untested consignment separately with distinct identification. If the GE is not satisfied with the storage/preservation of cement, he may order for any test(s) of cement as applicable for that consignment to ensure its conformity to the quality mentioned in the in the manufacturer‘s test certificate. The contractor shall bear the cost of necessary testing(s) in this regard and no claim what so ever shall be entertained.

(b) Stacking of cement shall be done as per relevant IS and as under:-

(i) Eachcement consignment shall be stacked separately and removal shall be made on the basis of First in First out.

(ii) Adequate top cover will be provided.

(iii) Stacks in no case shall be higher than 12 bags. The maximum width of each stack shall be 3.00m. If the stack is more than 7 or 8 bags high, the bags shall be arranged in header and stretcher fashion, i.e. alternatively lengthwise and crosswise so as to pile together and avoid topping over

(c) Adequate space shall be kept between two stacks.

(d) Cement godown shall be provided with two locks on each door. The key of one lock at each door shall remain with EIC or his representative and that of the other lock with the contractor‘s authorized agent at site of works so that cement is removed from the godown only according to daily requirement with the knowledge of both the parties. During the period of storage, if any cement bag (s) is found to be in damaged condition due to whatsoever reason, the same shall be removed from the cement godown on written orders of the GE and suitable replacement for the cement bag (s) so removed shall be made and no claim whatsoever shall be admissible on this account.

(e) In case more than one type of cement is used in the work ie Ordinary Portland cement or Portland Pozzolana Cement, both type of cement shall be stored separately as directed by the Engineer-in-Charge to avoid mixing of these type of cement. Separate record shall be maintained including the location/items where these type of cement are used. (e) Cement shall be removed from the store only according to daily requirement with the knowledge of both the parties and the EIC and the contractor shall record daily consumption of cement in cement consumption register, which shall be signed. Cement constants given in Appendix _A‘ to E-in-C‘s branch letter No. 19280/E8 dated 03 May 1976 shall from the basis of consumption of cement for various items of works unless specifically indicated otherwise.

(f) In case the consumption of cement as per cement consumption register is found to be more than the estimated quantity of cement due to whatsoever reason, the contractor shall not have any claim whatsoeverfor such excess consumption of cement.

6.1.4 SCHEDULE OF SUPPLY

The contractor shall procure the cement timely as required in accordance with CPM chart agreed between GE and the contractor. The contractor will forfeit his right to demand extension of time if the supply of cement got delayed due to his failure in placing order in time to the manufacturer

6.1.4.2 MEASUREMENTS AND PAYMENT OF CEMENT

a) The entire quantity of cement shall also be suitably recorded in the measurement book for record purposes as Not to be abstracted before incorporation in the work and shall be signed by the EIC and the contractor.

b) The payment shall only be allowed after production of original purchase voucher, certified copies of test certificates from manufacturer for each consignment and results of testing carried out in laboratory on receipt of cement (7 days compressive test) are found satisfactory after testing at site as per condition 64 of IAFW-2249. Rate of payment given in SSR shall be applicablefor cement irrespective of type of grade of cement specified for use in the work.

6.1.1.5. **DOCUMENTATION**

The contractor shall submit original vouchers, test certificates and test sheets from the manufacturer/producers for the total quantity of cement supplied under each consignment to be incorporated in the work. The GE alongwith the relevant documents before sample is taken for the testing shall inspect all consignments received at the work site. The original vouchers and test certificates (original/authenticated) shall be defaced by the Engineer-in-Charge and photocopy of the same shall be kept on record in the office of the GE duly authenticated and with cross reference to the control number recorded in the cement acceptance/rejection register. The cement acceptance /rejection register shall be signed by (JE Civil), Engineer-in-Charge, GE and the Contractor. The Accepting Officer may order a Board of Officers for verification of connected documents produced by the contractor. The entire quantity of cement shall also be recorded in the measurement book (not to be abstracted) for record purpose before incorporation in the work and shall be signed by the Engineer-in-Charge and the contractor.

Record of cement brought by the contractor shall be maintained on the following format

CEMENT SUPPLY/ ACCEPTANC EREGISTER

CA No & Name of Work _____Control No*

_____Name of Manufacture/ Brand Name/Gde of Cement

(A) Manufacture_____(b) Brand _____(c)Grade_____

Qty of cement & lot No/ Week No (inBags):Qty_____(b) Lot No/Week No _____

Manufacturer's test certificates No_____Random Test Details

(a) Physicaltest report from _____vide their letter No_____

(Nameofapproved Lab/EnggCollege)

(b) Chemical test report from _____vide their letter No

____(Name ofapprovedLab/EnggCollege)

Details of Physical & Chemical properties

	Physical Requirements (As per IS: 4031)										Chemical Requirements (As per IS: 4032)							
As per manufacture certificate	Specific Surface (m2/Kg)	Soundness by Le Chateliar	Soundness by Auto Clave	Initial Setting time (Minutes)	Final Setting Time (Minutes)	03Days	07Days	28Days	Temp during testing °C	Standard Consistency (%)	Lime Saturation Factor (Ratio)	Alumina iron Ratio (Ratio)	In soluble Residue (%)	Magnesium (%)	Sulphuric Anhydride (%)	Loss on ignition%	Alkailes (%)	Chlorides(%)

Remarks with Signature Accepted/Rejected

Contractor Junior Engineer Engineer-in-Charge GarrisonEngineer

Remarks of BOO/ InspectingOfficer/ CWE

*To be allotted serially by GE consignment wise.

PARTICULAR SPECIFICATION

AGGREGATES

6.3.1 FINE AGGREGATE

Fine aggregate (sand) for concrete work shall conform to materials specifications and grading within the limits of grading Zone I to III as specified in Clause 4.4.1 to 4.4.6 and 4.4.7(2) of MES Schedule Part-I and shall conform to approved sample kept in the office of GE. Mixing of sand for obtaining specified grading from two different sources shall not be permitted.

6.3.2 (a) COARSE AGGREGATE

Coarse aggregate (stone aggregate) for all cement concrete work such as PCC/RCC shall be of approved quality all as specified in clause 4.4.1 to 4.4.7 (1) of

MES Schedule Part-I. Mixture of two types of stones shall not be permitted

(b) Coarse aggregate (stone aggregate) 20 mm and below shall be machine crushed and above 20mm may be hand broken.

(c) Grading of coarse aggregate (stone aggregate) unless specified otherwise in the specifications hereinafter in various locations/work i.e. PCC/RCC shall be as specified in MES Schedule Part I.

(d) Stone for lean concrete in foundation shall be sand stone conforming to IS: 383 from approved source. Unless otherwise specified in Schedule ‘A’ stone for all RCC/ PCC/WMM BM&ASDC work shall be crushed Granite/Trap/ Basalt or any other igneous rocks stone from approved source irrespective of any distance from site of work.

6.3.2.1 WATER

Water to be used in this work shall be all specified in IS : 456 of 2000 and in clause 4.9 of SSR Part I (specification). When the water is not issued by the department, the water shall be got tested in Govt. approved laboratory and test certificate shall be submitted to GE. The water shall be used in the work only after receipt of satisfactory test results as per IS.

6.3.2.2 SURKHI

Surkhi shall be as specified in clause 4.7 of MES schedule Part I.

6.3.3 CEMENT CONCRETE MIX :

6.3.3.1 Grade of concrete shall be as shown in drawings. At situations where grade of cement concrete is not indicated in drawings the same shall be as under:-

PARTICULAR SPECIFICATION

Srl No	Situation	Mixofconcrete	Nominal size of graded stone aggregate
1	2	3	4
(a)	All RCC works in beams, cols incl footings, roof slabs including RCC bands, plinth beam, RCC walls, RCC lintels, chajjas, shelves except water retaining structure.	M-25 (DesignMix Concrete)	20mm
(b)	PCC in foundation of walls of build-ings lean concrete under plinth/grid/toe beam, ingaps between plinth beam/Column footing and foundation of brick steps under Column footings.	(1:4:8) (Volumetricmix)	40mm
(c)	Fillinginsunkenflooretc.	(1:5:10) (Volumetricmix)	40mm
(d)	PCCfillinginframes/chowkhat	(1:3:6) (Volumetricmix)	12.5mm
(e)	PCC floor topping, DPC, shelves etc having thickness less than 40mm.	(1:2:4) (Volumetricmix)	12.5mm
(f)	PCC floor topping 40 mm thick and above but not exc. 100 mm thick.	(1:2:4) (Volumetricmix)	20mm
(g)	AllPCCworksforwhichnomixis specified else where	(1:2:4) (Volumetricmix)	20 mm for works above 40 mm thick and12.5mmfor work below 40mm thick.
(h)	PCC coping, padding underbearing of RCC beams/lintels/slabs, splash stones PCC blocks for fixing clamps, bed blocks. Cills, plugging for scaffolding holes etc.	(1:3:6) (Volumetricmix)	20mm

PARTICULARSPECIFICATION

6.4.1 BATCHING, MIXING, DEPOSITING AND RAMING

6.4.2 Controlled concrete materials shall be batched by weight only. Combined batching with digital weighing system and mixing plant with auto cut off and computer print out facility shall be used for concreting. The capacity of mini batching plant provided at site shall be adequate enough to execute the work as per the CPM, or otherwise the contractor shall provide single batching plant of higher capacity or more number of batching plants of adequate ca-pacities as required at site, as asked by the GE. No claim what so ever arising out on this account is admissible. The decision of GE in this regard shall be final and binding. The plant/plants shall have the digital system of adding specified quantity of water into concrete mix as per the designmixrequirement.

6.4.3 Water shall be measured either by volume in calibrated tanks or weighed. All measuring equipments shall be kept in a clean serviceable condition and their accuracy checked periodically.

6.4.4 Provisions as in clause 4.11.3.2 to 4.11.3.5 of MES Schedule Part-I shall be followed. All batching of concrete and accuracy of batching shall be as per Clause 10.2 of IS-456: 2000.

6.4.5 The mixing shall be done for at least 2 minutes and until a uniform colour and consistency is achieved.

6.4.6 Quantity of concrete mixed in any one batch shall not exceed the rated capacity of the mixer. The whole of the mixed batch shall be removed before mate-rials for fresh batch enter the drum. Concrete mix as approved shall not be modified by addition of water or otherwise in order to facilitate handling for any other purpose. On ceasing of work and other stoppage exceeding 20 mi-nutes, the mixer and other plants used for handling wet mix shall be tho-roughly washed with clean water. Pickup and throw over blades in the drum of the mixer which are worn down 20mm or more in depth shall be replaced with new blades.

6.4.7 All cement concrete, both plain and reinforced shall be mixed in mechanical mixer as specified in para 4.11.5 and 4.11.5.1 of MES Schedule Part-I. However for small quantity of concreting (other than RCC works) i.e., the quantity of concrete required being less than one batch of mix, the contractor may after obtaining written approval of Engineer in charge which shall be exceptional, adopt hand mix subject to addition of 10% extra cement without price adjustment where hand mixing permitted, it shall be carried out on a concrete platform and care shall be taken to ensure that mixing is continued until the concrete is uniform in colour and consistency.

6.4.8 All cement concrete, both plain and reinforced shall be mixed in mechanical mixer as specified in para 4.11.5 and 4.11.5.1 of MES Schedule Part-I. However for small quantity of concreting (other than RCC works) i.e., the quantity of concrete required being less than one batch of mix, the contractor may after obtaining written approval of Engineer in charge which shall be exceptional, adopt hand mix subject to addition of 10% extra cement without price adjustment where hand mixing permitted, it shall be carried out on a concrete platform and care shall be taken to ensure that mixing is continued until the concrete is uniform in colour and consistency.

6.4.9 All cement concrete both plain and reinforced concrete, shall be deposited and compacted all as specified in Clause 4.11.10 and 4.11.11 of MES Schedule Part-I. However, RCC work in columns, foundation, beams walls, chajjas and slabs etc., shall be compacted using mechanical vibrator, compaction of lean concrete shall be carried out by ramming and consolidated by tamping and rodding as specified. In the event of break down of mechanical mixer and vi-brator, the contractor must have arrangements for standby mechanical mix-er and vibrator.

6.4.7 DESIGN MIX CONCRETE

6.4.10 Grade of design mix concrete shall be as specified here-in-before and shall be as referred to in IS-456-2000 and as specified hereinafter. Design mix concrete may also be referred to as controlled concrete. Mix design shall be done as per IS-10262 of 1982 (Recommended guidelines for design mix concrete) and as described in SP-23 of 1982 (an IS publication).

PARTICULAR SPECIFICATION

6.4.11 The requirement of cement per cubic meter of controlled concrete of grade M25 shall be as per IS-456 of 2000. The actual requirement of cement for the controlled concrete shall be ascertained by the tests as specified he- reinafter. The design mix for M25 concrete shall be carried out for MOD-ERATE environment conditions and good quality control. The tenderer shall ascertain the quantity of cement required and quote the lump sum accordingly. No claim whatsoever arising on account of quantity of actual cement incorporate in the work on ac- count of design mix is admissible.

6.4.12(a) Contractor shall use liquid admixtures (Super plasticizers) to achieve the workability and to reduce the water content in design mix. Admixtures shall confirm to IS 9103: 1999 shall be from approved manufacturers as given below: -

(i) FOSROC Chemicals (India) Ltd.

(ii) Roffe Construction Chemicals Pvt. Ltd.

(iii) STP Speciality Chemicals Ltd.

(iv) CICO Technologies Ltd

(v) Para 5.5 of IS 456: 2000 be also referred for quality of admixtures.

(vi) For maximum dose of admixtures, please refer para 10.3.3 of IS 456: 2000.

(vii) Various tests as specified in IS 9103: 1999 shall be carried out for each batch of Admixtures at contractor‘s cost.

(b) Contractor shall submit original purchase voucher and test certificate of manufacturer for complete quantity of admixtures used in the work before claiming payment for the same.

(c) Complete quantity of admixtures including name of manufacturer, its brand name, date of manufacturing, date of expiry, voucher No. and details of test certificates shall be entered in MB as — Not to be abstracted — duly signed by JE, Engineer-in- Charge, GE and representative of contractor before making payment in RAR.

6.4.13 As soon as possible after receiving the work order to commence the work, the contractor shall submit samples of the materials required for preparing design mix concrete viz cement, coarse aggregate, fine aggregate and admixtures for approval of GE and intimate the place out of the following where they propose to carryout the design mix and preliminary tests RCC M-25 grade concrete:-

(i) Indian Institue of Technology (IIT)

(ii) Any Govt Engineering College

(iii) Any National Engineering College

(iv) Command Test Lab (CTL), Lucknow

(v) National Test House

6.4.15 The cement used in the work shall be as specified here in before. Coarse aggregate shall be crushed stone aggregate. The gradation shall be followed as per clause 4.2 table 2 of IS 383 to obtain maximum density.

6.4.16 After the samples of all the materials are approved by GE in writing sufficient quantities of these materials shall be forwarded by GE at contractor‘s expense for carrying out design mixes.

6.4.17 PRELIMINARY TESTS

6.4.18 Preliminary tests are tests conducted on the trial mixes of concrete produced in the laboratory with the object of:-

(i) Designing concrete mixes before the actual concreting operation starts.

(ii) Determining the adjustments required in the design, when there is change in the materials used during execution of work.

(iii) Verifying the strength of concrete mix at 28 day.

PARTICULAR SPECIFICATION

6.4.19 The preliminary tests shall consist of 3 separate sets of tests covering possible variation of gradation of aggregates and each set of test using a minimum 7 cubes of size 150 mm x 150 mm x 150 mm and one slump test. Three cubes shall be tested at 7 days to get indication of minimum strength of 28days. Other 03 cubes shall be tested at 28 days and 01 cube shall be preserved for Government use for subsequent testing. The compressive strength tests of cubes shall be performed as per IS-516. Casting of cubes and testing of these cubes shall be carried out in the presence of contractor‘s representative, GE / GE`s representative. It will be contractor‘s responsibility to ensure that design mix is carried out at the earliest. Contractor shall ensure that designmix cal-culations, supporting trail mix (03 Nos) details and test results of trial mixes along with recommended trial mix are submitted to GE at the earliest for his further action. Based on test results, the GE shall approve the design mix in writing. Copy of approved design mix shall be submitted to Accepting Officer within 10 days of approval by GE. The testing charges for the design mix and the tests conducted shall be borne by the contractor. The cost of materials, labour and transport shall also be borne by the contractor.

6.5 WORK TEST:

6.5.1 The work tests shall be carried out at site Lab/Zonal lab of Chief Engineer, Jabalpur Zone / Command Test Lab (CTL), Lucknow.

6.5.2 Work test shall be conducted as per clause 15 of IS-456 of 2000. At the commencement of the concreting, samples of concrete shall be taken on each day as specified in clause 15 of IS-456 of 2000 and specimens made at the work site out of the concrete being used in the works, for the purpose of testing compressive strength.

6.5.3 From each of these samples, 7 test cubes of size 150 x 150 x 150 mm shall be taken to test 3 specimens at 7 days and 3 specimens at 28 days in MES laboratory. CA No., date of casting and location where concrete is being used shall be marked on each concrete cube. One test cube of preliminary and work test shall be preserved duly marking the date of casting and CA No. for verifica-tion / subsequent testing, if required. The cube shall be preserved by the GE / Engineer-in-Charge until the defects liability period of the work is over.

6.5.4 The testing charges for the work tests conducted in the Zonal Laboratory/Command Test Lab (CTL) shall be at the rate mentioned in Appendix _B‘ of Particular specifications and the same shall be effected from the payments due to the contractor in RAR / Final bill whichever is earlier. The cost of materials, labour and transport shall be borne by the contractor. The lump sum quoted shall include the cost of testing the concrete cubes both for design mix / volu-metric mix.

6.5.5 In the event of contractor setting up the laboratory at site as specified here in before in Special Conditions, the contractor shall carry out cube testing in site lab, in presence of Engineer-in-Charge and as specified here in before. Howev-er, random testing up to 5 percent of total tests to check the compressive strength of cube shall be carried out in zonal lab for which testing charges shall be recovered from the contractor at the rate mentioned in Appx _B‘. Con-tractor shall include this aspect in his lump sum while quoting his rates.

6.5.6 The Engineer-in-Charge shall maintain the record for all the test carried out in Site lab /zonal lab /CTL separately. The cost of testing including material, labour etc., incurred shall be borne by the contractor and the lump sum quoted shall be deemed to include this.

6.6 MIXING

6.6.1 The mix design and also execution of work shall be carried out by weigh batching. The quantum of cement for execution of work by weigh batching shall be as per mix design.

6.6.2 It shall be ensured that the grading characteristics as adopted in the mix design are followed throughout. Wherever the type and/or batch of cement/aggregate is changed, a fresh mix design shall be carried out. Nothing extra is payable on this account.

6.6.3 The contractor during the progress of work shall not change the mix design without the prior approval of the GE.

PARTICULAR SPECIFICATION

6.6.4 Engineer-in-Charge shall maintain a record of actual consumption of cement in proper register (other than the cement register mentioned in special conditions) and initial the entry for every day of quantity of materials used by contractor. The register shall be got checked and signed by GE. In case the consumption of cement as per cement consumption register is found to be more than the estimated quantity of cement due to what so ever reason, the contractor shall not have any claim, whatsoever for such excess consumption of cement.

6.7 WATER CEMENT RATIO

6.7.1 It is most important to maintain the water cement ratio constant and to it‘s correct value. To this effect determination of moisture content in both fine and coarse aggregate should be made as frequently as possible. The frequency for a given job shall be determined by the Engineer-in-Charge. According to weather conditions the amount of water to be added shall be adjusted to compensate any variations in the aggregate, IS-2386 method of test for aggregate, for con-crete, Part II specific gravity, density, voids, absorption and bulking of aggre-gates due to variation in their moisture contents shall apply. The maximum quantity of water to be added shall be determined by mix design to be carried out as specified herein before

6.7.2 Workability of concrete shall be checked at frequent intervals. The slump test or where facilities exist the compacting factor test in accordance with IS-1199 may be adopted for this purpose.

6.7.3 The slump for M-25 shall be between 25 to 75 mm and for medium degree workability as given in clause 7 of IS-456-2000.

6.7.4 Curing shall be carried out all as specified in MES Schedule Part I. 6.7.5Acceptance Criteria: As per Clause 16 of IS-456: 2000.

6.6.A NOMINAL MIX/VOLUMETRIC MIX:

Wherever nominal mix concrete is specified, it shall be as per IS: 456. The periodicity of mix design and other tests on materials/mix shall be as decided by the GE.

6.7. MIXING

6.7.1. All concrete shall be mixed in mechanical concrete mixer. Where only small quantity of plain cement concrete is involved, hand mixing may be adopted if approved by the Garrison Engineer. The contractor should arrange to wash out and clean the mixing drum on completion of work and or on stoppage of work if stoppage is for more than 20 minutes.

6.8. TRANSPORTING, DEPOSITING AND COMPACTING:

Transporting depositing and compacting generally shall be carried out as specified in Clause 4.11.9, 4.11.10 and 4.11.11 of SSR Part I 2010.

6.9 PROTECTION AND CURING OF CONCRETE:

This shall be carried out in accordance with specifications, given in clause No. 4.11.13 and 4.11.14 of MES Schedule Part I 2010.

6.10 PRECAST CEMENT CONCRETE:

6.10.1 Lintels (without chajja) with a span of less than 1.5 m clear and PCC bed blocks, copings and the like may be precast at the discretion of the contractor all as specified in Clause 4.20 of SSR Part I as applicable. All precast articles shall be set in CM (1:3) with joint to match.

6.11 STRIPPING TIME FOR FORM WORK:

The contractor‘s attention is invited to the stipulation in Clause 4.11.6.3 of SSR Part I regarding stripping of form work. The periods stipulated therein are for concrete using ordinary Portland cement. In case PPC cement used in the work, stripping time of form work applicable as per relevant IS.

6.12 LEAVE/FORM HOLES AND CHASES:

The contractor as the work proceeds should Leave/form holes/chases in concrete/masonry and RCC where and as directed by the Engineer-in-Charge and make good in cement and sand mortar (1:3) when ordered to do so.

PARTICULAR SPECIFICATION

6.13 FINISH TO CONCRETE SURFACES (refer clause4.11.16, 4.11.16.14.11.16.2 (a), (b), (c), (d), (e) and 4.11.16.3 of MES Schedule Part-I.)

(a) Exposed surfaces of RCC chajjas, cantilevers and the like (bottom, sides) and soffits and sides of beams, soffit of RCC roof slab/floor slab/waist slab of stairs, sides of isolated column, shelves, RCC parapets, fins/fascias and the like RCC walls shall be plastered not less than 5mm thick with cement mortar (1:3) and finished fair and even without using extra cement after removal of form work.

(b) Exposed surfaces of RCC columns, beams, lintels, bands etc. which are continuous with plastered surfaces of walls shall be plastered & finished in the same specifications as for adjoining wall surfaces.

(c) Exposed surface of columns, beams, lintels, bands and similar items which are not covered under clause 9.1 (a) and 9.1(b) above shall be finished with a thin layer of 5 mm thick plaster in cement mortar (1:3) and finished fair and even without using extra cement after removal of form work.

(d) If thickness of plaster in cement mortar (1:3) as specified above is required to be increased in excess of 5 mm to achieve fair and even surfaces, it shall be provided by the contractor without any extra cost to Government and the same shall be deemed to have been included in the lump sum amount quoted by the contractor against Schedule ‘A’ Part I.

(e) Surfaces of concrete which are required to be plastered as indicated above after removal of formwork shall be roughened with wire brush and hacked out closely.

6.14 FORM WORK FOR RCC/PCC WORK

(a) Refer to clause 4.11.6.1 to 4.11.6.5 and relevant clauses of 7.15 of MES Schedule Part I. Formwork shall be of steel with adjustable steel vertical props and other steel accessories and shall conform to IS 14687:1999 (Guide lines for false work for concrete structure). Release agents shall be applied so as to provide a thin uniform coating to the finished flat surface of shuttering before laying RCC for slabs. Deformed steel sheets shall not be permitted for use as formwork. However for the purpose of pricing DOs, the rates given in MES Schedule Part II for fair finished surfaces for concrete using steel formwork shall be applicable.

(b) Where steel shuttering is not feasible in locations such as soffits and sides of lintels, beams, chajjas and other locations as decided by GE, the plywood board of adequate strength may be used.

(c) Wooden Ballies/Bamboo & wooden planks shall not be allowed to use as shuttering for work.

6.15 SAMPLING AND TESTING OF CONCRETE:

6.15.1 Refer Clause No. 4.11.17 of MES Schedule Part I.

6.15.2 Tests will be carried out on 15 cm cubes all as per IS: 516. Minimum Six cubes per sample shall be supplied by the contractor for testing.

6.15.3 For relatively small and unimportant works even though testing may be waived off by the GE, the contractor shall be responsible to achieve the desired strength of concrete.

6.15.4 The contractor shall provide all necessary materials (including moulds etc) and labour for mixing and casting of cubes, transporting the cubes to the testing laboratory and bringing back in same and any other assistance that may be required for getting the samples tested in the laboratory without any extra cost. The charges for testing of cubes shall be borne by the contractor. The testing charges shall be as indicated in Appendix _B‘ to Special Condition. In case the tests are concluded in site laboratory set up by contractor, the recovery for testing charges shall not be effected.

6.15.5 The concrete which is not upto the desire strength shall be rejected and the same shall be made good by the contractor without any extra cost to Government.

PARTICULAR SPECIFICATION

6.16 PCC COVING : PCC coving with PCC (1:2:4) type B-0 using 12.5 mm graded stone aggregate mixed with integral water proofing compound shall be provided at junction of RCC chajja with wall/lintel /beam to the radius of 75 mm irrespective of whether shown or not shown on drawings. Coving shall be 15 cm high on wall and to the full length of chajja.

6.17. FILLETING:

6.17.1 Provide triangular filleting to skirting / dado as directed by Engineer-in-Charge with CM (1:3) finished even and smooth without using extra cement.

6.18 CONCRETE PADDING: Padding under bearing of RCC lintels to make up the height of opening in wall wherever required shall be done with PCC (1:3:6) type C-1 PCC blocks shall not be cut to under sized dimension and used to make up the required height in such positions.

6.19 RCC LINTELS: The bearing of lintels shall be 15 cm or effective depth (Whichever is more), unless otherwise shown on drawings.

6.20 Lintels (without chajja) for opening not exceeding 1.5 m clear span may be precast at contractor‘s option. However these shall be priced as cast-in-situ lintels in the event of deviation if any. All other lintels and bands shall be cast-in-situ.

6.21 DRIP COURSE/THROATING / WEATHERING: Following works shall be executed whether shown on drawings or not:-

i) Provision of proper drip course and / or throating and weathering to all chajja, roof, cills, coping and the like.

ii) All flat surfaces exposed to weather shall be finished with imperceptible slope for smooth run of rain water. (In cases no slope is mentioned in drawings or specified).

6.22 WINDOW CILL

6.22.1 PCC window cill shall be in Grade M-15 (Nominal mix) finished even and smooth without using extra cement as shown on drawing on external side of wall.

6.22.2 Irrespective of whatever shown on drawing or otherwise specified provide machine polished kota stone slab in one piece and thickness 18-22 mm on internal side of wall rising on walls over 10 mm thick bedding layer in CM 1:6 and jointed in cement slurry in cills of windows/ventilators. Each side bearing shall be 40 mm.

6.23 PLINTH PROTECTION:

6.23.1 Plinth protection shall be of 75 mm thick PCC (1:3:6) type C-1 using 20 mm graded stone aggregate with surface finished even & fair without using extra cement, laid over 75 mm thick [compacted thickness] of brick Soling as specified here-in- before, over rammed earth/finished surface. Construction joints shall be provided at every 3 m Interval and at every change of direction as specified in SSR and shall be filled with sealing compound grade A.

PARTICULAR SPECIFICATION

6.24 RCC JALI

6.24.1 RCC Jali with mix 1:2:4 using aggregate of size 6 mm and below shall be pre- cast with wire reinforcement as per drawings and jointed in cement mortar (1:3). Thickness where not shown on drawings shall be 50 mm. Faces of RCC jali shall have fair and even surface. In the event of deviation RCC jali shall be priced as pre-cast using aggregate of size 12.5 mm subject to contractor‘s percentage quoted in Schedule `A‘ part I.

6.25 DAMP PROOF COURSE:

6.25.1 Damp proof course shall be 40 mm thick in PCC (1:2:4) type B-0 using 12.5 mm graded stone aggregate mixed with integral water proofing compound conforming to IS: 2645 and covered with 100 micron thick polythene film. The percentage of water proofing compound shall be as per manufacturer‘s instructions. The same shall be taken as 3% (by weight of cement) while pricing of any deviation order in omit portion.

6.25.2 Damp proof course shall be laid to the full width of walls, including under door and other opening at plinth level. Damp proof course shall not be provided for portion below door for 100/ 115 mm thick walls, dwarf walls and at locations where RCC / PCC plinth band or RCC plinth beams are provided.

7. 0 BRICK WORK GENERAL

Irrespective of type / size of bricks shown on drawings contractor shall use old size bricks (non modular bricks) Fly ash bricks (IS 12894-2002) for building works. However in case of non availability of non modular bricks , contractor may use modular Fly ash bricks subject to following:-

(a) Centre line dimension of walls shown on drawing shall remain unchanged.

(b) No adjustment both _plus‘ or _minus‘ shall be made in the quoted lump sum on account of changes necessitated in foundations, floor, lintels, cills and the like and any other items of work effected.

(c) In case of difference of levels due to thickness of brick courses the level- ling shall be done by providing PCC (1:3:6) type C-1 levelling course and its cost shall be deemed to be included in the lump sum as offered by the contractor.

7.1 Cement - Cement shall be as specified for concrete work.

7.2 Sand for Mortar - Refer to clause 5.4 of MES Schedule Part I and Ganga sand shall not be used.

7.3 BRICK:

7.3.1 FLY ASH LIME BRICKS WITH CEMENT ADDITIVE : The fly ash lime bricks shall conform to IS 12894. These bricks shall be machine moulded/pressed and compacted by vibration in auto/semi-automatic plant using appropriate proportion of fly ash, sand, lime and cement as additive.

7.3.2 Visually the bricks shall be sound, compact and uniform in shape, free from visible cracks, warpage and organic matters. The brick shall be solid with frog one of its flat side. The brick shall have smooth rectangular faces with sharp corners and shall be uniform in shape and colour.

PARTICULAR SPECIFICATION

MATERIAL:

7.3.3 Pulverized Fuel Ash (Fly Ash) : Fly Ash shall be Grade-I or Grade-2 conforming to IS-3812.

7.3.4 Sand : Deleterious material such as clay and silt in sand shall be less than 5%.

7.3.5 Lime : Lime shall be class ‗C' hydrated lime conforming to IS-712.

7.3.6 Additives : Cement shall be used as additive to enhance durability of the bricks.

7.4 PHYSICAL CHARACTERISTICS:

7.4.1 Compressive Strength : The minimum average wet compressive strength shall not be less than 7.5 N/mm² when tested in accordance with IS-3495 (Part-I). The wet compressive strength of any individual brick shall not fail below the minimum average wet compressive strength of brick by not more than 20 percent.

7.4.2 Drying Shrinkage : The average drying shrinkage of the bricks shall not exceed 0.15 percent when tested in accordance with IS-4139.

7.4.3 Efflorescence test : Rating of Efflorescence shall not be more than moderate.

7.4.4 Water absorption : Water absorption of bricks shall not exceed 20% when tested in accordance with IS-3495 (Part-II).

7.5 Testing of Bricks :

7.5.1 Sampling and tests : Samples of bricks shall be subjected to the following tests.

(i) Dimensional tolerance.

(ii) Water absorption.

(iii) Efflorescence.

(iv) Compressive strength.

7.5.2 Sampling: For carrying out the above tests, the samples of brick shall be taken as per IS:5454 at random according to the size of lot given in Table 1 below.The sample thus taken shall be stored in a dry place until tests are made.

7.5.3 The samples shall be taken as below :-

(i) Sampling from a stack : When it is necessary to take a sample from a stack,the stack shall be divided into a number of real or imaginary sections and the required number of real and imaginary sections and the required number of bricks drawn from each section.Sampling from trucks shall be as per IS: 5454 Scale of sampling and criteria for conformity for visual and dimensional characteristics:-

(ii) Visual characteristics : The bricks shall be selected and inspected for ascertaining their conformity to the requirements specified in subsequent Para .The number of bricks to be selected from a lot and shall be in accordance of Col1 & 2 of Table-1 for visual characteristics in all cases and dimensional characteristics if specified for individual bricks.

7.5.4 Visual characteristics : All the bricks selected above in accordance with Col 1 & 2 of Table 1 shall be examined for visual characteristics as specified in Para-b. If the number of defective bricks found in the sample is less than or equal to the corresponding number as specified in col 3 of table 1 the lot shall be considered as satisfying the requirements of visual characteristics, otherwise the lot shall be deemed as not having met the visual requirements.

PARTICULAR SPECIFICATION

7.5.5 Dimensional characteristics : The number of bricks to be selected for inspecting the dimensions and tolerance shall be in accordance with col 1 & 4of Table 1. These bricks will be divided into groups of 20 bricks thus formed will be tested as per IS :1077 for all the dimensions and tolerance as given below:-

Type of bricks - Non-modular bricks Nominal size - 230X114x75mm Actual size - 230x111x70mm Dimensional for group of 20 bricks Length 442 to 478 cm (460+18cm) Width 213 ro 231 cm (222 + 9cm) Height 134 to 146 cm (140 + 6cm)

A lot shall be considered having found meeting the Requirements of dimensions and tolerance if none of groups of bricks inspected fails to meet the specified requirements.

7.5.6 Scale of sampling and criteria for physical characteristics shall as per IS 5454.The Lot, which has been found satisfactory in respect of visual and dimensional requirements, shall be next tested for physical characteristics like compressive strength, water absorption, and efflorescence.

7.5.7 The physical requirements of the bricks shall be tested as per IS:3495 (Part I to IV).

7.5.8 The bricks shall have physical characteristics as specified here under:-

(i) Water absorption : 20% Max.

(ii) Compressive strength : 75 Kg/cm2.

(iii) Efflorescence : Nil to Slight sign of efflorescence i.e 0-10% of surface area of the brick should show a thin deposit of salt.

7.5.9 A lot shall be considered having satisfied the requirements of physical Characteristics if the condition stipulated here in are all satisfied:-

(i) The average compressive strength shall satisfy the requirements specified in para 7.5.1.

(ii) The acceptance criteria shall be as per IS.

(iii) Average water absorption shall be as specified in Para 7.5.4.

(iv) The number of bricks failing to satisfy the requirements of the efflorescence should not be more than the permissible in IS 5454.

7.5.10 Handling and storage of bricks shall be as per Clause 5.6.9 of SSR Part I.

7.5.11 The general quality of brick shall be as per Clause 5.6.5 of SSR Part I

7.5.12 For pricing deviation if any (both plus or minus) involving brick work with old size brick pricing shall be made on the basis of rates for old size bricks sub class_B‘ bricks given in MES Schedule of rates subject to contractor‘s percentage quoted by the tenderer against that schedule. The preamble (e) and (f) given on page 54 of SSR Part II shall stand deleted.

7.5.13 BEARING PLASTER :

Irrespective of what is shown on drawings, two layers of indigenous bituminous craft paper shall be provided to the full width of brick wall under bearing of lintels, beams, slabs resting on load bearing walls, PCC bed blocks and floor/roof band where casted monolithic with RCC slab on 20mm thick bearing plaster in CM (1:4) finished even and smooth and given one coat of white wash. Each layer of indigenous bituminous craft paper shall weigh not less than 1.20 Kg per 10 Sqm. This provision, however, shall not

be made where RCC slabs rest on a levelling course of PCC and in RCC framed structure.

7.5.14 Size of PCC bed blocks for beam where not shown in the drawings shall be twice the width of beams in length, covering the entire thickness of wall and 200 mm thick (depending upon type of brick) below the RCC beams resting on masonry.

PARTICULAR SPECIFICATION

8. WOOD WORK AND JOINERY

8.1.1 (a) Timber for all wooden joinery and wood work shall conform to specifications given in clause 7.3 of the MES Schedule Part I and defects shall be within the permissible limits as specified in Clause 7.4 and 7.5 of the MES Schedule Part I. Unless otherwise specified timber for all joinery and woodwork shall be 2nd class hard wood

(Hollock/Sal/Benteak/Bijasal) and as per the approved sample kept in GE‘s office.

NOTE: - However, The contractor may provide Shisham wood in lieu of Hollock/Sal/ Benteak/Bijasal wood without any extra cost to Govt.

(b) Timber shall be well seasoned and chemically treated under pressure (kiln dried). Factory made shutters shall also be kiln seasoned only. The moisture content of timber shall not exceed the limits laid down vide clause 7.7 of the MES Schedule Part I.

(c) (i) Preservative anti-termite treatment shall be carried out to all woodwork and joinery. Contractor‘s quoted rates shall be deemed to cater for this provision. Factory made plywood, particle boards and factory made shutters are not to be further treated with any chemical against termites.

(ii) Wood work and joinery other than factory made, ply wood particle board and factory made shutters as described above shall be treated with copper Napthanate or any other chemical specified in IS: 401 of 1982 and shall be applied in any one of the manners specified in IS or as directed.

(d) Timber surfaces hidden or not exposed to view and in contact with brick work/masonry /concrete shall be left clean sawn and applied with two coats of creosote, unless specified otherwise hereafter.

8.1.2 WORKMANSHIP

(a) Dimensions of various parts of joinery shall be as shown on drawing and shall supersede those stipulated in clause 8.20& 8.24 of MES Schedule Part I. For the purpose of pricing deviations, factory made shutters shall be classified as second class hard wood and priced accordingly at the applicable rates in MES Schedule Part-II and no adjustment in SSR rates shall be done for change in sizes of stiles and rails

(b) Irrespective of what is shown on drawings, use of nails in the joinery is prohibited. Wood screw of appropriate size shall be used and shall be of ―Nettle fold‖ brand or any other equivalent brand as approved by GE.

(c) All joinery work shall be of full dimensions as shown on drawings. Overall sizes of joinery as shown on drawings shall be maintained. Tolerance in sizes and thickness shall be all as specified in SSR Part-I.

(d) Plugging/wooden gutties to walls wherever shown on drawings or where required shall be done with wooden plugs made out of hard wood (Hollock/Bijasal/ SAL/Benteak/Shisham/Teak) as per clause 7.29 of MES Schedule Part I.

8.2. PLY WOOD AND PARTICLE BOARD

(a) Un-veneered particle board where indicated shall be flat pressed, BWP grade bonded with phenol formaldehyde synthetic resin adhesive conforming to IS: 3087 (Type I Part I) all as specified in para 12.13 of MES Schedule Part I.

PARTICULAR SPECIFICATION

(b). Veneered particle board where indicated shall be three layers flat pressed with commercial or teak veneer (both sides) as indicated on drawing/specified and shall be BWP grade bonded with phenol-formaldehyde synthetic resin adhesive as per IS-3097- Grade I, Type-I for commercial veneer and decorative veneer all as specified in para 12.14 of MES Schedule Part I.

8.3 FRAME (CHOWKHAT) FOR DOORS EXEPT WC/BATH

8.3.1 PRESSED STEEL FRAME: -Where pressed steel frames/chowkhats are shown on drgs these shall be machine pressed steel rebated made out of 1.25 mm thick MS sheets (factory made). Pressed steel frame shall comply to IS: 4351(1976). Hollow space of chowkhat frames shall be filled with PCC (1:3:6) Type C-0 using 12.5 mm graded stone aggregate as the work proceeds. Refer Clause 10.27.1 of MES Schedule Part-I. For other specifications unless specified otherwise hereafter profile and section, size of frame and lugs/hold fasts shall be as shown on drawing and not as in IS. Back of chowkhat and MS

8.3.2 FACTORY MADE SOLID PVC DOOR FRAMES WC/BATH

Solid Single PVC Door Frame solid PVC Door Frame of the size 50 x 47mm made out of 5mm PVC sheet. The door profiles are to be reinforced with 19xl9mm M.S. Square tube. EPDM rubber gasket weather seal to be provided through out the frame. The profile shall be mitre cut at two corners and joined by inserting 2nos. of 150mm long brackets of 15x15mm M.S. square tube inside the 19mm x 19mm M.S. square tube and suitably screwed all as per table of clause 8B.3.1 of SSR Part II 2010

8.4. WOODEN SHUTTER

8.4.1. Paneled, partly paneled and wire gauge shutters for all wooden doors are shown on drawings shall be factory made as under:-

(a). Stiles, rails, sash bars: These shall be of 35 mm thick. Other details all as shown on drgs. ASCU Vacuum pressure treated out of kiln seasoned hard wood second class of species bijasal/hollock/ben teak all as per IS: 1003.

NOTE: - In case factory made shutters with aforesaid species of timber are not available due to any reason, the contractor shall provide factory made shutters made from Shisham wood without any extra cost to Govt.

(b). Panel for panelled doors :

(i) Unless otherwise shown on drawings, panel for panelled doors for Internal and external doors (OTHER THAN BATH AND WC) shall be of plywood 9 mm thick 7 ply of BWR grade .

(ii) FOR BATH, TOILET AND WC FACTORY MADE SOLID PANEL PVC DOOR SHUTTER

30 mm thick Factory Made Solid Panel PVC Door Shutter consisting of frame made out of M.S. tubes of 19mmx 19mm for stiles, & 15mm x 15mm for top & bottom rails. M.S. frame shall be covered with 5mm thick heatmoulded PVC sheet C‘ channel having a 5mm thick PVC sheet strip of 20mm width stuck inside with solventcement, forming stiles, and 5mm thick PVC sheets for top rail, lock rail & bottom rail on either side, and 10mm(5mm x 2) thick, 20mm wide cross PVC sheet as gap insert for top rail & bottom rail. Panelling of 5mm thickPVC sheet to be fitted in the M.S. frame welded / sealed to the stiles & rails with PVC sheet beading, and joinedtogether with solvent cement adhesive etc. complete as per direction of Engineer-in-charge, manufacturer‘s instructions and drawing. The width of stiles & rails will be in proportion to the width of the door shutter as per table of clause 8B.2.1 of SSR Part II 2010. Contractor may use PVC doorframe but at no extra cost to Govt.

8.4.2. Only one type of species of timber shall be used in stiles and rails. Shutters with mixed species will not be accepted

8.4.3. The factory made shutters shall be brought to site without application of primer. Filler coat on panel inserts shall be applied after approval of doors and shall be left for hardening before application of primer.

8.4.4. The manufacturer shall ensure testing of door samples from the lot as per IS specification from named approved lab and submit necessary testing report/ certificate for acceptance.

PARTICULAR SPECIFICATION

9. BUILDERS HARDWARE

9.1 (a) All articles of builder‘s hardware shall comply with the relevant IS specifications and the specifications given in the MES Schedule for the relevant item. Builders‘ hardware shall be provided all as specified in Schedule of fittings and as specified in relevant drawings and shall be ISI marked.

(b) In case the size of particular fittings is not shown in the drawings, it shall be of size as directed by the GE.

(c) Screws used for fixing items of Builders hardware shall be as specified in clause 9.2.6 of MES Schedule Part-I. Unless specified otherwise elsewhere, screws shall be dipped in adhesive before fixing on particle board.

(d) Irrespective of what is shown on drawings, all articles of builder‘s hardware (other than for steel windows, butt hinges and rising door springs with magnetic catches) shall be extruded sections, aluminium anodized white.

(e) Butt hinges shall be of SS, IS marked.

(f) The contractor shall produce samples of each article of builder‘s hardware (fittings) which he proposes to use and get it approved. The sample shall be labelled as such, signed by the GE and the contractor. These shall be kept in the custody of the GE till the completion of work. In case rat-tail spring door closers are mentioned on drawings these shall be changed to helical door springs, 150 mm long of an approved make.

9.2 ARTICLES

(a) Butt hinges for doors shall be SS conforming to IS and all as specified in clause 9.7.2 of MES Schedule Part I. Butt hinges shall be run welded to machine pressed steel sheet frame/angle frame and screwed to timber shutters.

(b) Size of sliding bolts unless otherwise specified shall be 300 mm with 16 mm dia shoot.

(c) MS tower/ Barrel Bolts shall be with bolt shoot 10 mm dia up to size 125 mm and 12mm dia for size 150mm and above. For steel window, the dia of bolt shall be 10 mm.

(d) All other articles of builders hardware shall comply with the requirement of relevant IS and as specified in MES Schedule Part I.

(e) Stainless steel wire cloth/gauge for mosquito/fly proofing shall be as specified in clause9.25.3 of MES Schedule Part I. It shall have 0.36 mm nominal dia of wire and 1.40 mm average with of aperture for mosquito/fly roofing and as approved by GE.

10. TMT BARS AND STRUCTURAL STEEL/ IRON WORKS

10.1 All steel and iron work shall be as per details shown on drawings & BOQ items. Steel work shall be carried out as described in MES Schedule Part I and as specified here-in-after.

10.2 TESTING OF STEEL:

(a) REINFORCEMENT STEEL AND STRUCTURAL STEEL All steel for reinforcement to be used in the work shall be contractor's supply. The contractor shall procure steel for reinforcement (TMT bars) directly from the following main producers of steel :-

Srl No	COMPANY NAME	BRAND NAME	ADDRESS	Type of Steel	REMARKS
1	2	3	4	5	6
1.	Rashtriya Ispat Nigam Limited (RINL)	RINL	Visakhapatnam Steel Plant Visakapatnam-530031, India Tel: (91891) 518226, 518376 Fax:(91891)518316 E-mail: cmdvsp@itpvis.ap.nic.in	All	
2.	Tata Iron & Steel Company (TISCO, or TataSteel)	TATA	Bombay House,2, 4, Homi modi Street Mumbai-400001, India Tel:(9122)2049131 Fax:(9122)2049522, 2770840 E-mail: corpcomm@jsr.tatasteel.com (Br office for North: Jeevan Tara Bldg, Patel Chowk, New Delhi)	All	
3.	Steel Authorityof India Limited, (SAIL)	SAIL	Central Marketing Organization, Northern Region 17th Floor, scope Minar, Laxmi Nagar Distt Centre, Delhi- 110092	All	
4.	M/S Jai Balaji Industries Ltd.,	Balaji Shakti	5 Bentek Street, Kolkata–700001 Delhi Office 510, Block-B, Navraung House, 21 Kasturba Gandhi Marg, New Delhi- 110001 Tel: (011) 43620219, 43620220 Mob:7838272772, 9958936103 E-mail:info@jaibalajigroup.com	TMT Bars of Gde Fe 500 Fe 500D	
5.	M/S Shyam Steel Industries Ltd	SHYAM	Shyam Towers EN-32, Sector-V, SaltLake, Kolkata-700091 Tel: (033) 40074007 Fax:(033) 40074010 E-mail:marketing@shyamsteel.com	TMT Bars of Gde Fe 500 Fe 500 D CRS	
6.	M/S Steel Exchange IndiaLtd.,	SIMHADRI TMT	My Home Laxminivas Apartments, Ameerpet, Hyderabad-500016, A.P. Tel: (040) 23403725 Fax:(040)23413267, E-mail:info@seil.co.in	TMT Bars of Gde Fe 500 Fe500D HSCRM	
7.	M/S Jindal Steel and PowerLtd.	JINDAL PANTHER	OPJindal Road, Hissar, Haryana-125005, Tel: +91 166 2222471-84 Fax:+91166220476	TMTBars of GdeFe500/ Fe 500D Fe550/Fe550D	

				& *CRS with sizes (8-40 mm)	
8.	M/S SRMB Srijan Ltd.,	SRMB	SRMB House, 7, Khetra das lane Kolkata-700012 Tel:033-66006600 Fax:033-22110483	TMT Bars of Gde Fe500/ Fe 500D Fe550/ Fe550D (Size8-32mm)	
9.	M/S Shri Bajrang Power & Ispat Ltd.,	GOEL TMT	Vill- Borjhara, Urla Industrial Area, Raipur-493 221, Chhattisgarh Tel: 07714288019/29/39	TMT Bars of Gde Fe 500 Fe 500D (Size 8-32mm)	
10.	M/S JSW Steel Ltd.,	NEO STEEL	JSW Centre, Bandra Kurla Complex, Bandra (East) Mumbai-400051, Maharashtra Phone: 022-42861000 Fax: 022-42863000	TMT Bars of Gde Fe 500 Fe 500D CRS (Size 8-40 mm)	
11.	M/S Shyam Metalics & Energy Ltd,	SEL	Viswakarma, 1st Floor, 86 C, Tapsia Roas, Kolkata - 700046 Ph:913322852212 Website:www.shyamgroup.com	TMT Bars of Gde Fe 500, Fe 500D (Size 8-32mm)	
12.	M/S Kamachi Industries Ltd.	KAMACHI	ABC Trade Centre, 3rd Floor (Inside Devi Theatre Complex) Old No. 50, New No.39, AnnaSalai, Chennai- 600002, India Tel:+91-044-42961100 Fax:+91-044-42961122 E-mail: sales@kamachitmt.com Website:www.kamachitmt.com	TMT Bars of Gde Fe 500, Fe500D Fe550, Fe550D HCRM (Size 8-40mm)	
13.	M/S Real Ispat & Power Ltd	GKTMT	Vrindavan, Near IDBI Bank Civil Lines Raipur - 492001, C.G. Tel:+91-771-4224000 Fax:+91-771-4224010 Email: real@realispat.com	TMT Bars of Gde Fe 500 Fe 500D (Size 8-	

			Wesite:www.realispat.com	36mm)	
14.	M/S Super Smelters Ltd. Kolkata	SUPER SHAKTI	Premlata, 39, Shakespeare Sarani, 3rd Floor, Kolkata - 700017 Tel/Fax: + 91-33-2289-2734/36 E-mail: info@supershakti.in Website:www.supershakti.in	TMT Bars of Gde Fe 500 Fe500D, Fe550 (Size 8-32mm)	
15.	M/S Electrotherm (India) Ltd	ETTMT	Survey No.72, Village: Palodia, Taluka: Kalol, Dist.: Gandhinagar-382115 Gujarat, India. Tel+91-2717-660550/+91-2717-660649	TMT Bars of Gde Fe 500 Fe500 D, Fe550 & CRS (Size 8-32mm)	
16.	M/S Adhunik Industries Ltd.	ADHUNIK Fe500SD	Lansdowne Towers, 2/1A, Sarat Bose Road 6th Floor Kolkata 700020, West Bengal, INDIA	TMT Bars of Gde Fe 500 D (Size 8-32mm)	
17.	M/S Gallantt Metal Ltd	GALLANTT TMX	Ward 10 BC, Plot No.123, Ground Floor, GandhiDham Kutch, Gujarat-370201 Tel:+91-2836-228164 Fax:+91-2836-235787 E-mail:gml@gallantt.com Website:www.gallantt.com	TMT Bars of Gde Fe500 Fe500 D CRS (Size 8-32mm)	
18.	M/S Rashmi Metaliks Ltd	RASHMI TMT	Premlata Building, 39, Shakespeare Sarani, 6th Floor, Kolkata-700017 Tel: 033-22894255/56 Fax: 033-22894254 E-mail: mkt.domesticdip@rashmigroup.com Website:www.rashmigroup.com	TMT Bars of Gde Fe500 (Size 8- 32mm) Fe500D, Fe550D (Size8- 25mm)	
19	M/S Rashtriya IspatNigam Limited (RINL)	RINL	Visakhapatnam Steel Plant Visakhapatnam-530031, India Tel:(91891) 518226, 518376 Fax:(91891) 518316 Email: cmdvsp@itpvis.ap.nic.in	Structural Steel (Angle, Beam, Column, Channel, Plate)	
20.	M/S Tata	TATA	Bombay House, 2, 4 Homi Modi Street	Structural	

	Iron & Steel Company (TISCO, or TataSteel)		Mumbai-400001, India Tel: (9122) 2049131 Fax: (9122) 2049522, 2870840 E-mail:corpcomm@jsr.tatasteel.com (Br office for North: Jeevan Tara Bldg, Patel Chowk, New Delhi)	Steel (Angle, Beam, Column, Channel, Plate)	
21.	M/S Steel Authority of India Limited (SAIL)	SAIL	Central Marketing Organization, Northern Region 17th Floor, scope Minar, Laxmi Nagar Distt. Centre, Delhi-110092	Structural Steel (Angle, Beam, Column, Channel, Plate)	
22.	M/S Jindal Steels and Power Ltd.	JINDAL	Jindal centre, Plot No 2, Sector-32, Gurgaon-122001, Haryana Tel:01246612000 Fax:01246612125 Wesite:www.jindalsteelpower.com	Structural Steel (Angle, Beam, Column, Channel, Plate)	

Note: Contractor can even procure TMT bars from the primary producer approved bythe AHQ E- in-Cs Branchduring currency of the contract work.

(b) The documents in support of purchase of steel shall be furnished to GE for verification. The particulars of the manufacturers of steel shall be furnished by the contractor for every lot of steel separately as per the Performa given in Annexure-I.

(c) **STRUCTURAL STEEL**: Structural rolled and tubular mild steel sections shall be procured directly from main producers i.e. SAIL/RINL/TISCO/Jindal.

(d) Reinforcement steel, structural steel and Galvanised steel sheets & fabric reinforcement for concrete may be permitted to be procured from authorized dealers of main producers in case the total requirement of steel is less than 5 tonnes and specifically approved by Accepting Officer in writing.

(e) Steel sections for railings, gates, fencing, guard bars, grills, steel chowkhat, holdfasts etc., which do not constitute structural members, can be procured from main producers/ secondary producers/ BIS marked manufacturers or their authorized dealers at the option of Contractor after written permission from the GE/AGE(I) and will be without any minus price adjustment.

(f) Contractor shall submit manufacturers test certificate for the each consignment of Steel section to be used for fencing /security & compound wall posts/strut.

(g) The documents including manufacturers test results in support of purchase of steel shall be furnished to GE for verification. The particulars of the manufacturers of steel shall be furnished by the contractor for every lot of steel separately as per the Performa given in Annexure-I.

PARTICULAR SPECIFICATION

10.3 TESTING OF STEEL

(a) The contractor shall submit the manufacturer‘s Test certificate in original alongwith the test sheet giving their results of each mechanical test as applicable and the chemical composition of the steel or authenticated copy thereof duly signed by the manufacturer with each consignment. The Engineer-in-Charge shall record these details in a steel Acceptance Register after due verification.

(b) It will be mandatory for the GE to carry out simple field test and record the findings before the contractor is allowed to use steel in the work. CWE may also carry out random checks and record his remarks in the Steel Test Register. Such simple testing includes sand papering the cross section of the TMT bar and dipping it in chemical solution to give a clearly defined annular ring of tempered steel. Cost of all such facilities for test shall be borne by the contractor.

(c) The Accepting Officer/ CWE may order a board of Officers for random check of steel and verification of connected documents. In case of such check, if it is found that steel brought on site/incorporated in work does not meet the requirements laid down in the contract, such steel/work in which such steel has been incorporated shall be rejected or devalued based on the defects. In this regard decision of the Accepting Officer shall be final and binding. Contractor shall have no claim whatsoever on account of such rejection.

(d) Independent testing of steel by the GE from random samples of steel drawn from various lots from National Test House, SEMT, CME, Regional Research Labs, Govt Engineering College, Zonal Lab or Govt approved Labs etc, as per the recommended frequency given here-in-after shall be optional at the discretion of the GE in case of procurement of steel from main producers and testing charges shall be borne in accordance with Condition 10 A of IAFW-2249 i.e. testing charges shall be borne by the Deptt, if the test results are found in order otherwise these shall be borne by the Contractor.

(e) Independent testing of steel by the GE from random samples of steel drawn from various lots from National Test House, SEMT, CME, Regional Research Labs, Govt Engineering College, Zonal Lab or Govt approved Labs etc as per the recommended frequency given here-in-after shall be mandatory in case of procurement of steel from secondary producers and testing charges shall be borne by the contractor irrespective of the outcome of test results.

(f) In both the cases at sub Para 10.3 (d) & (e) above, the Contractor at his cost shall provide all facilities required for the testing and cost of materials consumed in tests shall also be borne by the Contractor.

(g) The elongation of reinforcement steel shall not be less than 18%.

10.4 FREQUENCY OF NORMAL MASS, TENSILE, BEND AND REBEND TESTS OF STEEL.

	NOMINAL SIZE	QUANTITY
STEEL REINFORCEMENT		
(a)	Bars size less than 10mm.	One sample (three specimens) for each test for every 25 tones or part thereof).
(b)	Bars size 10mm to 16mm.	One sample (Three specimens) for each test for every 35 tones or part thereof).
(c)	Bars size over 16mm.	One sample (Three specimens) for each test for every 45 tones or part thereof.

PARTICULAR SPECIFICATION

STRUCTURAL STEEL

(a)	Tensile test including %age elongation	One test for every 25 tones of steel or part thereof).
(b)	Bend test	One test for every 10 tones of steel or part thereof.

NOTE: - APPLICABLE FOR ALL TESTS.

For various tests, acceptance criteria, tolerance etc. refer to steel supply/Acceptance form given here-in-after and relevant BIS Codes.

STEEL SUPPLY & ACCEPTANCE REGISTER

1. CA NO & Name of Work
2. Contract No
3. Name of Manufacturer's T.C. No
4. Manufacturer
5. Random Test Details

(a) Physical test report from _____ vide their letter No _____
(Name of approved Lab/Engg College)

(b) Chemical test report from_____ vide their letter No_____
(Name of approved Lab/Engg College)

6. Types of Steel, Dia & Qty (a) Type: TMT/CRS (b) Dia-____mm
(c) Actual Wt___MT (d) Conversion Wt ____MT.

	Chemical Test						Mechanical Test						
	Carbon%	Sulphur%	Phosphorous %	Maganese%	Silicon %	Corrosion Resistant Element	Wt per meter	Stress(N/mm2) 0.2% proof	Tensile Strength (N/mm2)	Elengation % (Not less than 18%)	Bend Test	Rebend Test	Remarks
As per IS-1786													
As per manufacturer's test certificate													
As per independent test													

Remarks with Signature Accepted/Rejected

Contractor Junior Engineer Engineer-in-Charge Garrison Engineer

Remarks of BOO/Inspecting Officer/CWE

PARTICULAR SPECIFICATION

10.5 DOCUMENTATION

(a) The contractor shall submit original purchase vouchers from the manufacturer for the total quantity of steel supplied under each consignment to be incorporated in the work. All consignments received at work site shall be inspected by the GE alongwith relevant documents before acceptance. The original vouchers and the test certificates shall be defaced by the Engineer-in-Charge and photocopy of the same shall be kept on the record in the office of GE duly authenticated and with cross reference of control number recorded in the steel Acceptance Register and will be signed by JE (Civil), Engineer-in-Charge, GE and the contractor duly endorsed CA No.& Year.

(b) The entire quantity of all steel items shall be recorded in Measurement Book indicating size, quantities, voucher numbers, dates brought on site for record purposes as "not to be abstracted" and signed by Engineer-in-Charge and the contractor before incorporation in the work.

(c) Irrespective of what is shown in drgs or specified else where, steel reinforcement to be used in this project shall be Thermo Mechanically Treated steel bars.

10.6 (a) STEEL FOR REINFORCEMENT

(i) High strength deformed steel bars produced by thermo mechanical treatment process (TMT steel bars of grades Fe-500 /Fe 500D/Fe550/Fe550D) meeting all other requirements of IS-1786. Note :- For Swimming pool, Diving pool, Balancing tank, Hazardous tank, RCC over Head water tank , High strength corrosion resistant deformed steel bars (from primary producers as approved by E-in-C's Branch) produced by Thermo Mechanical Treatment process (TMT steel bars) of grade Fe-500 /500Dmeeting all other requirements of IS-1786,shall be used.Minimum elongationshall be 18%.

(ii) Mild steel bars shall conform to IS-432 (Part-I) and Grade I.

(b) **STRUCTURAL STEEL**

(i) Structural steel standard quality shall conform to IS-2062-2006 (sixth revi-sion) and Grade Fe-410-W (Gde E-250), quality A. The above mentioned steel shall be provided in the location mentioned in drawings and in clause 10.4.1 of MES Sch Part-I.

(c) **Galvanised steel sheets (Plain and corrugated)** : Conforming to IS-277.

10.6.1 For pricing of deviation in respect of steel items, the quality of steel shall be taken as under:-

(a) Structural steel: Standard quality Fe410-W (Gde E-250) quality A As (other than mentioned in MES Schedule. general purposes)

(b) MS round bars for: IS-432(Part I) reinforcement Gde I mild steel/twisted bars and as mentioned in MES Schedule 6 mm can be wire drawn quality.

(c) TMT.Bars for : TMT bars Fe 500/Fe 500d/Fe 550d conforming to reinforcement IS 1786-1985.

(d) Steel (general purpose): Ordinary quality Fe 290 (Gde E-165)

(e) Fabric reinforcement for concrete: Conforming to IS -1566.

10.6.2 The steel shall be procured and brought to the site one month before it is required to be incorporated in the work taking into account the time required for it's inspection and acceptance as specified here-in-before.

10.6.3 Contractor shall be responsible for proper storage, preservation and maintenance of steel at site till it is consumed in the work. Steel rejected shall be stacked separately, so marked and removed as directed by GE.

PARTICULAR SPECIFICATION

10.7 **REINFORCEMENT**:

(a) Reinforcement of mild steel and deformed cold twisted bars shall be of TMT bars in lieu thereof without change of any spacing all as shown on drawings in accordance with the provisions of IS: 456 of 2000 without any extra cost to Govt.

(b) Reinforcement shall be fabricated, placed in position all as shown on drawings and specified in Clause 10.17 to 10.22 of MES Schedule Part- I.

(c) Laps and crossing shall be tied with mild steel wire (annealed) (soft drawn) of diameter not less than 0.90mm. Laps shall be staggered.

(d) Irrespective of whether corner reinforcement is indicated or not in RCC plan showing reinforcement of slabs, corner reinforcement shall be provided as shown in TD (typical drawings).

(e) The contractor shall be responsible for accurate fixing of reinforcement shown on drawings, and shall not pour any concrete until the reinforcement has been inspected in position and approved in writing by the GE. The contractor shall take necessary precaution to prevent any displacement of reinforcement bars during concreting.

(f) Mode of payment of RCC reinforcement bars under measurable/provisional schedules shall be net length of bars provided on site multiplied by the unit weight of respective dia of reinforcement bar(s). Lap length provided to use offcuts more than 1.5m length shall not be measured for payment. Only net length of bar (without lap) shall be measured to arrive weight of steel to be paid under respective item. Offcuts of reinforcement bars less than 1.5 m in length shall not be allowed to use with lap and are to be discarded / removed from the site.

Contractor to take into account above aspect while quoting rates for steel reinforcement.

10.8 **STRUCTURAL STEEL WORK**:

Structural steel work shall be executed as specified in relevant clauses of section 10 of MES Schedule Part I.

NOTE :-

(i) Steel for holdfasts, grills, guard bars, cooler brackets and gate shall be ordinary quality. Balance structural steel shall be of standard quality. Test certificate for ordinary quality steel for gate/grill/guardbars/hold fast (Except fencing posts/struts)is not required.

(ii) Each consignment of steel for fencing post/struct shall be accompanied with manufacturers Test Certificate.

10.9 **MS BLACK, BOLTS, NUTS AND WASHERS**

(a) All MS Black, Bolts and nuts shall conform to IS: 1353 part I to III of 1984 and round washer shall conform to IS: 5370, IS: 5372 or IS: 5374 where ever specification are not available, the American standard specifications to be followed.

(b) Bolts, heads, nuts, shall be of such length as to project and clear thread beyond the nuts when fixed in position and these shall fit in the holes without any shake. The nut shall fit in the threaded ends of bolts properly.

(c) The round washers shall be placed under the heads and nuts of permanent bolts. Maximum two washers for one nut and one for each bolt head shall be used. Bolts threads shall be out side limit of jointing members and un threaded portion of bolt shall not be out side the washer.

(d) Bolts, Nuts & washers shall be thoroughly cleaned and dipped in double boiled linseed oil before used. The bolts shall be tightened starting from the enter of joints towards the edge.

11. to 13 Blank

PARTICULAR SPECIFICATION

14. **MILD STEEL SHEET**: MS sheet /plates shall be of thickness as shown on drawings in various locations conforming to Fe-410-S.

15. **STEEL WINDOWS AND VENTS**:

(a) Steel windows and ventilators where shown on drawings shall be factory made box type and as per clause 10.25 of MES Schedule Part I except that the corners of frames shall be welded to form solid welded joints conforming to the requirements listed in IS- 1038 and the process of welding adopted shall be 'FLASH BUTT' welding.

(b) Steel windows frames shall be provided with horizontal glazing bars as per drawings. These glazing bars shall be welded. Fixing and glazing of windows/ventilators shall be in accordance with IS: 1081. Glazing shall be fixed to frames with oil putty. Special spring glazing clips shall be used if the size exceeds that specified in aforesaid IS.

(c) Steel windows shall be provided with MS grill as per pattern & design shown in drawings.

(d) Steel window/ventilator frames in contact with brick work/concrete shall be painted in mastic of one part of bitumen and three parts of sand or ready-made mastic supplied by the manufacturers. Painting with cement will not be allowed in any case.

(e) Hinges for side hung shutters shall be box type as specified in IS: 1038. Fittings (Builders hardware) for steel windows shall be of MS black japanned, pressed steel heavy duty, corrosion resistant without wrinkle finish. In lieu of lock handle MS Black japanned tower bolt two Nos on each shutter shall be provided.

(f) Steel windows/ vents shall be fixed to walls with lugs/ hold fasts embedded in PCC blocks. Lugs shall not be provided for fixing to lintels even though its provision exists in the Indian standard specifications and/or drawings. Lugs shall be embedded in PCC (1:3:6) type C-1 using 20 mm graded stone aggregate size 150 x 150 x 200 mm deep.

(g) The contractor shall procure steel windows and ventilators from the firms indicated in Appendix 'A' to PS. Steel windows and ventilators manufactured at site will not be accepted. Samples of the steel windows/vents and fittings accessories shall be submitted to the GE by the contractor for his approval in writing before placing bulk order for procurement.

(h) The dimensions of steel windows/ventilators unless otherwise indicated on the drawings, shall conform to IS: 1038. However, minor variations in the overall size to suit the standard practice of the manufacturer will be acceptable without any price adjustment either in the quoted rates or while pricing deviations.

(i) Steel windows/ventilators & lugs shall be given one shop coat of anti-corrosive zinc chromate primer. Two coats of synthetic enamel paint of approved shade shall be given after erection.

(j) In case of deviations, the rate given in the MES Schedule shall be deemed applicable for pricing steel windows and ventilators with above specifications.

(k) Wherever windows marked with 'F' double window are shown on drawings, two separate frames, one for glazed shutter/ paneled shutter and one for gauze shutter as shown on drawing shall be provided. Gap between two frames on cill shall be filled with cement mortar (1:4) and finished smooth.

PARTICULAR SPECIFICATION

16. **EXPANDED METAL:**
Expanded metal, where shown on drawings shall be of gauge as approved by the GE. Unless otherwise specified it shall weigh not less than 4 Kg per Sqm.

17. HOLD FASTS/LUGS:
Hold Fasts/lugs to steel frames shall be of section as shown on drawings but of length as under:-

(a) Frames chowkhats for doors /Six numbers to each frame and 250 mm long cup board (MS angle or pressed steel).

(b) Frames for windows, Four numbers to each frame and 200 mm long Independent grills fixed in openings frames for lobby shutters etc.

(c) Frames for vents etc. Two numbers, 200 mm long.

Hold fasts/lugs 250 mm long shall be embedded in PCC (1:3:6) blocks using 20 mm graded stone aggregate of size 1 Bk x ½ Bk x thickness of wall and with 200 mm length shall be embedded in PCC block 150 x 150 x 200 mm deep. Hold fasts/lugs in the RCC columns etc shall be fixed all as shown on drawings.

18. **MILD STEEL GRILLS**:
Grills shall be provided as per patterns & location shown on drawings. Where type of grills is not mentioned it shall be type 'B'. Grills shall be welded to the frame of glazed window. Where grill is to be fixed independent in the opening, it shall be fixed with 25 x 5 mm FI holdfasts four numbers embedded in PCC block as specified above.

19. Blank

20. **ROOFING**

20.1. **RCC SLABS**

20.1.1 RCC floor/roof slab shall be laid to the required thickness all as shown on drawings.

20.1.2 RCC roof slabs shall have a finished slope of all as shown on drawing. However where slope is not shown the same shall be finished to the slop of 1:40.

21.0 **WATER PROOFING TREATMENT TO ROOF SLAB**:

21.1 The Contractor shall carryout water proofing work over roof to ensure the entire roof water tightness during the guarantee period of 10years and he will be responsible to keep the entire surface of roof water tight for a period of 10years from certified date of completion of work. The security Deposit towards the guarantee for water proofing treatment shall be as per tabretained from the RAR/final bill amount. The amount shall be calculated as per given in table below and shall released to the Contractor after the satisfactory completion of TEN YEARS guarantee period. During the guarantee period if any leakage is found out same shall be got rectified by the Contractor without any extra cost to the Government. In any case during the guarantee period the Contractor who executed the WPT works should inspect and examine the treatment once in every year and make good any defects observed. Defects liability period of one year as laid down under condition 46 of IAFW-2249 is not applicable for water proofing work covered under this contract and shall deemed to be amended to the extent mentioned above for WPT work. The Garrison Engineer may accept Bank Guarantee Bonds from Schedule Bank or fixed deposit receipt (pledged in favour of GE) from any approved Bank for the above said amount for a period of 10years in which case no further amount will be recovered from the final bill on this account.

The amount of security deposit to be held back from the Contractor‘s bill against the guarantee for water proofing treatment shall be calculated on the amount of water proofing treatment at contract rates as per the scales given below enhanced by 1.25 times of the amount so calculated. This shall be worked out by GE and intimated to the Contractor. This amount shall be refunded to him after the expiry of the guarantee period. Alternatively, the Contractor may give a separate interest bearing security deposit to GE valid for 10 Years for this amount.

PARTICULAR SPECIFICATION

Cost at contract rates		Amount of Security Deposit
(i)	Upto Rs. 50 lakhs	2% of the amount subject to a minimum of Rs. 5000/-
(ii)	Over Rs. 50 lakhs and upto Rs. 100 lakhs	Rs. 1,00,000/- + 1.5% of the amount exceeding Rs. 50 Lakhs
(iii)	Over Rs. 100 lakhs and upto Rs. 500 lakhs	Rs. 1,75,000/- + 1% of the amount exceeding Rs. 100 Lakhs
(iv)	Over Rs. 500 lakhs and upto Rs. 1500 lakhs	Rs. 5,75,000/- + 0.5% of the amount exceeding Rs. 500 Lakhs
(v)	Over Rs. 1500 lakhs	Rs. 10,75,000/- + 0.5% of the amount exceeding Rs. 1500 Lakhs subject to Rs. 18.75 Lakhs

21.2 Tenderer shall submit 10years guarantee certificate (as per performa enclosed herewith) duly undertaking and accepting the responsibility to carryout all required rectifications at his own risk and cost. In case of Contractor‘s failure to carryout the rectifications, the Government shall be at liberty to get all such rectifications carried out at his risk and cost from security amount. The tenderer is required to visit the buildings in which the water proofing treatment is to be done in order to ascertain the exact nature and extent of work involved. His quoted rate in the Schedule-_A‘ Part-I against the respective items shall be deemed to include for the following aspects for effective water proofing treatment and no extra claim on whatever account will be entertained from the Contractor.

21.3 **WATER PROOFING MEMBRANE**

21.3.1 APP modified water proofing membrane to be incorporated in the work for WPT shall be obtained from one of the following manufacturers:-

(a) STP limited

(b) IWL India Ltd

(c) Fosroc Chemicals

(d) Sika India Pvt Ltd

(e) Tiki Tar Danosa

21.3.2 **PHYSICAL PROPERTIES** The physical properties of APP modified water proofing membrane thickness 4mm shall be as under:-

(i)	Membrane weight	Not less than 3.60 Kg/Sqm
(ii)	Not woven polyester weight	Not less than 160gms/sqm
(iii)	Softening paint of membrane	Not less than 150oC
(iv)	Joint tensile strength	Longitudinal > 650kg/sqcm at 230oC Transverse > 450 kg/sqcm
(v)	Tear strength (L/T)	Longitudinal > 300N/ sqcm, Transverse > 250N / sqcm
(vi)	Elongation of reinforcement	> 30%
(vii)	Impermeability under water pressure	> 3kg/cm2

The water proofing membrane treatment shall be carried out through the approved applicator of the manufacturer as approved by the GE.

PARTICULAR SPECIFICATION

21.4 CERTIFICATE BY SPECIALISED AGENCY

A certificate by the specialized firm, who has executed the work of water proofing treatment under this contract will be rendered to the department stating that the said work has been executed by them and shall be signed jointly with the main building contractor. This certificate however does not absolve the main contractor from his responsibility in respect of specialist work as per contract condition. The main contractor shall be responsible to ensure that the roof slab shall be water tight/leak proof during guarantee period.

21.4.1 APPLICATION AND WATER PROOFING WITH APP MEMBRANE OVER ROOF

(a) RCC roof slab shall be cleaned with water after removal of existing roof treatment.

(b) Cement bedding not less than in 10mm thick in CM 1:4 mixed with integral water proofing compound as per manufacturer‘s instructions shall be provided on RCC slab with smooth and true surface and slope and as specified under relevant item of Schedule'A'.

(c) Apply oil bound bitumen primer @ 0.30litre/sqm manufactured by the same agency as that of APP with a density of 0.87-0.89kg/litre at 250C on the prepared dry roof surface. Provide and lay APP modified water proofing membrane reinforced with non woven polyester mat one layer weighing > 3.5kg/sqm.

(d) APP membrane shall be laid using butane torch by torch-on application with minimum 10cm side overlap and 15cm end lap. The laying of membrane shall be got done through an authorized applicator of manufacturer of membrane.

(e) **For non accesssable roof**:-

(i) Top of torched APP membrane shall be provided with solar reflective bituminous aluminum paint @ 0.1 Kg/Sqm as prescribed by manufacture of membrane. All as specified in respective Sch ‘A‘ item.

(f) **For accesssable roof /Terraces**:-

(i) 20 to 22 mm thick Kota stone over torched APP membrane with joints grouted flush in cement mortar (1:3) mixed with 5% of crude oil by weight of cement over and including bedding layer of cement mortar 20 mm thick in 1:4 using WPC all as specified here-in after.

16.4.2 TESTING OF MEMBRANE

The following tests shall be carried out in approved Government Laboratories and necessary test reports shall be submitted to the Garrison Engineer for the materials before incorporation in the work:

(a) Thickness of membrane
(b) Weight of membrane
(c) Tensile strength (longitudinal/crosswise)
(d) Tear Resistance
(e) Softening point
(f) Weight of Carrier (Polyester).

Apart from above tests Contractor shall also produce manufactures test certificate. Test certificate shall be produced with every consignment of material brought at site by the Contractor.

16.4.3 **TEMPLATE (LABEL)**. Template (label) made of 15mm thick plaster in CM (1:3) of size 60x60cm showing the following particulars shall be provided on each bldg before the final completion of work without any extra cost to the Department:

PARTICULAR SPECIFICATION

(a) Name of Contractor
(b) CA Number
(c) Date of completion
(d) Guarantee period
The above information shall be written with white paint with 5cm size letters and numbers.

16.4.4 The entire quantity of materials such as water proofing membrane bitumen primer, bitumen and bituminous aluminium paint etc required for the whole work shall be brought at site in sealed position and shall be entered in the MB as _NOT TO BE ABSTRACTED‘ showing the date, total quantity brought. Each seal shall be opened in front of Engineer-in-Charge only. Paid vouchers from authorized dealer shall be submitted to the Engineer-in-Charge immediately after every consignment and copy of such paid voucher shall be forwarded to Garrison Engineer by the Engineer-in-Charge duly endorsed.

GUARANTEE TO BE EXECUTED BY THE CONTRACTOR FOR REMOVAL OF DEFECTS AFTER COMPLETION IN RESPECT OF WATER PROOFING WORKS

This agreement made this _____day of_____ two thousand_____between of____ _____(son of _____of_____ (hereinafter called the guarantor on the one part) and the PRESIDENT OF INDIA (hereinafter called the Government on other part).

WHEREAS THIS agreement is supplementary to a contract hereinafter called the contract) dated _____ and made between the ‘GUARANTOR‘ of the one part and the _GOVERNMENT‘ on the other part where by the Contractor interalia, undertook render the buildings and structures in the said contract recited completely water and leak proof.

AND WHEREAS THE GUARANTOR agreed to give a guarantee to effect that the said structures will remain water leak proof for ten years from the certified date of completion.

NOW THE GUARANTOR hereby guarantees that water proofing treatment given by him will render the structures completely leak proof and the minimum life of such water proofing treatment shall be Ten years to be reckoned from the certified date of completion.

Provided that the guarantor will not be responsible for leakages caused by earthquakes or structural defects or misuse of roof and sunken floor slabs or alterations and for such purpose.

(a) Misuse of and sunken floor slabs shall mean by operation which will damage roofing treatment, like chopping of firewood and things of the same nature which might cause damage to the roof.

(b) Alteration shall mean construction of any additional storey or any part of roof or construction adjoining to existing roof, where by roofing treatment is removed in parts. And removal of finishes of sunken portion of slabs.

(c) The decision of the Garrison Engineer with regard to cause of leakage shall be final and binding.

During this period of guarantee the guarantor shall make good all defects and in case of any defects being found, render the building water proof and finish the disturbed surfaces as existed to the satisfaction of Garrison Engineer at his cost and shall commence the work for such rectification within seven days from the date of issue of the notice from the Garrison Engineer calling upon him to rectify the defects, failing which the work shall be got done by the department by some other Contractor at the GUARANTOR‘S risk and cost. The decision of the Garrison Engineer as to the cost, payable by the Guarantor shall be final and binding That if the Guarantor fails to execute the water proofing or commit breach there under then the Guarantor will indemnify the Principal and his successors against all loss, damage, cost expense or otherwise which may be incurred by him by reason of any default on the part of GUARANTOR in performance and observing of this supplementary agreement. As to the amount of loss and/or damage and/or cost incurred by the Government, the decision of the Garrison Engineer will be final and binding on the parties.

IN WITNESS WHEREOF these presents have been executed by the obligator _____ and by _____and for and on behalf of the PRESIDENT OF INDIA on the day month and year first above written.

SIGNED sealed and delivered by OBLIGATOR in the presence of:

1.

2.

SIGNED ON BEHALF OF THE PRESIDENT OF INDIA BY _____in the presence of:

1.

2.

20.6 **FLOODING TEST TO RCC SLABS**

20.6.1 On completion of casting of RCC slabs of Building Works (Item Nos 1.01 of the BOQ) and sealing up pores the Contractor shall carryout flooding test of each RCC slab after 28 days of curing of RCC slab. Flooding test shall be carried out by forming earthen bunds over parts of the roof slab and at selected locations as directed by the GE. Water shall be ponded for minimum depth of 50 mm and shall be allowed to remain over slab for a period of 72 hours. During this period if any defect such as leakage/seepage etc is noticed it shall be rectified by the contractor by grouting to the satisfaction of the GE. After rectification of the defects more tests shall be carried out at the discretion and to the satisfaction of the GE.

20.6.2 Testing : The contractor will submit original purchase voucher along with the test certificate for each consignment/lot of material brought at site. Independent testing will also be carried out be GE for least one test sample consisting of three test specimen at random from each lot. The polymeric membrane shall be got tested from the test house as mentioned here under:-

Frequency of test At least one test consisting of three sample specimen of polymeric membrane at random from each lot. The polymeric membrane shall be got tested from the test house as mentioned hereunder:-

(a) Chemical and metallurgical, Services T.S. No. 63, SIDCO Industrial Estate Ekkaduthangal, Chennai-600097 Fax No: 44-2312944.

(b) Central Institute of Plastic Engineering and Technology (Ministry of Chemical and fertilizer, Govt of India) Guindy, Chennai-600032, Fax No : 94-44-2344673

(c) National Test House, Alipore, Kolkatta.

(d) Regional Testing Centre (Northern Region), NABL Accredited laboratory Shaeed Capt. Gaur Marg, Okhla Industrial Estate, New Delhi-20

20.6.3 **MAKES**

(a) Hyperplas polymeric water proofing membrane manufactured by IWL India Limited, C&D Laxmi Bhawan, 609, Anna Salai, Chennai - 600006.

(b) Super Thermolay APP PL/MT by STP Texsa limited 570 phase V Udyog, Gurgaon 122016 India.

(c) HYDROSTOP Waterproofing Membrane of Tiki Tar.

(d) M/s Torchtar Membrane & Bitumen Products Pvt Ltd

20.7 **Execution of work**

20.7.1 The work will be got executed through authorized applicator of the manufacture under strict supervision of Engineer-in-Charge.

PARTICULAR SPECIFICATION

21. **FLOOR FINISHES**:

21.1 **GENERAL**:

(a) Type of floor topping shall be provided to details as shown on drawings/ Schedule of finishes and as specified here in after. Flooring shall be laid as indicated in SSR Part -I.

(b) Floor topping shall be extended over dwarf walls, doors and other openings.

(c) Cement for floors shall be as described here in before.

(d) PCC sub floor may not be laid in panels/bays. For the purpose of deviation, the rates as given in SSR Part-II for flooring shall be applicable.

(e) Unless otherwise shown in drawings, the dividing line between the floors of different types, wherever they so meet between adjoining rooms, shall be determined on the basis of the finish visible when the doors are closed so that no dissimilarity appears and the finish shall accordingly be provided.

22. **FLOOR TOPPING**:

(a) Cast in situ PCC floor topping shall be laid in panels as per clause 13.32.1 and 13.38 of SSR Part-I respectively. The pattern/layout of panels for car & scooter garages shall be as directed by the GE.

(b) Joints in floor topping shall not be staggered while providing glass strips.

(c) Unless otherwise shown on drawings/Schedule of finishes and/or specified hereinafter, PCC floor toppings shall be broom finished.

22.1 **TYPES OF FLOOR**: Unless otherwise mentioned in drgs and in schedule of finishes floors shall be provided as specified here-in-after.

(I) **FLOORING FOR SCOOTER & CAR GARRAGES** :
PCC floor for car & scooter garrages shall be 100 mm thick of mix (1:2:4) type B-2 using 40 mm graded coarse aggregate over 75 mm thick sub base of mix (1:4:8) type D2 using 40 mm graded coarse aggragate over 75 mm thick hard core over rammed earth. Top of PCC floor shall be broom finished all as specified in PS clause 20.B.7.12.3 of SSR Part-I (2009).

(II) **NON SKID CERAMIC TILES FLOORING**

(a) Rectified /joint less coloured ceramic tiles shall be of uniform texture. Water absorption shall be less than 2%. Tiles shall be 7-8 mm thick hydra with breaking strength 350 to 400 Kg/Sqm and weighing not less than 14 Kg/Sqm. It shall be free from bubbles, distortion due to overheating. Size and shade of tiles shall be directed by GE.

(b) Tiles shall be set & jointed in cement slurry over 15 mm thick screed bed in CM(1:6), over 25 mm thick PCC(1:2:4), type B-1 sub floor, over 75mm thick sub base PCC(1:4:8), type D-2, using 40 mm graded stone aggregate, over rammed earth. Joints shall be cleaned immediately after laying tiles and filled with white cement mixed with coloring pigment to match with the shade of tiles.

(c) The size, colour, shade and pattern of tiles shall be got approved from the GE before procurement.The tiles shall be laid as per clause 13.40 of MES Schedule Part I.

PARTICULAR SPECIFICATION

(III) **KOTA STONE IN FLOORING/ TERRACES AND TREADS OF STEPS/STAIRCASE**

(a) Kota stone flooring/ terraces shall be provided as specified in para 13.47 of SSR (Part-1) 2009. Kota stone in flooring shall be 22 to 25 mm thick laid over 20mm thick screed bed in cement mortar (1:6) over 75mm thick sub base of plain cement concrete (1:4:8) type D-2 (using 40mm graded crushed stone aggregate), over 75 mm thick hard core over well rammed approved earth filling in Ground Floor. In upper floors kota stone shall be laid over 20 mm thick cement mortar screed in CM 1:6 over RCC slab. Kota stone shall be of size not less than 550X550mm except at borders & corners. Kota stone shall be hard, of uniform texture & without cracks flaws etc. Joints shall be filled in grey cement using pigment or epoxy grout of matching colour. Top surface shall be grinded and machine polished all as per PS clause No. 13.47.4 of SSR Part-I.

(b) Kota stone in treads shall be pre mirror polished 18 to 20 mm thick & of width equal to the tread of steps with bull nosing edge & two serration on top.Kota stone in treads shall be laid over 20 mm thick cement mortar bedding in cm 1:6.

(IV) **KOTA STONE SKIRTING**

Kota stone skirting shall be of 10 to 12 mm thick machine cut and pre mirror polished 100mm high & width matching with kota stone in flooring. Kota stone in skirting shall be fixed over 10 mm thick backing coat in CM.

(V) **GLAZED CERAMIC TILES DADO/SKIRTING**:

Irrespective of whatever shown on drawings the height of dado shall be 2100mm in toilet and bath. Glazed coloured ceramic tiles shall be as specified in Para 13.15 and shall be provided in accordance with Para 13.40 of SSR Part- I. Tiles shall be 7 mm to 8 mm thick of size, shape and colour as directed by the GE. Tiles shall be fixed over 10 mm thick backing in CM (1:4). Tiles shall be first quality free from bubbles & manufacturing defects.

NOTES:-

(i) All cutting and grinding shall be with machine except in situations where grinding by machine is not possible grinding by hand shall be permitted.

23. **GRANITE SLAB IN COOKING PLATFORM**

23.1 Granite slab shall be 18 to 20 mm tick jet black or Telephone black in single piece pre mirror polished laid over 1:4 cement mortar screed 20mm thick. The granite slab shall be hard, sound, dense and homogeneous in texture with crystalline and coarse grain, all as specified in clause 6.20.3, 6.20.4.2, 6.20.5 to 6.20.8 of SSR Part-I. Cost of rounding of edges of granite slab for coking platform including drilling hole for gas pipe in included in quoted rate.

24. **PRECAST INTERLOCKING PAVER BLOCK/ CHEQUERED TILE**

24.1 Thickness and size of precast interlocking paver/chequered tile shall be as specified in Sch ‘A’. Interlocking paver/ chequered tile shall be rubberized coloured (glossy non fadable) (using vibro compaction method) with 5mm thick toping in grey cement and non fadable glossy colour pigment. Interlocking paver shall be conforming to IS-15658-2006 and shall be laid over sand cushioning.

25. Stainless Steel Kitchen Sink with Draining board: Stainless Steel Kitchen Sink with Draining board shall be made out from non magnetic stainless steel sheet 1mm thick of over all size 915mm x 460mm with bowl size of 410mm x 330mm x 160mm if not mentioned in the TD. Contractor‘s quoted rate are inclusive of waste fittings brass chromium plated with fly nut and washer, PVC waste pipe 32mm bore & pair of cast iron brackets complete.

26. - BLANK-

PARTICULAR SPECIFICATION

27. **PLASTERING AND POINTING** :

27.1 **GENERAL**:

(a) All external plaster shall be carried out upto 150 mm below ground level except where steps, ramps, plinth protection are provided.

(b) Plaster and skirting/dado shall be returned in jambs, soffits of lintels and window cills etc, unless otherwise shown on drgs, height of skirting & dado shall be 100 mm & 2100 mm respectively.

(c) Thickness of plaster shall be the finished thickness exclusive of dubbing. Dubbing wherever required shall be executed without any extra cost.

(d) All corners, angles, junctions and edges unless otherwise specified shall be truly vertical or horizontal as the case may be and shall be carefully finished.

(e) Irrespective of what is specified and shown on drawing, 12mmx12mm continuous groove shall be provided in plaster throughout at the junction of wall & RCC slab. Trowel groove shall be provided at the junction of walls and RCC columns/RCC beams as directed. In other situations where it is not desirable to make groove, chicken wire mesh 30 cm wide shall be provided under the plaster.

27.2. **PLASTER TO INTERNAL SURFACES**
15 mm thick plaster in CM (1:6) finished even and smooth without using extra cement above skirting/dado in one coat.

27.3. **PLASTER TO EXTERNAL SURFACES**:
15mm thick in CM (1:4) mixed with water proofing compound as per manufacturer‘s instructions. Plaster shall be finished even and smooth using extra cement. Top and sides of parapet on roof shall also be plastered in CM 1:4 except wall cladding or where any other finishes are shown on drawings.

27.4. **PLASTER ON RCC CHAJJAS**:
Top surfaces of the RCC chajjas shall be plastered in CM (1:4) 15mm thick mixed with water proofing compound as per manufacturer‘s instructions. Irrespective of whether shown on drawings or not, the plaster shall be taken up to 200 mm height over adjacent vertical surface of wall in addition to external plaster for a length equal to length of chajja. Thickness of plaster at junction of wall and chajja shall be rounded to 50mm radius.

28. **WHITE WASH / DISTEMPER / CEMENT BASE PAINT**

28.1. **GENERAL**

(a) White wash/colour wash/distemper/cement paint shall be carried out as specified in section 15 of MES Schedule Part -I and as indicated in Schedule of finishes, after preparation of surfaces as specified. Unless otherwise specifieFFd all plastered surfaces shall be given three coats of white wash/colour wash to match with adjoining surfaces. Plastered surfaces above false ceiling and behind wall paneling shall be given three coats of white wash only.

(b) All exposed surfaces of parapets, sides of steps, risers, ramps, faces of columns, jallies etc and the like shall be treated with same finish as provided for the surrounding surfaces, plastered surfaces above false ceiling shall be given three coats of white wash only.

28.2. **WHITE WASH**:
White wash where indicated in Drgs/Schedule of finishes shall be in three coats using fat lime after preparation of surface as specified in Para 15.12 of SSR Part-I of 2009.

PARTICULAR SPECIFICATION

28.3. **OIL BOUND DISTEMPER**

Whatever shown on drawings/ schedule of finishes of Sch `A` Part I (Buildings) item No 1.01 wall surface shall be applied with two coats of oil bound distemper over a coat of cement wall primer from the same manufacturer of OBD over a layer of cement base wall care putty not less than 2 mm thick all as specified in Para 15.14 of SSR Part-I of 2009. Surface shall be prepared by applying one coat of cement base wall care putty to remove any unevenness above Skirting/Dados.

28.4. **CEMENT BASE PAINT**:

Whatever shown on drawings external schedule of finishes of Sch `A` Part I (Buildings) item No 1.01 & 1.07 shall be applied two coats of cement base paint over a coat of alkali resistance primer after preparation of surfaces, all as specified in MES Schedule ABOVE Plinth Protection. To prevent algae and moss growth and efflorescence, silicon base water repellent compound may be added to mixture at the rate as recommended by the manufacturer. Cement based paint conforming to IS: 5410-1969 and shall be ready mix & of make and shade as approved by GE.

After the surface is dry, primer shall be applied with brush followed by two consecutive finishing coats of paint/distemper.

Oil bound distemper/plastic emulsion /acrylic emulsion and polyurethane paint shall be standard make and shall be got approved from GE, before ordering for bulk supply.

29. **BLANK**

30. **PAINTING**:

30.1. **GENERAL**

(a) Refer Section 17 of MES Schedule Part I for paints and painting work.

(b) Three coats of paint wherever occurs/specified in these documents /drawings on wooden or steel surfaces shall consist of one priming coat, one under coat and one finishing coat and paint for these shall be of the same manufacturer. Tint of paint shall be as directed by the GE.

(c) Preparation of surfaces of woodwork and steel work for painting shall be carried out as specified in clause 17.6.1, 17.6.1.1 and 17.6.1.2 and 17.8.1, 17.8.1.1 and 17.8.1.2 of MES Schedule Part I.

(d) The contractor shall inform the GE within three weeks of the acceptance of the tender, the brand and names of the manufacturers of paint, which he proposes to incorporate in the works for his approval.

(e) Wooden or steel surfaces embedded in or in contact with masonry/brick work or concrete shall be treated with two coats of hot tar. Hold fasts, lugs shall be given two coats of tar and sanded all as specified in MES Schedule Part I.

30.2. **SURFACES OF WOOD WORK**:

(a) All exposed surfaces of wood work and joinery and other wood work including block boards/particle boards/plywood, unless otherwise specified or indicated on drawings shall be treated with two coats of synthetic enamel paint over a coat of pink primer.

(b) In case of un-veneered particle board /plywood panels, an additional coat of filler shall be provided before applying the primer and subsequent coats of paint.

30.3. **STEEL AND IRON SURFACES**:

(a) All steel and iron work, unless otherwise specified or indicated in drawings, shall be treated with two coats of synthetic enamel paint over a coat of red oxide zinc chromate primer.

PARTICULAR SPECIFICATION

(b) Reinforcement and galvanized surfaces (except wire gauge) shall not be painted. Wire gauge (mosquito proofing) shall however be given a coat of French blue paint.

(c) Exposed surfaces of MS/CI brackets and MS fan hook and boxes etc shall be painted with two coats of synthetic enamel paint over a coat of red oxide primer.

31. **MISCELLANEOUS**:

31.1. **FORMATS**:
Formats for numbering of buildings and display for anti-termite treatment shall be in 10mm thick setting coat in CM (1:4) over plastered surfaces of external walls, Size of letters and figures to be engraved shall be 50 mm x 100 mm (minimum) and painted black on white background of paint. Details for ATT and water proofing of roof shall be shown on the same format.

31.2. **TOWEL RAIL**:
Towel rail shall be of brass chromium plated 18/19mm dia (wall thickness shall not less than 1 mm) with heavy weight bracket at ends and fixed with chromium plated steel screws on wooden gutties. Unless otherwise shown on drgs the size of towel rail shall be 600 mm long.

31.3 **RCC PARAPET, RAILING, FACIA AND FINS**:

(a) RCC parapet, railing, fins etc shall be provided as shown on drawings and shall be cast monolithically with slabs/beam.

(b) Exposed surfaces of RCC parapet, railing, facia and fins shall be plastered not less than 5mm thick in CM (1:3) finished even and smooth using steel trowel without using extra cement.

31.4. **SOAP NICHES**:
These shall be provided where shown on drawings. Inside finish of the soap niche shall be same as that of dado. Where niche is not coming in dado, it shall have 5 to 7 mm thick ceramic tiles white. Where it is shown to be of vitreous china, it shall be got approved from GE before procuring.

31.5. **FAN HOOK AND BOX**
Fan hook shall be made out of 12mm dia MS round bars of shape and size complete with box all as shown on drawings. It shall be given a coat of steel primer before placing in position.

31.6. **OVERHEAD SERVICES WATER TANKS**:
HDPE service overhead water tanks having triple layered wall shall be of capacity and placed at location as shown on drawings/as directed by GE depending upon load distribution on roof slab. It shall have lockable cover. In case tanks of capacity shown on drawings are not available, tank of next higher capacity shall be provided without any price adjustment. Tanks shall be placed on 75 mm (average) thick PCC (1:3:6) Type C-1 padding. For each tank GI light grade 300mm long, over flow pipe of same dia as that of inlet pipe with PVC anti-mosquito cap and 25mm dia GI light grade wash out pipe extended 300mm from outer edge of PCC padding shall be deemed to be included in the cost of respective building in addition to tank fittings such as connection, float valve, ball valve and stop cock for inlet, outlet and washout pipes.

31.7 & 31.8 **BLANK**

31.9 **BRICK STEPS**:
Brick steps shall be provided all as shown on drawings. Brick work shall be in CM (1:4) sides and risers of steps shall be plastered CM (1:4), 15mm thick and treads of steps shall have 40 mm thick PCC (1:2:4) type B-0 using 12.5 mm stone aggregate finished even and smooth without using extra cement and having XPM impression.

PARTICULAR SPECIFICATION

31.10. **Blank**

31.11. **PCC CILL**:

18-20 mm thick granite/green marble of full length and width (one piece) over PCC (1:3:6) 40 mm thick type C-I. Top finish of cill shall be same as for dado or as for the wall as the case may be.

31.12. **NITCHES BOXES FOR MAIN SWITCH/ELECTRIC METER/WATER METER**:

(a) Boxes shall be fabricated out of MS Sheet, MS angle, MS flat etc all as shown in drawings. MS work shall be painted as specified here-in-before.

(b) Niches shall be provided as per details on drawings and be finished with 15mm thick plaster in CM (1:4)

(c) Wood work in nitches shall be of first class hard wood Teak.

31.13. **CURTAIN ROD**:

Curtain rod where indicated in the drawings shall be provided with 25 mm dia stainless steel grade 304 16 gauge with decorative finials and SS brackets etc fixed with wall complete Curtain rods and Initials shall be approved by PMG/GE.

31.14. **Blank**

31.15. **PEGS**:

Stainless steel grade 304 Pegs fancy type (cast or extruded section) shall be provided on wall/ wood with cadmium plated screws.

31.16. **MIRROR**

Mirrors shall be 5.00 mm thick with square corners polished fixed to 5 mm thick plywood (kitply) backing and with aluminium channel frame fixed to walls with brass chromium plated screws to wall all as shown on drgs. Mirrors shall be SWASTIC or ATUL make as approved by GE.

31.17. **OPENING FOR EXHAUST FAN**:

Opening for exhaust fan shall be provided wherever shown on drawing. Provn of exhaust fan and making connection shall be payable separately.

31.18. **STEEL DOORS**:

Mild steel doors shall be provided to details all as shown on drawings. These shall be given two coats of synthetic enamel paint over a coat of red oxide zinc chromate primer.

31.19. **GLAZING FOR STEEL WINDOW**:

(a) Unless otherwise shown on drawings/specified elsewhere, the sheet glass in pans up to 0.5 Sqm (each) shall be 3mm thick and exceeding 0.5 Sqm (each) it shall be 4 mm thick plain sheet glass B quality as specified in MES Schedule conforming to IS: 1761

(b) Pin headed figured glass shall be conforming to IS: 5437 for glazing. Where thickness of pin head glass is not shown on drawings it shall be 3mm thick.

PARTICULAR SPECIFICATION

(c) Glass will be fixed as specified in Clause 16.9.2 (for wood surrounds) and clause 16.10.1 (for steel surrounds) of SSR Part I. Putty shall be as specified in Clause 16.5 of MES Schedule Part I.

31.20. **RAIN WATER PIPES**:

(a) Irrespective of what is shown on drawings, rain water pipes shall be of HDPE medium density (ISI marked) 160 mm dia for hanger and 160 mm dia for other buildings. RWP fixed to the walls with MS holder bat clamps embedded in cement mortar (1:3) conforming to IS: 13512-1992 or in RCC as the case may be. These shall be provided all as specified in MES schedule Part I and given same finish as for wall. Pipes shall be jointed properly as per manufacturer's instructions or as specified.

(b) Grating shall be of PVC round type for PVC pipes and CI grating for CI pipes not less than weighing 1.50 kg each at the inlet to RW/down take pipes.

31.21. **DRIP COURSE**:

Drip course shall be provided under the projection of roof slab, projected balcony and chajjas etc, It shall be made in CM (1:4) and shall of size 40 mm x 20 mm thick (average).

31.22 **CONSTRUCTION JOINTS/CRUMPLE JOINTS**:
These shall be provided to details as shown on drgs Gap shall be filled with 25 mm thick preformed filler fiber board. Covering sheet shall be fixed with stainless steel screws in wooden gutties /plugs to wall at 150 c/c.

31.23. **SPLASH STONE**:
Splashed stone shall be provided below the shoe of each rain water pipe as shown on drawings. In case details not shown on drawings, it shall be 500 x 750 x 75 mm thick self faced Shahabad/Dholpur stone slab set on 10mm thick CM (1:6) screed.

32. **SANITARY FITTINGS AND PLUMBING WORKS**:

32.1. **GENERAL**
The work of supplying, fixing/laying and jointing of water closets, wash hand basin, all pipes and fittings with accessories/traps and GT for building is included in schedule ‘A‘ Part-I and shall be carried out upto first manhole but excluding the manhole.

32.1.1. Water closets, wash hand basin etc shall be vitreous china first/standard quality white glazed and shall conform to relevant part of IS: 2556 (Vitreous china sanitary appliances) and shall be ISI marked or of superior quality and finish. Tolerance in the size of fittings as given in IS shall be permissible. Also refer relevant clause of MES Schedule Part-I.

32.1.2. Flush pipe and sockets of flushing rim of WC shall be jointed with Epoxy Putty manufactured by Unique Enterprises or M-Seal or any other equivalent Epoxy putty.

32.1.3. ‘P‘ or‘S‘ trap shall be jointed to WC pan with Epoxy Putty.

32.1.4. Plumbing work shall be carried out as specified in Section 18.13 to 18.23 of MES Schedule Part-I and the Contractor shall employ licensed plumber.

32.1.5. Where not indicated elsewhere angle iron bracket shall be fixed to walls with PCC (1:3:6) Type C-1 block of size 100 mm x 100 mm x 75 mm.

32.1.6. Cast iron brackets shall be fixed on walls with wooden plugs (built in walls) or plugged to walls. Size of PCC blocks shall be 100mm x 100mm x 75mm and shall be in PCC (1:3:6) Type C-I.

32.1.7. Excavation and earth work up to first manhole shall be deemed to be included in the cost of respective building in Schedule ‘A’ Part-I.

PARTICULAR SPECIFICATION

32.2. **WATER CLOSET**

32.2.1 **WATER CLOSET (EUROPEAN)**

Water closet European type (pedestal pattern) where shown on drawings shall comprise as under:-

(a) Vitreous china white water closet apparatus (pedestal pattern) of height 400 to 410 mm rear outlet, conforming to IS with integral ‘P’ or 'S' trap having minimum 75 mm water seal. The closet shall be screwed to and including wooden plugs embedded into floor.

(b) Seat and cover shall be of thermostat material conforming to IS: 2548 (Part-I) black plastic closed solid pattern flat bottom, hinged with chromium plated brass hinges, rubber buffers of suitable size and ISI marked.

(c) PVC low level flushing cistern of 10 litres capacity with internal fittings, make commander/ champion white.

(d) Flush pipe of adequate length shall be chromium plated brass tube, 32 mm dia.

(e) Toilet paper holder shall be box type vitreous china.

32.3. **CI SOIL WASTE/ VENT PIPE AND FITTINGS**:

(a) All soil & waste shall be of centrifugally cast (spun) pipe bearing IS: 3989 of 1994 spigot and socket ends and shall bear ISI marking. Vent pipe shall be 75 mm dia PVC pipes fixed with slotted cowl. CI fittings and accessories shall be as suitable for spun pipes as specified in IS-3989.

(b) CI bend where resting on ground for soil pipes shall be with duck foot/heal rest. All other CI bends, tees junctions etc shall be of required angle and shall be with oval access door for maintenance.

(c) Jointing of CI pipes and fittings shall be with run lead joints as specified for concealed pipes and cement joints in CM (1:1) as specified for exposed pipes.

(d) All soil, waste and vent pipes and their fitting shall be fixed along vertical face of wall and shall be kept 50 mm to 75 mm away from face of wall while fixing.

(e) Exposed surfaces of soil, waste and vent pipes, their fittings and accessories shall be given two coats of paints to match adjoining wall finish.

(f) Soil/waste pipe from WC to first manhole/GT shall be 100/75mm dia CI pipe.

(g) Cast iron pipe and fitting shall be secured with cast iron holder bats or mild steel clamps all as specified in clause 11.37 & 11.58.5 of MES Schedule Part-I.

32.4. **CI NAHANI / FLOOR TRAP**:

Cast Iron Nahani trap with long arm shall conform to IS 3989 with 75 mm outlet, Nahani trap shall be with CI grating and shall be given two coats of black bituminous paint all over before fixing. Nahani trap shall have long outlet where required to cross thickness of wall and to avoid joint in the middle of the wall.

32.5. **GULLY TRAP**:

Gully trap shall be Salt glazed stone ware, square body as per IS: 651 size 150 mm x 150 mm with ‘P’ type 100 mm outlets and MS/CI grating. The gully trap shall be set in PCC (1:3:6) type C-1 with foundation, side brick wall and kerb. Cover to the gully trap shall be solid type with single seal all-round and jointing shall be done in CM (1:1).

PARTICULAR SPECIFICATION

32.6. **TESTING**:
All soil, vent and waste pipe shall be tested as specified in section 18 of MES Schedule Part-I. All CI Pipes shall be tested by filling with water to check leakage if any before fixing in position.

33. **Stainless steel grade 304 pipe railing in lieu of MS pipe railing**

33.1 **PEG SET OF TWO/ THREE**
Peg set of two/ three where indicated/shown on drgs shall be and shall be provided all as shown on drgs. These shall be fixed to first class hardwood boarding all as shown on drgs with matching screw. Boarding shall be fixed to hardwood plugs embedded in the walls. All exposed surface of the boarding shall be painted with two coats of synthetic enamel paint over a coat of primer and surfaces in contact with brick work shall be treated with two coats of tar all as specified.

34 **INTERNAL WATER SUPPLY**:

34.1 **SCOPE**
The work covers internal water supply as described in Schedule 'A' under respective parts, Notes to Schedule 'A', as shown on drgs and as specified here-in-after.Work shall be carried out through licensed plumbers.

The following Indian Standard shall be applicable in addition to those mentioned in MES Schedule Part-I.

34.2 **STANDARD OF QUALITY AND WORKMANSHIP**
Refer condition 25 of General Conditions of Contracts (IAFW-2249). All water supply works shall be carried out by the licensed plumber and in the supervision of a qualified supervisor. The contractor shall on demand by Engineer-in- Charge produce such evidence of qualifications of his workmen/supervisors to GE either at commencement of work or at any time thereafter during the currency of contract. Such record shall be kept in works dairy with reference to license of plumbers. The entire work shall be of high class with best workmanship **and to the entire satisfaction of Engineer-in-Charge**.

All equipment and materials incorporated in this work shall be of standard make and shall be ISI marked.

IS NO	SUBJECT
1172	Code of basic requirement for water supply drainage and sanitation (second revision)
2065	Water supply in buildings, Code of practice (First revision)

34.3. All pipe runs shall be truly vertical/horizontal. At all corners, either bend or elbow shall be provided. Suitable sleeves shall be provided for full width of the wall where the pipe is passing through wall. The sleeves shall also be of medium grade GI pipe.

34.4. Steel water tubes (pipes) galvanised shall be medium grade all as specified in MES Schedule. It shall be given two coats of oil paint white over a coat of primer.

34.5 Copper alloy bib cocks and stop cocks shall comply to IS: 781 high pressure qualities. Crutch handles, spindles, glands etc of stop cocks and bib taps shall be of cast brass (extruded rolled or forged) in lieu of brass rod and as specified in IS: 781 without any price adjustment. Ball valve shall be as specified in para 18.19 of MES Schedule Part-I. The sample of the following shall be submitted to GE for their approval within 30 days of the date of issue of the work order, before any bulk order is placed:-

PARTICULAR SPECIFICATION

35. SAMPLES

35.1 Samples of fittings shall be produced by the contractor for approval of the GE before commencement of the work and his approval shall be obtained in writing. The sample of the following shall be submitted to GE for their approval within 30 days of the date of issue of the work order, before any bulk order is placed:-

(a) GI pipe fittings.
(b) Stop cocks, bib cocks and pillar cocks
(c) Polythylene connection.
(d) Float valves.
(e) Shower rose.
(f) Half turn valve of make ‘L &‘K‘ or“KING’’ as approved by GE.
(g) Gate valves

36. GI PIPES/FITTINGS

36.1 Mild steel water pipes galvanised except those issued under schedule ‘B‘ shall be of medium grade and shall conform to IS-1239 (Part-I) and shall be distinguished by colour bands i.e. light grade with yellow, medium grade with blue and heavy grade with red. The tubes shall have the ISI certification mark at standard length.

36.2 Mild steel galvanised iron fittings shall be as per IS-1239 (Part II). The fittings may be welded or seamless. Alternatively, the pipe fittings shall be of galvanized malleable cast iron conforming to IS-1878.

37. BIB TAPS AND STOP VALVES

37.1 Brass bib taps and stop valves (cocks) shall be marked IS-781 high-pressure quality. The crutch handle, spindles, glands, etc of stop valves and bib taps shall be of cast brass (rolled or forged) in lieu of brass rod as specified in IS- 781 without any price adjustment. Fancy bib taps and stop valves shall be of approved make and shall be nickel-chromium plated and conform to IS-8931- 1978.

37.2 Pillar taps shall be of cast copper alloy, chromium plated conforming to IS:1795-1974. The minimum finished mass of pillar taps shall be as specified in clause 18.16.2 of MES Schedule Part I. Fancy pillar taps shall be as specified in clause 18.16.1 and 18.16.2 of the MES Schedule Part I.

37.3 Ferrules- These shall conform to the specifications given in MES Schedule Part I and will be as per IS-2692-78. The materials for different parts shall be as per table I given in IS. For taking out branch connection up to 50mm dia from main, ferrule will be used and for branch connection above 50mm dia, Tee will be used.

38. LAYING OF PIPE IN TRENCHES

38.1 Pipe shall be laid in trenches not less than 60cms below ground level or as directed by Engineer-in-Charge. Refilling of trenches with soft earth shall not be carried out untill the laying and jointing have been completed and tested.

38.2 Excavation and preparation of trenches shall be as per clause 18.42 of MES Schedule (Part I).

39. WATER TANKS

39.1 Cost of service water tanks (HDPE polythene tanks as shown in the drawings) including fittings and hoisting in position is included in the unit rates of buildings mentioned in Schedule ‘A‘ Part-I .

PARTICULAR SPECIFICATION

39.2 Rotational moulded polyethylene multi layered water storage tanks shall be as per IS-12701-1991 and BIS certification marking, make shall be as per Appendix.

39.3 The inlet connection shall be provided with a plunger type ball valve of brass of the dia of inlet pipe with polythene float ball. 25 mm bore GI Over flow pipe shall be taken from roof top to ground level with perforated PVC mosquito cover screwed to the pipe, whether shown on drawing or not.

39.4 Ball valve for water tank shall be of high pressure grade of brass with polythene float ball conforming to IS-1703 and as specified in clause 18.19 of MES chedule (Part-I). The rod of the ball valve shall be cast or extruded brass at the option of the contractor.

40.0 **POLYTHENE CONNECTORS**

40.1 Polythene pipe connector with two sets of mild steel chromium plated adaptors to connect steel tubing to the flushing cistern/WHB shall be as per samples kept in the office of GE and of approved make. The connector shall be of adequate length suitable for 15mm nominal bore GI pipe complete with brass unions at tails and rubber washers. Longer lengths up to 40 cm may be used without any price adjustment.

40.2 The weight of the pipe shall not be less than 150 gms per metre and gross overall weight of the polythene pipe used in polyethylene pipe and connector not less than 80 grams. The polyethylene pipe used in polyethylene pipe connector shall conform to IS-3076. 40.3 Cost of polyethylene connectors required for wash hand basins and cisterns etc is deemed to be included in the lump sum cost of buildings mentioned in Schedule ‘A‘ Part I.

41. **TESTING**:

41.1 Testing pressure shall be the maximum working pressure as specified in IS and there shall not be sweating or leakage in joints and fittings. A test certificate will be prepared and signed both by Engineer-in-Charge and contractor. The test should be carried out before pipe are covered by plaster/concrete etc.

41.2 All Equipment for testing shall be provided by the contractor.

41.3 **DRAWINGS**:
Contractor shall submit 6 copies of drawing for internal water supply layouts duly signed for records.

42. **INTERNAL ELECTRIFICATION**:

42.1. **SCOPE**

The work covers Internal Electrification as described in Schedule ‘A‘ under respective parts, Notes to Schedule _A‘, as shown on drgs and as specified here in after.

42.2. **WORKMANSHIP**:

(a) The work carried out shall be of the highest standard conforming to latest edition of Indian Electricity Act and rules framed there under as applicable and Para 19.2 of SSR 1991 (part I) read in conjunction with relevant IS codes for Internal electrification and code of practice for electric wiring installation as per IS: 2274 with amendments. All cable connection to the MCBs shall be by using copper lugs/thimbles only.

(b) Refer condition 25 of General conditions of contracts (IAFW 2249). All electric work shall be carried out properly by skilled and licensed electrician under the supervision of qualified electrical supervisor. The contractor shall on demand by Engineer in Charge produce such evidence of qualifications of his workmen and supervisor during the currency of contract. This provision shall be complied with even if the contractor himself is qualified and enlisted for electrical work.

(c) All the work shall be executed as per the code of practice and as listed in the IS. All electrical work shall be governed by the requirement as specified under the Indian electricity Act/Rules with latest amendments. The code of practice for electric wiring installation as per IS: 2274 with amendment No. 1 will be strictly followed.

Internal wiring shall conform to IS: 732 code of practice for electrical wiring installation. Point wiring shall be with 1100 Volts grade cable with stranded copper conductor all as

(d) described in Schedule 'A' items:-

(a) Screws: All screws shall be of steel chromium plated.
(b) Link clips : These shall be of aluminium conforming to IS: 2412
(c) Terminal sunken boxes shall be of suitable sizes required to mount switches, regulators and 3 pin sockets made up of galvanized iron sheet of thickness not less than 1.5 mm.

wiring shall be carried out all as described in Schedule 'A' and as specified in MES Schedule part I.

42.3. **SAMPLES**

The GE in writing before incorporation in the work shall be first approved samples of all electrical fittings. Utility light fittings shall be of the best quality available and conform to the samples kept in GE's office duly approved by him. Samples of the following shall be supplied within 30 days of the date of issue of work order:-

(a) All types of cable

(b) Fittings, wires and accessories used for Internal Electrification.

(c) The contractor shall ensure that the materials used in the works are identical with the approved samples. The materials shall be brought to the site by the contractor in the makers' original packing with the seal intact or with wrapper and shall not be installed unless approved by Engineer in Charge.

(b) Height of switch boards, socket outlet, light fittings and fans etc shall be as shown in drawings or as directed by the GE.

(c) All electrical fittings & wiring shall run clear of doors, windows & other openings.

(d) Arrangement for lighting circuit shall be as decided by Engineer in Charge. Permissible number of points per circuit shall not be more than 8.

(e) Loop in system of wiring shall be followed throughout the Installations.

(f) No burrs or twisted joints shall be made at intermediate points in the through run of cable. If inevitable, such joint shall be made through proper cut out.

(g) In any condition cables shall not come in contact with metallic pipe or non earthed metal work.

(h) Not more than two power socket out let points shall be wired on one circuit.

(j) Diagonal run of wiring shall not be permitted.

(k) Suitable inspection boxes shall be provided for periodical inspection or to facilitate removal of wiring. In concealed wiring these shall be mounted flush with the walls.

(l) For conduit wiring colour coding and colour of insulation of wires shall be got approved before the wires/cables are procured.

PARTICULAR SPECIFICATION

(m) Where conduits are concealed, these shall be held in chases with MS hooks except in RCC work where these can be secured to reinforcement with binding wire.

(m) Cutting and forming chases in brick work/concrete where directed shall be done while the building work is in progress prior to the commencement of plastering to brick work. Gaps/voids of all cut/formed chases including recesses formed for terminal boxes for housing power plugs, switches, sockets etc shall be filled flush with cement concrete (1:3:6) type C-1, using 20 mm graded stone aggregate or in cement mortar (1:3) mix after fixation of pipes/boxes etc prior to the plastering of walls at no extra cost.

(o) Clause 19.101 to 19.111 and 19.114 of SSR Part I will be referred to for workmanship. Special attention is drawn to clause19.111 of SSR Part I for method of passing conductors through walls and floors.

42.4 **MATERIALS**:

(a) All materials to be incorporated in the work shall be of the highest standard and unless specified otherwise, shall strictly comply with the relevant ISS or BSS for which no ISS has been published.

(b) Unless otherwise specified point wiring shall be with 1100 volt grade cable conforming to IS: 694 with stranded copper conductor all as described in different items of Schedule ‘A‘.

(c) Wooden battens/blocks/Boards shall be of well seasoned hard wood wrought on all exposed surfaces and coated with shellac varnish. The finished minimum thickness of batten shall be 10 mm.

42.5. **CONDUIT & CONDUIT FITTINGS**:
PVC conduits shall be of PVC medium grade of 25mm dia ISI marked. Conduit fittings shall be of un plasticised PVC conforming to IS: 3419.

(a) **SUNKEN GI BOXES**:-
For housing the various fittings such as switches, sockets, fan regulators etc, shall be made of galvanised steel sheet not less than 1.6 mm thick conforming to IS: 5133. The boxes shall be embedded in walls in cement mortar 1:3 and need not be fixed with plugs or screws. The minimum depth of these boxes shall be 60 mm and each box shall have an earth dolly. Corner of boxes shall have 3 mm thick MS sheet corners with screwed holes for fixing laminated sheet, 3 mm uniform thickness. Laminated sheet of approved colour shall be fixed with brass screws & washer to box.

(b) **PLASTIC LAMINATED SHEET TO COVER SUNKEN BOXES FOR POINT WIRING**: ‘Hylam / Wood lite / Anchor brand with the make embossed on the sheet.

MATERIALS - The materials shall be as follows:-

Sl No.	Materials	Reference to Para of SSR (Part-I)
(a)	Cable cords and earthing leads	19.24
(b)	Cable for internal wiring for light power	19.25
(c)	Wooden battens, blocks and boards	19.28
(d)	Conduit and conduit fittings and accessories	19.29
(e)	Plug/Gutties	19.30
(f)	Screws and fastenings, link clips	10.31
(g)	Ceiling rose	19.32
(h)	Shades	19.33
(j)	Bulkhead fittings	19.34
(k)	Socket outlets	19.40

PARTICULAR SPECIFICATION

(l)	Lamp holders	19.41
(m)	Main switch	19.45
(n)	Switch boards	19.10
(o)	Distribution fuse board	19.44
(p)	Miniature circuit breaker	19.46

42.6 **EQUIPMENTS AND MATERIALS**

Piano type switches, lamp holder, main switches and socket outlets/switch fuse, cable for internal wiring, miniature circuit breaker etc shall be of the make as approved by GE. Flexible chord 23/.0076 shall conform to IS: 434. Steel Conduit shall be stove enameled black rigid steel conduit conforming to IS: 9539 (Part I).

(a) The following tests shall be carried out to the entire satisfaction of the Engineer -in-Charge before the work is finally handed over by the contractor and completion certificate is issued :-

(a) Polarity test

(b) Insulation resistance test — Test results shall be lower then the minimum specified in IE rules/ISS.

(c) Continuity test sectional and overall

(d) Earth resistance test — This is not to exceed one ohm

(b) Clause 19.146 of SSR (Part I) is also applicable.

42.7 **LAYOUT PLAN**

(a) Locations of various light fittings/fans/socket outlet, main switches and DBS are tentatively shown on drawings.

(b) The layout of equipment, fittings etc and alignment of cables shown on drawings are tentative. The exact position of these shall however, be approved by the GE in wiring before the contractor shall take the work in hand without any price adjustment.

42.8 **TYPE OF ELECTRIC SUPPLY**

Type of electric supply will be 4 wire, 50 cycles, 415/230 volts, AC.

43. **EARTHING**

43.1 The earthing shall be strictly in conformity with MES Schedule Part I electrical plate No.3 Earth plate shall be galvanised iron pipe shall light grade, concrete PCC type D-2 (1:4:8), wire mesh 0.63 mm and funnel of PGI sheet, PCC cover and frame shall be 10mm thickness: -

(a) All metal parts shall be provided with a continuous earth wire stranded PVC insulated copper conductor of size not less than main phase conductor, conforming to IS-732.

(b) Supply and fixing of switch fuse distribution boards shall include for the earth continuity conductor as specified above.

(c) No extra payment shall be made if the contractor is required to provide extra depth of pit and extra quantity of charcoal dust to achieve specified test result.

(d) Necessary tests as per SSR part I/Relevant IS shall also be conducted by the contractor without any extra cost and test results thereon shall be kept on record.

PARTICULAR SPECIFICATION

43.2 . **RECORD DRAWINGS**:

The contractor shall submit 6 copies of the following drawings duly signed by him to the Engineer-in-Charge without any extra cost :-

(a) Internal wiring diagram to include :-

(i) Lights, switches, sockets, fans etc and circuit diagram as per symbols adopted in the extract drawings.

(ii) All outlets marked with circuit number to which they belong.

(iii) Typical section indicating mounting heights adopted.

(b) Drawing for each panel board shall be submitted by the contractor before execution for approval of the Engineer-in-Charge.

(c) One copy of the above drawings shall be submitted to the Engineer-in- Charge within 15 days of the placing of the work order for his approval and remaining with any alterations as directed by the Engineer-in- Charge within 15 days of the completion of work.

44. **SEWAGE DISPOSAL**:

44.1. **GENERAL**

The work of external sewage disposal includes external sewage disposal beyond internal sanitary installation as described in relevant items Schedule 'A', as specified here-in-after, as specified here-in-before for the relevant trades of works, as shown on drawing as directed.The work of sewage disposal shall be carried out as described in Schedule 'A' items, as shown on drawings and as directed by Engineer-in-Charge. Layout of sewage line and manhole shall be approved by GE before execution of work. For general works like excavation and earth works, concrete, brick work, plaster etc refer relevant clauses here-in-before.

44.2 **SUB SOIL WATER LEVEL**

The contractor is advised to ascertain the maximum and minimum sub soil water level as excavation for sewers may extend beyond sub soil water level and deeper foundations may be necessitated due to site conditions. No extra payment on this account shall be admissible to the contractor, if he has to work under sub soil water level.

44.3 **PIPE TRENCHES**

Trenches for pipe shall be excavated to the lines, level and gradient as directed by the Engineer-in-Charge. Filling to a depth of 150mm above the top of the pipe shall be made with earth free from clods, etc, and above this level filling shall be done as described in MES Schedule. Surplus spoil shall be disposed off as directed.

44.4. **CLASS NP -3 REINFORCED CEMENT CONCRETE PIPE**:

Reinforced concrete pipes for drains shall be non-pressure type, class NP-3 and shall conform to IS 458-2003. Specification for concrete pipes with or without reinforcement. Bends, junctions and specials shall be of cast iron and of suitable size. The ends of concrete pipes shall be suitable for butt end joints. The butt ends shall be prepared for collar joint with grooves. The pipe joints shall be capable of withstanding the same pressure as the pipe. Concrete pipes shall be straight and free from cracks excepting craze cracks. The ends of pipes shall be square with their longitudinal axis so that when placed in a straight line in the trench, no opening between ends in contact shall exceed 3 mm in pipes upto 600 mm diameter and 6 mm in pipes greater than 600 mm in diameter. The outside and inside surfaces of the pipes shall be smooth dense and hard and shall not be coated with cement wash or other preparation. The pipe shall be free from defects resulting from imperfect grading of the aggregate, missing or moulding. Pipe shall be free from local dents or bulges greater than 3 mm in depth and extending over a length in any direction greater than twice the thickness of barrel.

44.5 **MANHOLES**

Manhole shall be built all as specified in clause 18.78 of MES schedule Part I with the following modifications.

PARTICULAR SPECIFICATION

(i) Cover slab RCC M-25 design mix to be provided.

(ii) In lieu of cast iron cover, precast RCC manhole covers as described in clause 27.6 below shall be provided.

(iii) RCC precast manhole covers circular or square as shown in the drawing of RCC M-35 grade reinforced shall be as per relevant or as per manufacturers instruction. Joints of pipe and manholes shall be filled with PCC 1:2:4 type B-0 with WPC.

(iv) Steps in manholes shall be cast iron (CI Rungs) and shall be coated with a black bituminous composition

44.6 **RCC MANHOLE COVER**

44.6.1 SFRC manhole covers with frame confirming to 1912592:2002 HD 20 grade design mix as per pattern approved by the GE. These shall have MES markings as per design approved by GE.

45. **TESTING**

Water and smoke tests shall be carried out to ensure that the manholes, sewers are without any leakage. Results of the tests carried out on various sections of the sewer shall be signed both by Engineer-in-Charge and contractor or their representative. All material and labour etc required for testing shall be provided by the contractor without any extra cost to the Govt.

46 & 47 **Blank**

48 **CI RUNGS/STEPS**

48.1 The rungs shall conform to IS: 5455-1969. The step shall be clean, well cast and they shall be free from air and sand holes, cold shuts and warpings which are likely to impair the utility of castings.

48.2 The portion of the step which projects from the wall of the manhole shall have a raised chequerred design to provide an adequate non-slip grip. Any ribs, cheque ring, lettering of other projection for thick purpose shall be raised above the eneral plane of top surface of step and shall be placed particularly along the edges of treads. It is considered that in this position the most protection given s against slip.

48.3 Rungs shall be provided in all manholes over 0.8m in depth shall preferably be of cast iron and suitable dimensions (see IS : 5455-1969*). May be 300 mm apart horizontally as well as vertically and shall project a minimum of 100mm beyond the finished surface of the manhole wall. The top rung shall be 450 mm below the manhole cover and the lowest not more than 300 mm above the benching. Footrests shall be painted with coal tar, the portion embedded in masonry on cement concrete block being painted with thick cement slurry before fixing.

49. **EXTERNAL WATER SUPPLY**:

49.1 **GENERAL**

(a) Work shall be carried out by licensed plumbers

(b) Layout of pipes shall be generally as shown on drawings.

(c) The holes through walls, floors etc, around the pipes shall be made good with cement and sand mortar (1:4) for full thickness and finished as per surrounding surface.

PARTICULAR SPECIFICATION

(d) The location and distribution of various items on the drawings are tentative. Alteration can be made by GE as per the site requirements and there shall be no price adjustment on this account. The rates quoted by the contractor against each item of Schedule 'A', shall be deemed to include expenditure on account of all extras which are not specifically shown on drawings or given in particulars specifications but are essential for the execution of work. In case of difference of opinion in this respect between the contractor and the GE, the decision of the Accepting officer shall be final, conclusive and binding.

(e) Material and workmanship required to be incorporated in these works and services and testing shall be as given under specification and workmanship in the trade section of MES Standard Schedule of Rates Part-I & II duly modified by these particular specifications and notes on the drawings.

(f) Layout of water supply pipeline including fittings shown on drawings is tentative. The exact layout/alignment of water supply pipe line including their fittings etc shall however be as directed and approved by the Engineer-in-Charge in writing before commencement of work.

(g) On completion of the entire work, the contractor shall submit on tracing cloth to the Engineer-in-Charge, layout plans drawn to a suitable scale, showing actual routes/alignment of various pipe lines(with indication of their sizes) and fittings and location of various valve, valve chamber etc without any extra cost. The said layout plans shall indicate the distance of pipe lines from the main buildings and roads. All sizes and dimensions shall be shown in the above plans in metric units (i.e. in CGS system) only. All work shall be carried out by licensed plumbers/fitters/skilled labourers.

49.2. **EXCAVATION AND EARTH WORK**:

Excavation for pipes/cables shall be done all as specified in MES Schedule. Workmanship shall be as specified in clause 18.42 of MES Schedule Part I.

(a) Before commencement of excavation the contractor and Engineer-in- Charge shall jointly take levels at suitable intervals decided by Engineer-in-Charge along the alignment marked on the ground and point the same in longitudinal section.

(b) The proposed levels shall be signed by the contractor and GE as a token of acceptance, the minimum depth of excavation for pipes shall be 70 cms below ground level. The quantity of excavation shall be calculated from the prismoidal formula as far as possible.

Note: Invert levels for the purpose of this shall be taken as bottom most levels of pipe and not the flow.

(c) The contractor shall excavate to levels as shown in longitudinal sections. Excavation made deeper than required level shall be made good by the contractor at his own expense with approved earth by ramming and watering in 150mm layer. No payment shall be made for extra depth excavated and filled by the contractor

(d) The width at bottom of excavation in trenches for various bores of pipe shall be as specified in section 3 clause 32.3 on page 21 of MES Schedule Part II at joints of pipes and fittings. Width of excavation shall be sufficient for a person to make proper joints. Extra excavation if required on account of this and any excavation in addition to the authorised widths will not be measured and the lump sum rate quoted against the item shall be deemed to include for these allowances.

(e) When roads have to be crossed, half the width shall be dug at a time and proper working notices, signs and light shall be displayed and a watchman shall be posted to avoid accidents.

PARTICULAR SPECIFICATION

Note : Invert levels for the purpose of this shall be taken as bottom most levels of pipe and not the flow.

(f) The contractor shall excavate to levels as shown in longitudinal sections.Excavation made deeper than required level shall be made good by the contractor at his own xpense with approved earth by ramming and watering in 150mm layer. No payment shall be made for extra depth excavated and filled by the contractor.

(g) The width at bottom of excavation in trenches for various bores of pipe shall be as specified in section 3 clause 32.3 on page 21 of MES Schedule Part II at joints of pipes and fittings. Width of excavation shall be sufficient for a person to make proper joints. Extra excavation if required on account of this and any excavation in addition to the authorised widths will not be measured and the lump sum rate quoted against the item shall be deemed to include for these allowances.

(h) When roads have to be crossed, half the width shall be dug at a time and proper working notices, signs and light shall be displayed and a watchman shall be posted to avoid accidents.

49.3 RETURN, FILLING IN TRENCHES AND DISPOSAL

(a) After excavation is carried out to required depth, the bottom of trenches shall be evenly dressed and well rammed before pipes are laid down. Unless otherwise specified, all trenches shall be filled back with spoil obtained from excavation, watered, rammed and consolidated to the satisfaction of Engineer-in-Charge and shall be laid in regular layer not exceeding 150mm thick. The returning filling specified above shall be done up to height of 80mm above ground level. Settlement of earth in trenches which may take place during the execution of work or during the maintenance period shall be made up to date by the contractor at his own cost.

(b) Joints of pipes shall be kept open and no filling over the joints shall be done till pipe is tested. In case these are filled up or get filled up during the course of time, these shall again be opened before commencing the testing of the pipe line and after ward, the trenches shall be properly filled.

(c) Pipes shall not be laid directly over rock [where rock excavation is done but 150mm cushion of approved earth, well rammed shall be provided below the pipes.

(d) All surplus spoil obtained form excavation shall be removed to a distance not exceeding 100 metres and spread and levelled at places as directed by Engineer- in-Charge without any extra cost to Govt.

49.4 GI PIPES:

(a) GI pipes and fittings shall be medium grade, conforming to IS 1239 (Part I) and IS – 1239 (Part II) respectively with tolerance as specified in Clause 18.42 of SSR Part I.

(b) GI pipes and fitting shall be laid at a depth not less than 225mm and not more than 450mm below ground level all as specified in Clause 18.50 of SSR Part I.

(c) Jointing of GI pipes and fittings shall be done as specified in Clause 18.50.3 of SSR Part I.

PARTICULAR SPECIFICATION

49.5 CAST IRON PRESSURE PIPES

49.5.1 Centrifugal cast iron pipes shall confirm to IS 1536-2001 : Specification for centrifugally cast (spun) iron pressure pipes for water and vertically cast iron pressure pipes shall conform to IS 1537-1976 : Specification for vertically cast iron pressure pipes for water. Spigot and socket pipe (centrifugally) shall be of Class LA, Class A and Class B and flanged pipes (centrifugally) shall be of Class A& Class B as indicated. Vertically cast iron pressure pipe shall be of Class 'A' or Class 'B' with spigot and socket ends or flanged ends as indicated or specified. Cast iron fittings shall conform to IS-1538-1993. Also refer Clause 18.2, 18.2.1 to 18.2.4.6 of MES Schedule 2009 Part I. The fittings shall withstand hydrostatic test pressure without showing leakage, sweating or defects of any kind as laid down in IS Manufacturer Test Certificate shall be produced by the contractor.

49.6 DUCTILE IRON PIPES AND FITTINGS

49.6.1 All Ductile iron centrifugally cast spun) pressure pipe shall conform to IS- 8329-2000 and fittings shall conform to IS 9523-2000. Spigot and socket pipes and flanged pipes shall be as described Schedule 'A'. Laying, jointing and testing of D.I. pipes shall be as per IS-12288-1987. The fittings shall withstand hydrostatic test pressure without showing leakage, sweating or defects of any kind as laid down in IS Manufacturer Test Certificate shall be produced by the contractor.

49.7 FLANGED JOINTS

49.7.1 Jointing material used between flanges of pipes/fittings shall be of thickness 3.00mm and samples shall be got approved from the GE before procurement in bulk. Flanges shall conform to IS-6392‖ Specification for steel pipe flanges‖ shall be welded or screwed (at the option of contractor without any price adjustment on the pipe/fittings as per IS requirement. The numbers and position of holes on the flanges shall be as per IS requirement. Nuts, bolts and washers shall be of MS as per IS.

49.8 JOINTING OF CI PIPES AND FITTINGS

49.8.1 (a) No flange or spigot and socket joint shall be buried in the masonry of the valve chambers.

(b) Cast iron fittings and specials such as bends, tapers, tees, connecting pieces etc. shall be carefully and accurately aligned and jointed to pipes all as described under relevant clause of IS 3114 of 1985 as amended.

(c) While laying pipes in trenches care shall be taken that no earth etc gets inside pipe. The work when completed shall be tested as per IS requirements

(d) (i) In run lead joints for pipes, fitting and specials, molten lead and spun yarn will be used. The spun yarn shall be handled with care in order to prevent contamination and shall be dry when put into place in the joints. The yarn shall be free of oil, fat or greasy substances.

(ii) When a single strand of yarn is used, it shall have an over lap at the top not more than 5 cms. When more than single strand is required for a joint, each strand shall be cut to sufficient length so that the ends of pipe will meet without causing overlap. The end of the strand shall meet on opposite side of the pipes and not on the top or the bottom. Successive strand of yarning material shall be driven home separately.

(iii) The lead for caulking purposes shall conform to IS 782-1978. The quantity of lead and spun yarn required for different sizes of pipes shall be as specified in clause 18.48.24 of MES Schedule Part I.

PARTICULAR SPECIFICATION

49.9 **DETECTION OF CRACKS OF PIPES/TUBES**

49.9.1 The pipes/tubes and fittings shall be inspected for defects and rung with a light hammer preferably while suspended to detect cracks. If doubt persists further confirmation shall be obtained by pouring little paraffin on the pipe at the suspected spot. The pipes/tubes and fittings shall be inspected for defects and rung with a light hammer preferably while suspended to detect cracks. If doubt persists further confirmation shall be obtained by pouring little paraffin on the pipe at the suspected spot.

49.10 **SLUICE VALVES/REFLUX OR NON RETURN VALVES**

49.10.1 Sluice valve shall comply with the requirements of IS 14846-2000 and shall be of class PN 1. The make of sluice valve shall be as mentioned in Appendix 'D'.

49.11 Non Return Valves (Reflux Valves)

49.11.1 These shall be flanged, swing check type valves, single door pattern conforming to IS – 5312 (Pt I). The valves shall be with or without by-pass arrangement and of the makes as mentioned in Appendix.

49.11.2 The valves shall be fixed as specified in clause 18.57 of MES Schedule Part I. 49.12 Testing of Pipes and pipe lines;

49.12.1 All pipes and fittings shall be inspected carefully before being laid. Broken or defective pipes shall not be used and removed from the site of works. Pipes shall be rung with light hammer preferably white suspended to detect cracks. If doubt persists, confirmation may be obtained by powering a little paraffin on the inside of the pipe at the suspected spot, if crack is present the paraffin seeps through and shows on the outer surface.

49.13 Testing of Distribution pipeline: -

49.13.1 (a) Pressure and leakage test:- After a new pipe line has been laid, jointed,the test shall be carried out as directed at in double the working pressure for period of two hours.

(i) A record of the above test shall be maintained and separate register duly signed by the contractor and Engineer-in-charge.

(ii) All facilities and equipments for testing will be arranged by the contractor at the own cost.

(iii) Any cracked or defective pipes, fittings valves and joints discovered as a result of the above test shall be removed and replaced by the contractor at this own cost and the test shall be repeated till the results are obtained to the satisfaction of the Engineer-in-charge.

(iv) During the leakage test should any pipe or joint fitting disclosed leakage pipe should be removed/replaced until the leakage is removed to the satisfaction of Engineer-in-charge and as specified.

49.14 Testing of pressure pipe rising main:-
49.14.1 The pressure pipe rising main only will be subjected to testing in accordance with para 18-48.7 of SSR Part-I.

50. **VALVE PIT CHAMBER**:
The unit rate for valve chamber in Sch 'A' and as shown on drawings shall include for:-

(a) Excavation and earth work shall be in any type of soil. Surplus soil shall be disposed off to a distance not exceeding 50 metres.

(b) PCC in foundation shall be 1:4:8 type D-2 using 40mm graded stone aggregate and top exposed surface finished fair without using extra cement.

PARTICULAR SPECIFICATION

(c) Brick work shall be one brick thick wall built in CM(1:4).

(d) Coping shall be in PCC (1:3:6) Type C-1.

(e) Plaster 15mm thick in cement mortar (1:4) shall be provided on all brick and concrete surface internally, surface finished fair and even without using extra cement.

Cover shall be of 6.0 mm thick mild steel sheet fixed with MS angle iron 40mm x 40mm x 6mm thick and locking arrangement all as shown on drawing. All steel and iron work shall be given two coats of synthetic enamel paint over a coat red oxide primer.

50. EXTERNAL ELECTRIFICATION

50.1 GENERAL

(a) Specifications of materials and workmanship shall be all as described in MES Schedule (Part-I) except where specifically mentioned otherwise in these documents. General rules, preambles to various sections, rates, special conditions, methods of measurements etc given in MES Schedule shall apply to this contract unless otherwise mentioned in these tender documents. The location and distribution of various items in the drawings are tentative. Alterations can be made by the GE as per site requirements and there and there shall be no price adjustment on this account. The lump sum quoted by the contractor against Schedule 'A' shall be deemed to included expenditure on account of all extras which are not specifically shown on drawings or given in particular specifications but are essential to the execution of work.

(b) In case of difference of opinion in this respect between the contractor and GE/CWE, the decision of the Accepting Officer shall be final, conclusive and binding.

(c) On completion of entire work the contractor shall submit in triplicate to the Engineer-in-Charge layout plan showing actual position of poles, overhead lines, routes, cable run and all other information that will be necessary for maintenance and operation.

50.2 LAYOUT

The layout of OH lines/UG cable shown on site plan is tentative. The electric lines and cables shall be laid by the route as actually directed by the Engineer-in-Charge. The Contractor shall measure on the ground the actual length along with route and work out the exact requirement of electric lines and cable before placing order for the same.

50.3 EXCAVATION AND EARTH WORK

(a) Excavation for laying of underground cables/poles/struts and fixing of stay, etc shall be restricted to dimensions shown in the drawings and as directed by the Engineer-in-Charge. The minimum depth and width of the trenches shall be as laid in paras 19.74, 19.74.1 and 19.76 of SSR Part-I respectively, In case of excavation of poles/struts and stays, excavation in excess of required depth shall be made good by the contractor in cement concrete 1:4:8 type D-2 without any extra cost to the Govt.

(b) The trenches for cables shall be excavated to levels/gradients as directed by the Engineer-in-Charge.

(c) All excavated earth be deposited at least one m clear to the edge of the excavation. When roads have to be crossed, trench up to half the width of road shall be dug at a time and proper notice, sign boards and lights shall be displayed and watchman posted to prevent accident.

(d) Refilling of earth in trenches shall be dome in 25cm thick layers with excavated soil approved by the Engineer-in-Charge in a manner and to ensure greater compactness and solidity. Each layer shall be well rammed, surplus soil removed as provided in Schedule 'A'.

PARTICULAR SPECIFICATION

50.4 **CONCRETE, BRICKS, AGGREGATE SAND ETC**

Concrete work shall be carried out all as specified here-in-before. For other general specifications such as bricks, aggregates, sand etc refer relevant items in Schedule 'A' and Particular Specifications given here in before.

50.5 **CABLE PROTECTION**

(a) Cable shall be protected in trenches by providing sand cushioning and cable covers all as described in Schedule 'A' items.

(b) On poles/walls cable shall be protected by GI protection pipes all as specified. The pipes shall be fixed to pole by means of proper MS clamps.

(c) Pipe for laying of cable and earth wire shall conform to IS: 1239.

(d) Ends of pipe shall be properly sealed with wooden bush and bitumen compound after drawing the cables through pipe.

50.6 **CABLE END BOX**

Cable end box shall be epoxy cold resin filled type suitable for relay voltage grade as described and jointed as per relevant IS/manufacturers instruction.

50.7 **CABLE GLANDS**

Cable glands shall be of brass chromium plated of appropriate size as per IE Rules.

50.8 **SUPPORTS, CLAMPS ETC**

(a) The clamps shall be made out of MS flat iron of size not less than 50 x 6 mm. All nuts, bolts, washers used in fixing clamps shall be of galvanised mild steel and of size 19 mm dia.

(b) While calculating the weight of steel, the weight of nuts/bolts/washers etc shall be included for payment and weight of paint will not be included for payment.

50.9 **EARTH WIRE**

(a) Earth wire shall be of 4 mm dia galvanized iron and shall be stretched on poles with clamps or eye body hooks.

(b) The earth wire shall be paid by weight either by calculating from table or actual, should these differ only lesser shall be paid. No extra allowance for sag and jointing shall be made while computing the weight of GI wire.

(c) Earthing to metal supports with and accessories shall be done by 2.5 mm dia wire, connected to main continuous earth wire running on poles. Nothing extra shall be paid on this account.

(d) The contractor shall furnish the test certificate from the manufacturers of the following equipments/instruments, material and accessories to GE, in accordance with the relevant ISS.

PARTICULAR SPECIFICATION

50.10 LT CABLES

(a) LT cables shall be of 2/3/3½ or 4 core aluminium conductors, XLPE insulated Heavy duty and suitable for 1100 volts grade and conforming to IS: 7098 (Part-I). Cross section area of these cables shall be armoured unless specified otherwise in Schedule 'A'.

(b) The aluminium conductors used should be stranded, compacted and circular/shaped and the main insulation should be of cross linked polyethylene (XLPE) with inner sheathing PVC wrapped or PVC extruded and each core of the cable should have colour identification all as specified in IS.

(c) Armour over the inner sheath should be either of strip or wire type and outer sheath should be of extruded PVC conforming to IS: 5831 of 1970.

50.11 LAYING OF HT/LT CABLES

(a) The work of underground cables shall be carried out as specified in IS: 1255-1983 or latest edition and as described in section 19 (Electrical works section) sub clause 19.74 to 19.82.2 and 19.83 to 19.86.1 of MES Standard Schedule of Rates 2009 (Part–I specifications). The cables under road crossing, railway tracks etc (existing or proposed) shall be enclosed in GI pipes. The sizes of the pipes shall be as decided by the Engineer-in-Charge or as described in Schedule 'A'. Size of the pipe shall be such that the suitable space around the cable is available. The pipe shall be provided through out the width of roads, drains and extended for 1200mm on either side of the road. The joints of GI pipe shall be as specified in standard schedule of Rates 2009 (Part-I specifications). GI pipes shall be of light grade conforming to IS: 1239. The both ends of pipes shall be sealed with wooded plug/pegs and bitumen compound after drawing the cable through pipe. Where the cables are to be laid in the trenches, 80 mm thick cushion of clean dry sand shall be provided in the trench before laying the cable. After the cable has been uncoiled and laid properly into the trenches, it shall be covered with dry sand to a depth of 150mm. This sand will gently punned down thus providing good bedding for the protective cable covers. The punning shall be done by hand and mechanical punners shall not be used. The depth of cable trenches shall be minimum 600 mm and 800 mm respectively for LT and HT cables and width of the trenching shall not be less than 450 mm.

(b) The reinforced pre cast concrete cover or bricks for protection of cable for covering the cables shall be as specified herein before. The pre cast cover shall be laid flat (width of protection being equal to length of brick). After the layer of pre cast cover/brick is laid the remaining trenches will be filled with excavated earth all as described in Schedule 'A'.

(c) Care should be taken during laying of cable to avoid sharp bending and twisting. Cable shall be unwound from the drum by lifting the drum on the center shaft supported both ends with suitable Jacks/stands. Under no circumstances the cable winding shall be lifted off a coil or drum lying flat at the flanges to avoid serious twist and damages. Further the minimum bending radius of 15 x diameter of the cable should be ensured.

(d) The cable shall be snaked to provide a 4 m length of cable for further joints at all straight joints (both side of the joint i.e. 4 m on each side) of cable laid in trenches and a 3 m lengths at terminal cable boxes. The above additional length of cable specified for snaking shall also be measured and paid for under the respective items of cables in Schedule 'A'. Laying pre cast cover/brick protection and sand cushion at places where cables are snaked shall be done as directed by the GE.

PARTICULAR SPECIFICATION

(e) Cable path indicators shall be provided at distance of every 100 meters and at cable joints and turning of all cables except for street light cables and 2 core cables. The cost of these indicators shall be deemed to be included in the unit rates quoted for the respective cables. Cable path indicators shall be of CI plate of minimum 150 mm dia and 4mm thick indicating cable size and voltage written on it. The CI path indicator shall be fixed on MS angle iron with minimum two numbers of nuts and bolts. The nuts shall be tucking welded with bolts after full tightening. The size of MS angle iron shall be 30 mm x 30 mm x 6 mm and the length of angle iron should be 900mm adequate to ensure minimum planting depth of 400 mm below ground and 300 mm above GL. The angle iron, at the location of planting should be splitted and bent to shape to provide better grip and should be embedded in PCC (1:3:6) type C-1 using 20 mm graded stone aggregate and of size 200 x 200 x 300 mm. MS angle iron shall be painted with one coat of red oxide primer throughout length and thereafter it will be painted with two coats of synthetic enamel paint and bituminous paint in the portion above ground and below ground respectively. The cable path indicators shall be erected at the time of refilling the trenches and the cable path indicator shall be painted as under:-

(i) For HT 11 KV cables - “BROWN”
(ii) For LT Cables - “BLUE”

(f) While laying cables in ducts, care should be taken to ensure that the cables do not touch each other and minimum clear gap equivalent to diameter of the bigger cable should be provided. Further the cables should be fixed with suitable MS Flat iron (Minimum size 20 mm x 3 mm) clamps in the cable ducts to avoid jumbling. If necessary, cables may be fixed with clamps on the walls of the duct. Where cables are to be fixed by means of suitable clamps or bracket or on cable racks fixed along poles/steel structure, they shall be drawn through pipe as directed. Wherever the cables are embedded in the floor, it should be laid in the GI pipe all as directed. The cost of clamps etc shall be deemed to be included in the unit rate quoted for the respective cables.

(g) Lay out of cables shown in the drawings is for guidance only. The contractor shall, however, verify the cable lengths actually required at site before procurement of the same.

(h) Joints in cable shall not be at distance less than 300 meters. When large length of cable are to be laid, exceeding 500 m, which is a standard drum length, the joints will be done at interval of 500 m. Straight through joints shall be measured and paid separately under the respective items of Schedule ‘A‘.

(j) The unit rates quoted for respective LT cables, shall be deemed to include the cost of cable glands, lugs/thimbles, preparation and making cable end joints including crimpling etc all as specified. The termination of LT cables in to Transformers/Switch gears/Feeder pillar-boxes/Distribution boxes shall in-variably be done through lugs or thimbles only to avoid loose connections. No termination of cables without suitable lugs/thimbles will be accepted. However, cable termination in case of HT cables shall bemeasured & paid separately under the respective items of Schedule ‘A‘.

(k) The metal sheath, metal screen (if any) and armour of all the LT/HT cables should be effectively earthed at both the ends all as specified in IS: 1255-1983 and SSR part-I.

(l) The cables should be purchased directly from the manufacturers or their authorized dealers only. Contractor should also submit the test certificate from the manufacturers besides carrying out the field tests. All new cables should be tested for insulation resistance using 1000 V, 2.5 KV and 5.0 KV IR Tester for LT, 11 KV and 33 KV grade cables respectively. Cables should be tested for insulation resistance between all the phase to phase and from each phase to earth before decoiling the cable from the drum and also after laying the cable in the trenches. Only after satisfactory results are obtained, cable jointing and termination work should commence. The test results of completed cable installation should also be measured and entered into record book for comparison purposes during service life of the cable installations. Arrangement for carrying out all these tests shall be made by the contractor without any extra cost to Govt in the presence of Garrison Engineer/ Engineer-in-Charge.

PARTICULAR SPECIFICATION

The record of all such tests shall also be signed by Engineer-in-charge and contractor and countersigned by Garrison Engineer. At the closing of the work on every day, cable endsshould be covered with suitable rubber/plastic caps and sealed effectively by insulating tap/bitumen to avoid any ingress of moisture into the cable.

(m) Unless otherwise specified in Schedule 'A' the cost of cutting through roads/drains for laying cable under roads/drains and making good shall be deemed to be included in the unit rate of cables laid in pipes.

(n) Wherever HT/LT cable will be crossing or running parallel to existing telephone cables, suitable measures will be taken and distance will be observed as per relevant ISS/IE Rules on the subject.

50.12 CABLE END TERMINATION AND STRAIGHT THROUGH JOINT FOR 1100V GRADE XLPE CABLES

(a) Cable and termination Indoor/ outdoor type and straight through joints for LT cables shall be based on cast resin system. The casting resin should be polyurethane based having compatibility with the cable components. The cast resin should be suitable to provide adequate electrical insulation, mechanical protection and moisture imperviousness to the joint. The joint should be suitable for compression connection/ mechanical connector/soldering of cable conductors to meet the technical requirement as required at site. The termination kit should be designed for easy installation and have a crutch sealing arrangement for environmental protection on the core trifurcation.

(b) All the jointing of cables in joint boxes etc and filling in of the compound shall be done strictly as per the instructions furnished by the manufacturers of the cables and joint boxes.

(c) The joints should be suitable to withstand mechanical impact, heat shock tests, load cycle test and water penetration test (under pressure) without damage to the outer sheath.

(d) No separate payment for providing above joints for LT cables shall be admissible. Rates quoted for cables shall be deemed to include for this provision.

50.13 STANDARD OF QUALITY AND WORKMANSHIP

(a) The supply and installation of all electrical equipment and accessories shall strictly comply with provisions contained in the latest edition of Indian Standard Code of practice/specifications as applicable to above mentioned works, except where such regulation and rules are modified by these Specifications.

(b) Refer condition 25 of General Conditions of Contracts IAFW-2249. All electrical work shall be carried out by properly skilled and electrical supervisors/ Engineers. The contractor shall, on demand by the Engineer-in-Charge produce such evidence of Qualifications of his workmen/ supervisors/ Engineers, either at commencement of work or at any time thereafter during the currency of the contract. The entire work shall be of high class with the best workmanship and to the entire satisfaction of GE/Engineer-in-Charge.

(c) The Kiosk, transformer, cables, accessories, etc. incorporated therein shall be capable of giving satisfactory operation under the climatic conditions.

(d) The design and manufacture of the equipment shall be of highest standard to ensure continuous and trouble free service.

50.14 EARTHING

(a) Earthing shall generally conform to the requirement of IS: 3043-1987 (Code of practice for earthing) and IE Rules Nos 32, 51, 61, 62, 67, 69, 88 (2) and 90 and all as specified in clause Nos 19.137 to 19.146 of SSR Part I 2009.

PARTICULAR SPECIFICATION

(b) The neutral conductor of a three phase four wire system shall be earthed by not less than two separate and distinct connections each having its own electrode both at the generating station and at the substation. For earthing of neutral of 33/11 KV transformers, copper plate electrode shall be used.

(c) The frame of every stationery motor, metallic parts of transformers, HT/LT switchgear panels shall be earthed by two separate and distinct connections with earth. This electrode used shall be galvanised iron or steel plate type.

(d) The earth electrode shall not be situated less than 1.5 meter away from any building and its location should be selected such that the soil has reasonable chance of remaining moist. Entrance, pavements and roadways shall be avoided for locating the earth electrode. The distance between any two-earth electrodes shall preferably be not less than 8 m.

(e) The type of earthing shall be as described in relevant item of schedule ‘A‘. The contractor shall obtain approval of Engineer-in-charge for layout of earthing before commencement and shall be executed in the presence of his representative. Excavation shall be passed by Engineer–in-charge before filling.

(f) Galvanised steel pipe electrode shall be of medium grade 40 mm dia and 2.5 m in length. Pipe electrodes shall be cut tapered at the bottom and provided with 12mm holes (Staggered) at 75mm center to center up to 2 m of length from bottom. The electrode shall be buried in the ground vertically with its top not less than 1.25 m below ground level all as specified in Electrical plate No. 2 of SSR part I of 2009 except the modification mentioned in these particular Specification and Schedule ‘A‘.

(g) Plate electrode shall be of 600 x 600 x 6 mm or 600 x 600 x 3 mm in case it is made of galvanised iron steel and copper respectively all as specified in schedule ‘A‘. The plate electrode shall be surrounded with charcoal dust and common salt mixture to a packed thickness of 300 mm on two sides perpendicular to the face of the plate and 150 mm below, above and out ward on remaining sides. The earthing shall be executed as shown in Electrical plate No. 3 of MES Standard Schedule of Rate 2009 (Part I) except modification mentioned in these particular Specifications and Schedule ‘A‘.

(h) The charcoal dust and common salt mixture and returning filling shall be done in layers not exceeding 15 Cm and shall be properly watered and rammed. Surplus soil shall be disposed off and site left cleared on completion. Common salt and charcoal dust as available in the market shall be used for earthing.

(i) The depth of earth electrode shall be as specified in Schedule ‘A‘ but the ultimate depth of earth electrode would depend on the nature of soil and contractor is required to take it to a depth where sufficient moisture exist and natural conditions are suitable for resistance as specified below.

(j) Main earthing lead shall be of minimum size 4 mm dia GI wire or 32 x 6 mm GI strip or 20 x 3 mm /32 x 6 mm copper strip as described in Schedule ‘A‘. The cost of earthing quoted in schedule ‘A’ (unless otherwise mentioned in Schedule ‘A‘) shall be deemed to include the main earthing lead of 7.5 m from the electrode.

(k) The earthing lead and protection pipe from electrode onwards shall be suitably protected from mechanical injury by a 15 mm dia GI pipe in case of GI wire and by 40 mm dia GI pipe in case of GI or copper strip from earth electrode to the test point. Protection pipe (unless otherwise mentioned in Schedule ‘A‘) shall be measured and paid separately beyond 7.5 m length from earth electrode. The protection pipe used shall be of light grade.

PARTICULAR SPECIFICATION

(l) The earthing system shall be mechanically robust and joint shall be capable of retaining low resistance even after many passage of fault current. Inter connections of earth continuity conductors and main and branch earth wire shall be made in such a way that reliable and good electrical connections are permanently ensured. All the joints of earth strip shall be made by welding and brazing with suitable overlap in case of GI earth strip and copper earth strip respectively. The overlap should not be less than twice the width of the earth strip. Before welding/brazing, the joints should be prepared and levelled properly. In case of welding, the welded area, after jointing, should be cleaned thoroughly by steel wire brush and sand papers, thereafter it shall be painted with two coats of bituminous paint to arrest the rusting. Further earth strip where exposed shall be painted with two coats of aluminium paint over one coat of red oxide paint.

(m) In the case of plate earth electrode, the earthing lead shall be properly levelled and welded in case of steel plate electrode and brazed in case of copper plate electrode and thereafter it shall also be securely bolted to the plate with two bolts, nuts, check nuts, and washers. The length of GI/ Copper earth strip to be welded/ brazed with earth plate shall not be less than 150mm. In case of pipe electrode, it shall be connected by means of through bolts, nuts and washers and cable sockets as indicated in IS: 3043. All material used for connecting the earth lead with electrode shall be galvanised iron in case of GI pipe or GI plate earth electrode and of tinned brass in case of copper plate electrode. The welded area shall be properly cleaned by steel wire brush and emery papers and thereafter it shall be painted with two coats of bituminous paint to arrest the rusting of the earth plate and electrode.

(n) Each earth system shall be so devised that the testing of individual earth is possible. The earth test point terminal block shall be provided in the circuit of each earth pit at a suitable location as directed by Engineer-in-Charge. Earth test point terminal block shall be made of gun metal/phosphorous bronze or galvanised iron or steel of size 75 x 75 x 25mm where earth lead is of copper or GI steel plate/GI wire respectively. Test point terminal block shall be drilled and screwed including 3 Nos of 8 mm dia 25 mm long hexagonal head screws for fixing of earth strip or wire. These screws shall be made of tinned copper in case of gun metal/phosphorous bronze terminal block else where it will be of Galvanised steel. In case of GI wire earth lead, it shall be connected to test terminal point using hexagonal headed screw and cable socket.

(o) Concrete chamber including coping shall be PCC (1:2:4) mix using 20mm graded stone aggregate. All the internal surfaces of concrete chamber shall be given 12 mm thick plaster in CM (1:4) finished fair without using extra cement. Funnel in the chamber shall be with wire gauge of 22 meshes per linear inch. The frame for earth pit shall be of MS angle iron of size 40 x 40 x 6 mm with RCC cover. The angle iron frame shall be firmly fixed into the PCC earth pit and RCC cover shall be 40 mm thick (1:2:4) type B-1 using 20mm graded stone aggregate with 8 mm dia TMT bars @ 75 mm C/C both ways with MS handle 12 mm dia. The other details as described in the schedule ‘A‘.

(p) On completion of the Installations, the earth resistance of each earth electrode shall be measured as specified in IS: 3043. The earth resistance of the whole system shall be as near to Zero as possible but shall not exceed one ohm in case of Earthing for Neutral of transformers/DG sets and HT system. Further in respect of other installations, value of earth system resistance shall not be more than 5 ohms.

(q) Earth resistance test shall be carried out by the contractor using his own measuring instrument and in the presence of Engineer-in-Charge and the test results shall be recorded and signed by contractor and the Engineer-in-charge. However, on testing after completion, if desired earth resistance is not obtained, the additional number of earthing may be made by the contractor and connected in Parallel with suitable earth strip all as directed by Garrison Engineer at no extra cost to Govt.

PARTICULAR SPECIFICATION

50.15 MOULDED CASE CIRCUIT BREAKERS (MCCBs)

(a) Moulded case circuit breakers (MCCBs) shall be suitable for operational voltage of 415 Volt AC, 50 Hz, 3 phase, 4 wire system for a rated current and ultimate breaking capacity as mentioned in Schedule 'A' and it shall conform to IS: 13947 (Part 2) and IEC: 60947 (Part 2). All MCCBs shall be of four-pole version unless mentioned other wise in Schedule 'A'.

(b) MCCBs shall be of Compact and elegant design suitable for reversible load and line terminations without affecting its performance. MCCBs shall be suitable for fixing flush on the panels and shall be provided with handle operating mechanism including Rotary Handle vari-depth type.

(c) The insulating case and cover of MCCBs shall be made of high resistant and flame retardant thermosetting insulating materials. The switching mechanism shall be quick make, quick break and trip free. The position of the operating knob/handle shall clearly indicate ON, OFF and TRIP position.

(d) Each pole shall be provided with a pair of contacts, which shall open at a high speed over a large distance under short circuit faults. The special designed arc cut outs of insulating materials shall be provided to contain the arc by providing effective arc quenching device. The tripping mechanism shall be thermal release or Electronic release or thermal magnetic release for protection for over load and short circuit as per the details given below: -

Capacity of MCCBs	Ultimate breaking Capacity	Overload release	Short circuit release
1	2	3	4
Upto 125 Amps	16 KA	Adjustable Thermal release (range 70% to 100 % of rated current)	Fixed magnetic type suitable for 10.0 In
160 to 630 Amps	36 KA	Adjustable Thermal release (range 70% to 100 % of rated current)	Adjustable magnetic release (3.5 to 10 times of rated current)
800 and 1250 Amps	50 KA	Adjustable electronic release (over load Zone range 40% to 100 % of rated current)	Adjustable electronic release (range (5 to 10 times of rated current)

(e) Under voltage trip, mechanical interlocks etc shall be provided as per standard practice and ISS. Under voltage trip shall be designed to operate when the control voltage drops below a tripping threshold i.e. 20% to 70% of rated voltage and shall be suitable for operation on 230V/415V AC.

(f) The terminals shall be suitable for both copper and aluminium terminations as specified. The MCCBs shall be one of the makes mentioned herein after in these Particular Specifications.

50.16 MCBs, DISTRIBUTION BOARDS AND PLUG & SOCKET DBs

(a) Distribution boards shall be factory made and conforming to IS: 8623 suitable for universal mounting copper bus bar, Neutral bar, and Earth bar, Standard DIN bar Rail and cable ties for cable management. Top and bottom shall have removable gland plates with knock outs. The copper bus bar shall be suitable for 100 amps unless specified other wise in Schedule 'A'.

PARTICULAR SPECIFICATION

(b) SPN Distribution boards shall be suitable for provision for DP/SPN MCBs/ Isolators as incomer and SP outgoings all as specified and shall be flush mounting type. The degree of protection shall be IP-43 protection with acrylic door unless specified otherwise in Schedule 'A'.

(c) TPN distribution boards shall be suitable for FP/TPN MCBs/Isolator as in comer and TPN/SP outgoings all as specified and shall comply with IP-43 protection with metal door unless mentioned otherwise in Schedule 'A'.

(d) All distribution boards shall be fixed flushed on the wall at locations as directed by Garrison Engineer.

(e) MCBs: - All MCBs shall be conforming to IS: 8828 of 1996 and shall be ISI marked. These shall be suitable for _C' curve and 10KA breaking capacity and shall be provided with box terminal on top and bottom both suitable for adopting cable size up to 35 Sqmm. The enclosure of MCBs shall be of _Molded Self Extinguishing thermo set plastic and these shall be suitable for snap fixing on standard Din Rail. The current capacities and poles shall be all as mentioned in Schedule 'A'.

(f) MCBs Isolators – All MCB type isolators shall conform to IS: 13947-3 and shall be suitable for impulse voltage of 6KV and short time with stand capacity of 1000 Amps for 0.3 seconds.

(g) Plug and Socket DBs :- Plug and socket distribution Boards including SP/DP MCBs for protection of appliances like window type/split type Air conditioners and Geysers shall only be provided all as specified. Similarly distribution boards with MDS Legrands type P-17 Tempera plugs and sockets including TPN/FP MCBs shall be provided to make available 3 phase 415 volts, 50Hz AC electric supply in Technical Buildings. All these plug and socket DBs shall be of universal mounting type. The makes of Distribution Boards, Plug and socket DBs and MCBs shall be all as specified here in after and the models/Type shall be as indicated in the respective Schedule.

51. DRAINAGE

51.1 Drainage work shall be carried out all as directed at site by the Engineer-in- charge and as described in Schedule 'A' and as specified in these particular specifications .

51.2 Materials such as coarse aggregate, fine aggregate shall be all as specified here-in-before.

51.3 The drains shall be constructed as shown in the drawings and as directed by the Engineer-in-Charge to the required slope and shall be connected to the existing drains to clear the rain water collected therein.

52. FIRE POINT

52.1 Fire Point work shall be carried out all as directed at site by the Engineer-in- Charge and as described in Schedule 'A' and as specified in these particular specifications.

52.2 Materials such as coarse aggregate, fine aggregate shall be all as specified herein-before.

52.3 The Fire Point shall be constructed as shown in the drawings and as directed by the Engineer-in-Charge

53. ROAD WORK

53.1 FORMATION FOR ROAD WORK

Preparation and compaction of formation surfaces shall be done with 8 to 12 tonne power roller all as per clause 20-A, 22 of MES schedule Part I.

PARTICULAR SPECIFICATION

53.2 **SOLING**

Soling shall consist of one layer of 150 mm thick spread thickness with approved quality locally available broken boulders/quarried stone and shall be laid & levelled in one layer watered & rolled to required surface camber all as specified in clause 20.A.20 and 20.A 20.1 of MES Schedule Part I 2009.

53.3 **MATERIAL FOR ROAD WORK**

(a) All materials required for road work shall be stacked for measurements before incorporation in the work.

(b) The quantities collected shall be sufficient for works required and in addition (at no extra cost to the Govt) also adequate for repairing to fill hollows appearing during the execution of works.

(c) Measurements of stacks shall be jointly taken by the Engineer-in-Charge and the contractor‘s representative and shall be recorded in the measurement book as a check of the quantities brought for the work. The measurement of stacks recorded in measurement book shall be signed by the contractor and the Engineer-in-charge.

53.4 **AGGREGATE FOR ROAD WORK**

(a) **Coarse Aggregate:** Coarse Aggregates shall consist of crushed rock, crushed gravel or other hard materials. They shall be clean, hard, and durable, of cubical shape, free from dust and soft or friable matter, organic or other deleterious matter. The specifications for binder and granular course shall be as per MORTH specifications, Forth Revision (reprinted Apr 2005) or as per subsequent revisions from time to time.

Note:-

(i) Before sample of material for aggregates is approved, these shall be tested for stripping. Where aggregates have poor affinity for bitumen, these shall be treated with approved antistripping agents.

(ii) Where crushed gravel is proposed for use as aggregate, not less than 90% by weight of the crushed material retained on 4.75 mm sieve shall have at least two fractured faces.

(iii) The aggregates shall satisfy the physical requirements & grading requirements as indicated hereinafter & shall be procured in bulk only after obtaining sample approval from the GE.

(iv) The aggregates shall be hydrophobia & of low porosity. Sources of supply of aggregates once approved by GE shall not be changed.

(b) **Physical requirements**

The coarse aggregate shall satisfy the following physical requirements: -

(c) The fine aggregate shall consist of crushed or naturally occurring material and be fraction passing 2.36 mm sieve and retained on 75-micron sieve consisting of crushed screening, natural sand or mixture of both. It shall be clean, hard, durable uncoated and dry, free from injurious, soft or flaky pieces and organic or deleterious substance.

Test	Percentage Max		
	Water Bound Macadam	Bituminous Macadam	Semi Dense Asphaltic Concrete
Flakiness Index	250 Combined Flakiness & Elongation Index	15	15
Impact Value	30	30	27
Los Angeles Abrasion Value	35	40	35
Strippig test	-	25	25

PARTICULAR SPECIFICATION

Test	Percentage Max		
	Water Bound Macadam	Bituminous Macadam	Semi Dense Asphaltic Concrete
Water Absorption Test	-	1%	1%
Soundness Test Sodium Sulphate Magnesium Sulphate	- - -	- 12% 18%	- 12% 18%
	*Aggregate impact value may be upto 35% if bituminous macadam is used as base course.		

Srl No	Test	Test Method	Requirement
1.	Gain size analysis	IS 2386 Part 1	Max 5% passing 0.075 mm sieve
2.	Los Angeles Abrasion value*	IS :2386(Part -4)	35 Percent Maximum
3.	Aggregate impact value*	IS :2386(Part -4)	30 Percent Maximum
4.	Flakiness and Elongation** Indices	IS :2386(Part -1)	30 Percent Maximum
5.	Coating and stripping of bitumen aggregate Mixtures	IS :6241	Min retained coating 95%
6.	Soundness	IS :2386(Part -5)	
	(i) Loss with sodium sulphate	5 cycle	12 Percent Maximum
	(ii) Loss with Magnesium Sulphate	5 cycle	18 Percent Maximum
7.	Water absorption	IS :2386(Part -3)	2 Percent Maximum

(d) **FILLER**

Refer Para 20.B.2.5 of SSR Part I. The filler materials shall be lime stone dust or cement as specified in SSR if recommended in design mix.

54. **CHAIN LINK FENCING**: Chain link fence shall be made of galvanised wire and shall conform to IS 2712-2003, specification for galvanised steel wire chain link wire fabric. The mesh wire and line wire of the fabric shall be manufacture from galvanised steel wire confirming to IS 28-2006 having zinc coating as laid down in IS 4826. Tensile strength of wires shall be not less than 400 mpa. Dia of wire and mesh size shall be as indicated in respective item in the BOQ. The wire shall be free from scale, irregularities imperfection, flaws, sand splits and other defects. Zinc coating shall be smooth, even and bright. Chain link shall be fixed to the angle iron post all as directed.

55 to 60 **Blank**

PARTICULAR SPECIFICATION

61. **QUALITY CONTROL PLAN**

PART-I

1.	Contract Agreement Reference No	
2.	CPM Net work prepared and approved by GE	
3.	Resource scheduling done base on CPM	
4.	Site Laboratory (with equipments) set up as per Contract Agreement (CA)	
5.	Concrete mix design submitted and approved.	
6.	Preliminary works completed to standard engineering practice.	
7.	Arrangements for water made (including testing of water).	
8.	Arrangement for electric supply made	
9.	Materials	
10.	List of all T & P, make and numbers that the contractor would deploy at site of work.	
11.	Name of person nominated by contractor for exercising quality control.	
12.	Qualifications / Experience of person at Serial No. 11 above.	
13.	Names of supervisors with their qualifications/ experience employed by contractor.	
14.	Confirmation that contract requirements relating to quality of all materials and quality standards of workmanship and finishes and acceptance criteria are explained and understood by all.	
15.	Confirmation that requirement of tests to be conducted on materials before approval of samples and during execution, tests on workmanship, tests before acceptance including the testing procedure, sampling techniques, frequency and agencies responsible for testing are understood and shall be complied with.	
16.	Method to be adopted for maintaining records of test result.	
17.	Certificate that contractor shall maintain a log of all materials received at site as per the following format :-	
18.	General Remarks by contractor of his plan of actions to ensure that quality standards.	

(To be completed by GE before forwarding for approval by CWE)

1.	Verification of Serial No 2 to 8 of Part I.	
2.	Verification of Serial No 9 to 18 of Part I.	
3.	Confirmation that Stage Passing Register laying down the stages and authority responsible for approving the same has been prepared, shown to contractor and kept at site.	
4.	Confirmation that all sites as required by contract had been handed over to contractor on the date fixed in the Work Order No 01.	
5.	Confirmation that arrangements for Govt liability in supply of water electricity have been made and no hold up on this account is expected.	

(Signature of Contractor) Approved by CWE DY Director (Contracts)

Date (Signature of GE) for Accepting Officer

Particular Specifications

Appendix "A"

1. APPROVED CEMENT MANUFACTURERS FOR MES WKS ON PAN INDIA LEVEL

(Updated by E-in-C's branch letter No. A/90154/ 27/E2 Des-3 dated 12 Jun 2024)

Sr No	Company Name & Brand	Address	Type of Cement
1	The Associated Cement Companies Ltd. Brand: "ACC"	414-421, Splendor Forum (4th Floor), 3, District Centre, Jasola, New Delhi-110044 Ph- 011 46583600	ALL
2	M/s Ultra Tech Cement Brand: "ULTRATECH"	B Wing, 2nd Floor, Mahakali Caves Road, Andheri (East) Mumbai – 400093 Ph : 022-66917800	ALL
3	The India Cement	Dhun Building, 827, Anna salai, Chennai-600002	ALL
4	M/s Dalmia Cement (Bharat) Ltd Brand: "DALMIA INFRA PRO"	Dalmiapuram Distt- Truchirappalli Tamil Nadu – 621651	ALL
5	Century Cement Brand: "CENTURY"	Industry House, 159 Church gate Reclamation. Mumbai-400020 Ph-022 22023936	ALL
6	Saurashtra Cement Brand: "SAURASHTRA"	Gala No A-1, Ground Floor, Udhyog, Sadan No 3 MIDC, Central Road, Andheri (East), Mumbai – 400093 Ph-022 32955557/67 Mob.9320290081	ALL
7	The Ramco Cements Ltd. (Formerly Madras Cement) Brand : "RAMCO"	Aurus Corporate Centre, 98-A, Dr RadhakishnanSalai , Mylapore, Chennai-600004 Ph: 044-28478666	ALL
8	Mangalam Cement Ltd Brand: "MANGALAM"	PO Adityanagar, Morak, Dist-Kota, Rajasthan-326520 Ph-9351468076	ALL
9	Birla Corporation Ltd. Brand: "BIRLA"	Birla Building (3rd&4th floor), 9/1, R.N. Mukherjee Road Kolkata-700001 Ph-033 30573700	ALL
10	Orient Cement Brand: "ORIENT"	5-9-22/57/D, 2nd and 3rd Floor, GP Birla Centre, Adarsh Nagar, Hyderabad-500063 Pin-044 23688600	ALL
11	Nuvoco Vistas Corporation Ltd (Formerly Lafarge Cement) Brand: "NUVOCO"	Equinox Business Park Tower-3, East Wing, 4th Floor, LBS Marg ,Kurla (west), Kurla Mumbai, Maharashtra-400070	ALL

APPROVED CEMENT MANUFACTURERS FOR MES WKS ON PAN INDIA LEVEL contd….

Sr No	Company Name & Brand	Address	Type of Cement
12	Shree Cement Brand: “SHREE”	Bangur Nagar, Beawar, Disst- Ajmer, Rajasthan-305901 Ph-01462 228101-06	ALL
13	J K Cement Brand: “J K”	Kamla Tower, Kanpur-208001	ALL
14	JK Lakshmi Cement Ltd Brand: “J K LAKSHMI”	Jaykaypuram, Dist-Sirohi, Rajasthan Ph-02971 244409/10	ALL
15	JaypeeRewa Cement Brand: “J AYPEE”	Jaypee Nagar, P.O- Jaypee Nagar, Rewa-486450 M.P	ALL
16	Ambuja Cement Ltd Brand: “AMBUJA”	Kodinar, PO- Ambujanagar, Taluka- Kodinar, Disst- Junagadh, Gujarat-362715 Ph-02795 237000	ALL
17	M/s Kesoram Industries Ltd Brand “BIRLA SHAKTI”	Unit 12th 1201 of T 19 Towers, Indira Nagar Ginwala Compound, MG Road, Secundarabad Hyderabad Telangana-500003 Tel: +91-40-4334-4555, 2335672 Fax: +91-40-4334-4534 Website: www.kesocorp.co	
18	M/s Sagar Cements Ltd, Hyderabad Brand “SAGAR”	M/s Sagar Cements Ltd Plot No 111, Road No 10, Jubilee Hills, Hyderabad-500 033 Tel: +91-40-23351571, 2335672 Fax: +91-40-2335673 Website: info@sagarcements.in	(a) OPC 43 (b) OPC 53 (c) PPC (d) PSC
19	M/s Chettinad Cement Corporation Pvt Ltd., Chennai Brand: “CHETTINAD CEMENT”	4th Floor, Rani Seethai Hall Building, 603, Anna Salai Hall, Chennai-600006 Tel-No 044-42951800 Fax-No.044-28291558 E mail:- shares@chettinadcement. com Website- www.chettinadcement.com	(a) OPC 43 Gr (b) PPC
20	M/S Star Cement Meghalaya	Lumshnong, PO-Khaliehriat, Dist- Jantia Hills, Meghalaya, Ph-03655 278215/16/18	(a) OPC (b) PPC (Only for CE Shillong Zone & CE (AF) Shillong Area)
21	M/S Meghalaya Cements Ltd Brand: “ TOP CEM”	Lohia House, M. G. Road Factory Bazar, Guwahati-781001 Tel 0361-260367/671	(a) OPC (b) PPC (Only for Eastern Region)

APPROVED CEMENT MANUFACTURERS FOR MES WKS ON PAN INDIA LEVEL

contd….

Sr No	Company Name & Brand	Address	Type of Cement
22	M/s Hill Cement Company Ltd Brand “Taj”	Shivam Complex, Bharalumukh, AT Road, Guwahati, Assam-781009 Mob-7086085118 Ph-(0361) 2735527, 2735552 Email: kchakraborty@tajcement.com	(a) OPC-43 (b) OPC-53 (c) PPC (Only for Eastern Region)
23	M/s Green Valley Industries Ltd Brand”Max Cement”	4th Floor, LB Plaza G S Road, Bhangagarh Guwahati-781005 Tele +91-361-2465482/84 FAX- +91-361-2465481 Email: info@greenvalley.com Website – www.maxcement.co.in	(a) OPC-43 (b) OPC-53 (c) PPC (Only for Eastern Region)

2. APPROVED TMT STEEL MANUFACTURERS FOR MES WKS

(Updated by E-in-C's branch letter No. 83044/Steel/ 72/E2 Des-3 dated 12 Jun 2024)

Sl No.	Company Name & Brand	Address	Type of Steel
1	RashtriyaIspat Nigam Limited (RINL) Brand: " RINL"	Visakhapatnam Steel Plant Visakhapatnam – 530 031 India Tel : (91 891) 518226, 518376 Fax : (91 891) 518316 Email : cmdvsp@itpvis.ap.nic.in	For all types &dia of TMT bars
2	Tata Iron & Steel Company (TISCO or Tata Steel) Brand: "TATA"	Bombay House, 2-4 HomiModi Street Mumbai – 400 001, India Tel : (91 22) 204 9131 Fax : (91 22) 204 9522, 287 0840 Email : corpcomm@sr.tatasteel.com (Br office for North: Jeevan Tara Bldg. Patel Chowk. New Delhi)	For all types &dia of TMT bars
3	Steel Authority of India Limited (SAIL) Brand: "SAIL"	Central Marketing Organisation, Northern Region,17th Floor, scope Minar, Laxmi Nagar Distt.Centre Delhi – 110092	For all types &dia of TMT bars
4	M/s ShyamMettalics&Energry Ltd Brand: "SEL"	Vishwakarma, 1st Floor, 86 C Topsia Road, Kolkata-700046 Ph: +913340113000 Fax: +91 33 22852212 Website: www.shyammetallics.com	For all types &dia of TMT bars
5	M/s Electrotherm (INDIA) Ltd, Gujarat Brand: "ET TMT"	Survey No 72, Palodia, (Via-Thaltej, Ahemdabad) Gujarat-382115, India Tel : +91-2717-234553/660550 Fax : +91-2717-234866 Website :www.electrotherm.com	TMT Bars of Gde Fe 500 Fe 500D CRS (Size 8-36mm)
6	M/s Incredible Industries Ltd Brand: "ADHUNIK Fe 500SD"	Landsdowne Towers 2/1A, Sarat Road 6th Floor, Kolkata-700020 Tele: 033-66-38-4700 Fax- +91-33-22890285 Website: www.adhunikindustries.com	TMT Bars of Gde Fe 500D CRS (Size 8-32mm)

APPROVED TMT STEEL MANUFACTURERS FOR MES WKScontd…..

Sl No.	Company Name & Brand	Address	Type of Steel
7	M/s Gallantt Metal Ltd, Gujarat Brand: "GALLANTT TMX"	Ward 10BC, Polt No. 123 Ground Floor, Gandhi Dham, Kutch, Gujarat - 370201 Tel : 91-2836-228164 Fax : 91-2836-235787 E-mail : gml@gallantt .com Website : www.gallantt .com	TMT Bars of Gde Fe 500 Fe 500D CRS (Size 8-32mm)
8	M/s RashmiMetaliks Ltd, W.B. Brand: "RASHMI TMT"	Premlata Building, 39 Shakespeare Sarani, 6th Floor, Kolkata -700017 Tel : 033-22894255/ 56 Fax : 033-22894254 E-mail : mkt.domesticdip@rashmigroup.com Website :www.rashmigroup.com	TMT Bars of Gde Fe 500 (Size 8-32mm) Fe 500D Fe 550D (Size 8-25mm)
9	M/s Shri Bajrang Power &Istpat Ltd. Brand: " GOEL TMT "	Vill – Borjhara, Urla Industrial Area,Raipur – 493 221, Chattisgarh Tel – 0771 4288019/ 29/ 39 Fax: 0771-2323601/602, 4288123 E-mail : infobir@giekgriyo.co.in Commercial: bir@goekgroup.co.in	TMT Bars of Gde Fe 500 Fe 500D (8mm -32mm)
10	M/s Real Ispat& power Ltd, Chattisgarh Brand: "G K TMT"	Vrindavan, Near IDBI Bank Civil Lines, Raipur -492001,CG Tel : 91-771-4224000 Fax : 91-771-4224010 E-mail : real@realispat.com Website : www.realispat.com	TMT Bars of Gde Fe 500D
11	M/s Steel Exchange India Ltd. Brand: "SIMHADRI TMT"	D. No.: 1-65/K/60, Plot No. 60 AbhisHiranya, 1st Floor, KavuriHills,Hyderabad Telengana-500081 Tele: 040-23403725/23413267 Email Id: info@seil.co.in	TMT Bars of Gde Fe 500D Fe 500D HSCRM
12	M/s Super Smelters Ltd Kolkata. Brand: "SUPER SHAKTI"	Premlata, 39, Shakespeare Sarani, 3rd Floor, Kolkata -700017 Tel/ Fax : 033-2289-2734/ 36 E-mail : info@supershakti.in Website :www.supershakti.com	TMT Bars of Gde Fe 500D Fe 550 (Size 8-32mm)

APPROVED TMT STEEL MANUFACTURERS FOR MES WKScontd…..

Sl No.	Company Name & Brand	Address	Type of Steel
13	M/s Shyam Steel Industries Ltd, W.B. Brand: "SHYAM"	Shyam Towers EN-32, Sector- V, Salt Lake, Kolkatta - 700091 Tel-033-40074007, Fax 033-40074010 E-mail : marketing@shyamsteel.com	TMT Bars of Gde Fe 500D Fe 500D CRS (Size 8-32mm)
14	M/s Jai Balaji Industries Ltd, W.B. Brand: "Balaji Shakti"	5 Bentek street, Kolkata-700001 Delhi Office 510, Block-b, Navraung House 21 Kasturba Gandhi Marg. New Delhi-110001 011-43620219, 43620220 Mob: 7838272772/9958936103 E- mail: info@jaibalajigroup.com	TMT Bars of Gde Fe 500 D Fe 500D CRS (Size 8-32mm)
15	M/s MSP Steel & Power Ltd, Chattisgarh Brand: "MSP TMT 500D"	16/S, Block-A, New Alipore Kolkata 700053 Tele: 033-4005 7777/ 2398 2239 Mob: 7381028976 E- mail: contactus@mspsteel.com Website: www.mspsteel.com	TMT Bars of Gde Fe 500 D (Size 8-32mm)
16	M/s SPS Steels Rolling Mills Ltd. Brand "ELEGANT TMT"	"Diamond Prestige", 41A, A.J.C Bose Road, 8th Floor, Room No 801, Kolkata-700017 Mob: 9831055568/8910797649 Telefax: 033-66255252 Email: Info@spsgroup.co.in	TMT Bars of Gde Fe 500D

3. APPROVED STRUCTURAL STEEL MANUFACTURERS FOR MES WKS

(Updated by E-in-C's branch letter No. 83044/Steel/ 72/E2 Des-3 dated 12 Jun 2024)

Sl No.	Company Name & Brand	Address	Type of Steel
1	RashtriyaIspat Nigam Limited (RINL) Brand: " RINL"	Visakhapatnam Steel Plant Visakhapatnam – 530 031 India Tel : (91 891) 518226, 518376 Fax : (91 891) 518316 Email : cmdvsp@itpvis.ap.nic.in	Structural Steel (Angle, Beam, Column, Channel, Plate)
2	Steel Authority of India Limited (SAIL) Brand: "SAIL"	Central Marketing Organisation, Northern Region, 17th Floor, scope Minar, Laxmi Nagar Distt.Centre Delhi – 110092	Structural Steel (Angle, Beam, Column, Channel, Plate)
3	Tata Iron & Steel Company (TISCO or Tata Steel) Brand: "TATA"	Bombay House, 2-4 HomiModi Street Mumbai – 400 001, India Tel : (91 22) 204 9131 Fax : (91 22) 204 9522, 287 0840 Email : corpcomm@sr.tatasteel.com (Br office for North: Jeevan Tara Bldg. Patel Chowk. New Delhi)	Structural Steel (Angle, Beam, Column, Channel, Plate)
4	M/s Jindal Steels and Power Ltd. Brand: "JINDAL"	Jindal Centre, Plot No. 2, Sector- 32, Gurgaon – 122001 Haryana Tele : 0124 6612000 Fax : 0124 6612125 Website : www.jindalsteelpower.com	Structural Steel (Angle, Beam, Column, Channel, Plate)

4. LIST OF PRODUCTS APPROVED BY HQ CE CC

B/R RELATED PRODUCTS

Sr No	PRODUCT CATEGORY	PRODUCT	MAKES/BRANDS	Remarks
1.	*FLOOR AND WALL TILES*			
		Ceramic & Vitrified tiles	KAJARIA	
			ORIENT BELL LTD (BRAND- ORIENTBELL)	
			Johnson	
			Somany Ceramics Ltd (Brand- Somany)	
		Wooden Flooring	Greenpanel Industries Ltd	
			Action Tesa	
			Armstrong	
2.	*PAINT*			
		Exterior Paint	Asian Paints Ltd	
			Berger paints India Ltd	
			Kansai nerolac Paints Ltd	
		Interior Paint	Asian Paints Ltd	
			Berger paints India Ltd	
			Kansai nerolac Paints Ltd	
		Enamel Paint / Oil Paint	Asian Paints Ltd (Apcolite Premium Gloss)	
			Berger paints India Ltd (Luxoi Hi-Gloss)	
			Kansai nerolac Paints Ltd (Synthetic Hi-Gloss Enamel	
		Cement based Paint	Super Snowcem	
			Duracem Plus	
		High Build Textured & Wall finishes	Asian Paints Ltd	
			Berger paints India Ltd	
			Kansai nerolac Paints Ltd	
3.	*FACTORY MADE DOORS / WINDOWS*			
	Doors, Windows, Ventilators & Frames	UPVC shutters for doors windows, ventilators & Frames	FENESTA	
			KOEMMERLING	
			ALUPLAST	
			PROMINANCE	
			NCL Veka Ltd (Brand- NCL Veka	
		WPC Door, Window & Frames	RajshriPlastiwood	
		Pre-paid steel windows ventilators & door frames	Elixir Met Form Pvt. Ltd	
			Tata Pravesh	
			NCL Buildtek Limited	
			ASHISH INDUSTRIES GAZIABAD	
		Wooden flushdoor/ panel/ gauge shutters for door & window	Century Plyboard (India) Ltd. (Brand- Century Doors)	
			Greenply Industries Ltd.	
			Greenpanel Industries Ltd	
		Steel windows, ventilators, Door frames, Shutters Z Section Steel Windows (Fabricators)	ASHISH INDUSTRIES GAZIABAD	
			NCL BUILDTEK LTD	

LIST OF B/R RELATED PRODUCTS contd….

Sr No	PRODUCT CATEGORY	PRODUCT	MAKES/BRANDS	Remarks
4.	Ply /veneers, laminates/false ceilings/ particle boards/wooden boards sheet/ wooden floor etc			
		UPVC/PVC False Ceiling	RajshriPlastiwood	
		Metal false ceiling	P R Ceiling Products	
			ManrajCeiling Products Pvt Ltd (Brand-Rayal Kraft)	
			Armstrong	
		Gypsum Board False Ceiling/ Gypsom Plaster	Everest Industries Ltd	
			Gyproc	
			Armstrong	
			USG Boral	
		Non-Asbestos Fibre cement board (Confirming to IS: 14862/2000) False Ceiling	Everest Industries Ltd	
			Shera	
			Kalsi	
			Prima	
			Duragreen	
		Glass Reinforced Gypsum (GRG) False Ceiling	Gyproc	
			Armstrong	
			USG Boral	
			Green panel Industries Ltd	
			Century Ply	
			Greenply	
			Kitply	
		Aluminium Composite Panels (ACP)	Aludecor	
			Alstrong	
			Alucobond	
			Alstone	
			Viva	
			Virgo Laminates Ltd	
		Laminates/ Veneers	Merino Laminates	
			Royale Touche Laminates	
			Greenlam Laminates	
			Century Laminates	
			Kitply	
			Sunmica	
			Formica	
			Greenpanel Industries Ltd	
			Virgo Laminates Ltd.	
		Plywood for general purpose	Century Ply	
			Greenply	
			Kitply	
			Greenpanel Industries Ltd	
5.	Sanitary Fitting / Fixtures			
		CP Bath Fittings & accessories	Grohe	
			Kohler	
			Jaquar& Co Pvt Ltd (Brand – Jaquar)	
			Cerasanitaryware ltd	
		WHB/ WC/Urinals	JAQUAR & CO PVT LTD (BRAND- JAQUAR)	
			CERA SANITARYWARE LTD	
		Metropole& Concealed Flushing Cistern	JAQUAR	
			Grohe	
			Kohler	

LIST OF B/R RELATED PRODUCTS contd….

Sr No	PRODUCT CATEGORY	PRODUCT	MAKES/BRANDS	Remarks
		Seat Cover for EWC	JAQUAR	
			Grohe	
			Kohler	
			CeraSanitaryware Ltd	
			Parryware	
		Stainless Steel Sinks	Johnson	
			Nirali	
			Hindware	
5	*Pipes*			
	Internal Water Supply	CPVC PIPES & fittings	Astral Ltd Brand- CPVC Pro	
			Ashivad	
			Supreme	
			Savour Fair Manufacturing Co	
		UPVC PIPES & fittings	Astral Ltd (Brand- Aqua Safe & Aquarius)	
			Ashivad	
			Supreme	
			Savour Fair Manufacturing Co Pvt. Ltd	
		HDPE PIPE/MDPE PIPE/ PVC PIPE & fittings	Astral Ltd (Brand- HDPE Pro)	
			Finolex	
		PPR Pipes & fittings	Prince	
			Finolex	
			Supreme	
			Savour Fair Manufacturing Co Pvt. Ltd	
		ERW MS Pipe	Surya Roshni Ltd	
			TATA	
			Jindal	
		SWR Pipes as per IS-13592 for pipes & IS-14735 for fittings	Astral Ltd (Brand- Drain Master)	
			Supreme	
		PVC UG drainage pipes as per IS: 16098 Part-1	Astral Ltd (Brand- Foam core)	
			Prince	
			Finolex	
		DWC UG drainage pipes as per IS: 16098 Part-2	Astral Ltd (Brand- D-rex)	
			Prince	
			Finolex	
		PP low noise SWR Pipe	Astral Ltd (Brand- Astral Silencio)	
	EXTERNAL WATER SUPPLY			
		DI Pipes & Fittings	TATA	
			Electrosteel	
			Jindal Saw Ltd	
		GI Pipes	TATA	
			Jindal Hisar	
			Surya Roshani Ltd	

PARTICULAR SPECIFICATION (Contd….)

Part-II

E/M Products

Sr No	PRODUCT CATEGORY	PRODUCT	MAKES/BRANDS	Remarks
1.	Lifts/Elevators			
		Lift /Elevators	Otis Elevator Company India Ltd	
			Kone India	
			Schindler India Pvt Ltd	
			Hitachi Lifts India Pvt Ltd	
2.	*DG Sets*			
		Engine	Kirloskar	
			Cummins	
			Greeves cotton	
			Caterpillar	
			Eicher engines	
		Alternator	Kirloskar	
			Stemford	
			Crompton	
			Jyoti	
		DG Set assembled with sound proof canopy	Kirloskar	
			Cummins	
			Greeves Cotton	
			Sudir	
3.	*Crane*			
		(a) Single/Double Girder EOT Cranes	Safes Energy Pvt Ltd	
		(b) Goliath Cranes	Pioneer Cranes & Elevators Pvt Ltd	
		(c) Jib Cranes	Hercules Hoists Ltd	
4.	*Transformers*			
		Compact Sub station	Seimens	
			Schneider	
			Crompton	
			ABB	
			Indian Transformers & Electricals Pvt Ltd (Brand-ITE Gurgaon)	
		Transformers 11/0.433 KV upto 500 KVA copper wound	Crompton	
			Schneider	
			Indian Transformers & Electricals Pvt Ltd (Brand-ITE Gurgaon)	
		Transformers step down 11/0.433 KV above 500 KVA copper wound	Crompton	
			Schneider	
			Indian Transformers & Electricals Pvt Ltd (Brand-ITE Gurgaon)	
		Transformers step down 33/11 KV upto 10 MVA & 33/0.433 KV copper wound	Crompton	
			Schneider	
			Indian Transformers & Electricals Pvt Ltd (Brand-ITE Gurgaon)	
		Isolation Transformer	Indian Transformers & Electricals Pvt Ltd (Brand-ITE Gurgaon)	
		HT CT & PL	Schneider	
			Crompton Greeves	
			ABB	
			Siemens	

PARTICULARSPECIFICATION (Contd….)

Part-II

E/M Products

Sr No	PRODUCT CATEGORY	PRODUCT	MAKES/BRANDS	Remarks
5.	Central AC Plant and all types of Air Conditioners			
	Central AC Plants			
		Chiller for Central AC Plant	Voltas	
			Carrier	
			Bluestar/ Daikin	
		Cooling Coil/AHU	Voltas	
			Bluestar	
			Zeco	
			Accel	
			Carrier/Daikin	
		Cooling Tower	Paharpur	
			Mihir	
			Delta	
			Advance	
		Expansion Valves/ Thermostat/Humidistat	Danfoss	
			Sporlon	
			Indfoss	
			Honeywell	
	Package type AC	Compressor/ Condenser/Cooling Coil	Bluestar	
			Hitachi	
			Daikin	
			Voltas	
	Window/Split ACs	Window/Split ACs	O General	
			Mitsubishi Electric	
			Hitachi	
			Daikin	
			Voltas	
			Carrier	
6.	*HT Circuit Breakers*			
		VCB/SF6 Circuit Breakers/HT Switchgear Panels up to 36 KV	Schneider	
			Crompton	
			ABB	
			Seimens	
			Indian Transformers & Electricals Pvt Ltd (Brand-ITE Gurgaon)	
		HT Ringmain Units (RMU) up to 36 KV	Schneider	
			Crompton	
			ABB	
			Seimens	
7.	Automatic Power Factor Correction Panels			
		APFC Panel (Fabricators)	Indo Asian	
			Shalabh (India) Industries Roorkee	
			Power Systems & Control Jalandhar	
			REW Constructions Pvt Ltd	
		Power Factor Improvement Capacitor	L&T MD XL Ultra	
			Schneider	
			Seimens	
			ABB	
			Neptune	
			Havells	
		APFC	L&T	
			Schneider	
			Ducati	

PARTICULARSPECIFICATION (Contd….)

LIST OF E/M RELATED PRODUCTS contd….

Sr No	PRODUCT CATEGORY	PRODUCT	MAKES/BRANDS	Remarks
8.	*HT Cables*			
			Polycab Wires Pvt Ltd.	
			Gloster Cables Ltd	
			KEI Industries Ltd	
	XLPE/PVC LT Cables			
			V guard Industries Ltd Indo Asian Havells Finolex KEI	
9.	Switch, Socket, Fan Regulator, Lamp holder, Ceiling rose, Bell push, Call Bell			
		Modular Switch/ Socket/Fan regulator/Bell push & mounting accessories	Legrand	
			Schneider Zencelo	
			Anchor Electrical Pvt Ltd	
		Non modular Switch/Socket /Fan regulator/Lamp	SSK	
			Anchor Penta PC	
10.	*Fans*			
		Ceiling fans IS marked & not less than 4 BEE Star rating	Crompton	
			Orient	
			Khaitan	
			Usha	
			Bajaj	
		BLDC Ceiling fan	Atomberg	
			Havells	
			Bajaj	
			Crompton	
			Orient	
			Khaitan	
		Air Circulator/Wall mounted fan/Cabin fan/Fresh air fan	Havells	
			Bajaj	
			Crompton	
			Orient	
			Usha	
			Khaitan	
		Exhaust fans	Crompton	
			Orient	
			Khaitan	
			Usha	
			Bajaj	
11.	*Lights and light fittings*			
		LED light fittings for indoor/outdoor use	Crompton	
			Orient	
			Bajaj	
			Halonix	
		Solar based Led street light fittings	Philps Havells	
			Bajaj	
			Crompton	
			Halonix	
		Steel Tubular Pole	Bansal pole Calcutta pole Jindal poles Kwality steel Kolkata National Tubing Company	
13.	Voltage Stabilizer Servo Type			
		Servo Voltage Stabiliser as per BIS 9815	Indian Transformers & Electricals Pvt Ltd (Brand-ITE Gurgaon)	
14.	*PVC CONDUIT*		Prince Plaza AKG M/S Richa Cable Pvt ltd Presto plast	
15.	*PVC insulated cables ISI marked upto 1100 volts copper/alumuniumconducter sheathed/ unsheathed as per IS 694*		Indo Asian/Havells /Finolex/RR Kabel Ltd/KEI/Polycab Gloster cables Ltd M/s Larsen & Toubro Ltd	

PARTICULAR SPECIFICATION (Contd….)

16.	*MCB/MCCB/DBs*	Legrand Siemens ABB Indo Asian (NovateurElec& Digital Sys Pvt Ltd) Havells India Pvt Ltd Schneider GE C&S Electric Ltd M/s Larsen & Toubro Ltd	
17.	*LT Panel*	Adhunik Switch gears Pvt Ltd Universal switch gear control M/s Sterling Generators Pvt Ltd Stelmec Ltd PowertechSwitch Gears L&T C&S Schneider M/s IndianTransformers and Electronics Pvt Ltd or any make having CPRI approved	
18.	*CI Sluice Valve*	Kirloskar(KBL) L&T Leader	
19.	*Gate Valve*	Kirloskar L&T Leader Zoloto	
20.	*Wall decorative Light / Post Top light*	Phillips GE Wipro	
21.	*Junction Box*	Powertech switchgear Hensel Schneider C&S Electric Ltd	
22.	*Voltage Stabalizer*	AE Brentford ITE Gurgaon Vintek V guard	
23.	*Desert Cooler*	Bajaj Symphony Khaitan Crompton Greaves	
24.	*Refrigerator*	Bluestar Bosch Croma Electrolux Whirlpool LG	
25.	*Water cooler*	Bluestar Voltas Usha	
26.	*Rigid steel or Metal conduits/ERW*	Zenith Bansal Swastik BST AKG Prakash Tata Jindal Kalinga	
27.	*UPS*	Numeric Schneider Mitsubishi	
28.	*Oxygen Manifold*	Active MR MK MEDITECK/MAP	
29.	*Oxygen Control Panel with audio visual indicator*	Active MR MK MEDITECK/MAP	
30.	*Emergency Oxygen regulator*	Active MR MK MEDITECK/MAP	

PARTICULARSPECIFICATION (Contd….)

31.	*Oxygen outlet with probe*	Active MR MK MEDITECK/MAP	
32.	*DPC flow meter with humid fire bottle*	Active MR MK MEDITECK/MAP	
33.	*Nitrous oxide manifold system*	Active MR MK MEDITECK/MAP	
34.	*Fully automatic control panel for nitrous oxide*	Active MR MK MEDITECK/MAP	
35.	*Nitrous oxide outlet with probe*	Active MR MK MEDITECK/MAP	
36.	*Compressed air plant/ Vaccum Plant*	ANEST IWATA INDIA FS CURTIS INGRSOLL RAND INDIA	
37.	*Medical Air outlet with probe*	Active MR MK MEDITECK/MAP	
38.	*Vaccum meter*	Active MR MK MEDITECK/MAP	
39.	*Copper pipe medical grade*	MAX FLOW MAHTA TUBES MANDEB TUBES JAYNA	
40.	*Valve medical grade*	RB RM IBP CONNEX METRIX PAX	
41.	*Area Alarm Panel*	Active MR MK MEDITECK/MAP G-FLOW	
42.	*Medical gas supply Panel*	Schneider Siemens ABB	
43.	*Centrifugal Pump*	Crompton Kirloskar KSB Grundfos	
44.	*Expansion tank*	Emerald Xylem Precision	
45.	*Hotwater Generator*	KEPL Emerald Rapid Cool	
46.	*Hvac Panel*	Legrand ABB Schneider Siemens	
47.	*Strainer/Air Vent Valve*	Sant Hindustan Advance	
48.	*Temperature gauge/ Pressure Gauge*	General Wika TSI	
49.	*Fire and smoke Damper*	Ruskin-titus Lloyd Caryaire	
50.	*Aluminium sheet ducting*	Ruskin Titus Systemair Trox Rola Star	
51.	*Volume control damper/ Air terminal/ Exhaust Air Grill*	Ruskin Titus Systemair Caryaire Air Master	

PARTICULAR SPECIFICATION (Contd….)

52.	*Thermal Insulation*	Armaflex Bidoflex Kaimenflex	
53.	*Wall Mount Digital temperature cum humidity controller*	Honeywell HTC General	
54.	*Control Cable*	Havells Polycab Finolex	
55.	*Pan type Humidy Fire*	KEPL Emerald Rapid cool	
56.	*Starter*	L&T Legrand Schneider	
57.	*First Aid Hose Reel/ Fourway fire brigade Inlet/ Lockable hose/ Landing Valve/ Orifice Plate/ RRR Hose/ Branch pipe for fire fighting*	AAAG Coffco Minimax Newage safeguard	
58.	*Fire Alarm Control Panel/ Repeater Panel*	Esser JC Notifier	
59.	*Multi sensor detector*	Honeywell Morley IAS Ravel	
60.	*Manual Pulll station/ Addressible control module/ remote power supply/ ceiling speaker/ digital audio amplifier/ addressable telephone jack/ telephone hand set*	Esser JC Notifier Hornstrobe	
61.	*Fire Extinguisher*	AimEx Coffco Kanex Minimax Safeguard Variex	
62.	*Electrical UPS power distribution board/ peripheral light/ operation theatre touch screen central panel/ two plate X ray view box*	MK Mediteek Mediline Digiline MR Engi	
63.	*Single Arm Pendant*	Active MR MK Mediteek Mep Medirail	

Particular Specifications
LIST OF DRAWINGS

SL No	Description	Ref Drawing No	Sht No	Date of Drg	Revision up-to
1	2	3	4	5	6
	SITE PLAN				
1.	EXTERNAL B/R SERVICES	CEJZ/SP/212	1/2	10.07.2024	-
2.	EXTERNAL E/M SERVICE	CEJZ/SP/212	2/2	10.07.2024	-
	ARCHITECTURAL DRAWING				
	MODULAR OT, CSSD & ICU				
3.	GROUND FLOOR PLAN SECTIONS	CEJZ/DAN/99	1R/40	01.10.2024	-
4.	FIRST FLOOR PLAN & SECTIONS	CEJZ/DAN/99	2R/40	01.10.2024	-
5.	ROOF AND MUMTY PLAN	CEJZ/DAN/99	3R/40	01.10.2024	-
6.	ELEVATION AND SECTIONS	CEJZ/DAN/99	4R/40	01.10.2024	-
7.	DETAILS OF STAIRCASE-1	CEJZ/DAN/99	5R/40	10.07.2024	-
8.	DETAILS OF STAIRCASE-1	CEJZ/DAN/99	6R/40	10.07.2024	-
9.	MISC & DETAILS OF TOILET	CEJZ/DAN/99	7R/40	01.10.2024	-
10.	DETAILS OF DUCT, DUMB WAITER, NURSE DUTY ROOM, RECEPTION, PANTRY, SHELVES	CEJZ/DAN/99	8R/40	01.10.2024	-
11.	DETAILS OF FIXED GLASS WINDOW	CEJZ/DAN/99	9R/40	01.10.2024	-
12.	DETAILS OF RCC FIN	CEJZ/DAN/99	10R/40	01.10.2024	-
13.	MAJOR OT ROOM DETAILS	CEJZ/DAN/99	11R/40	01.10.2024	-
14.	MAJOR OT ROOM DETAILS	CEJZ/DAN/99	12R/40	01.10.2024	-
15.	OT ROOM DOOR DETAILS	CEJZ/DAN/99	13R/40	01.10.2024	-
16.	SCHEDULE OF FINISHES	CEJZ/DAN/99	14(R)/ 40	23.09.2024	-
17.	SCHEDULE OF FINISHES	CEJZ/DAN/99	14A(R)/40	23.09.2024	-
	E/M DRAWINGS				
18.	INTERNAL ELECTRIFICATION-GROUND FLOOR	CEJZ/DAN/99	15/40	10.07.2024	-
19.	INTERNAL ELECTRIFICATION- FIRST FLOOR	CEJZ/DAN/99	16/40	10.07.2024	-
20.	SINGLE LINE DIAGRAM OF ELECTRICAL POWER DISTRIBUTION	CEJZ/DAN/99	17/40	10.07.2024	-
21.	SINGLE LINE DIAGRAM OF ELECTRICAL POWER DISTRIBUTION	CEJZ/DAN/99	18/40	10.07.2024	-

Particular Specifications

LIST OF DRAWINGS

SL No	Description	Ref Drawing No	Sht No	Date of Drg	Revision up-to
1	2	3	4	5	6
22.	SINGLE LINE DIAGRAM OF ELECTRICAL POWER DISTRIBUTION	CEJZ/DAN/99	19/40	10.07.2024	-
23.	SINGLE LINE DIAGRAM OF ELECTRICAL POWER DISTRIBUTION	CEJZ/DAN/99	20/40	10.07.2024	-
24.	SINGLE LINE DIAGRAM OF ELECTRICAL POWER DISTRIBUTION	CEJZ/DAN/99	21/40	10.07.2024	-
25.	SINGLE LINE DIAGRAM OF ELECTRICAL POWER DISTRIBUTION	CEJZ/DAN/99	22/40	10.07.2024	-
26.	FENERAL NOTES OF FIRE FIGHTING SYSTEM	CEJZ/DAN/99	23/40	10.07.2024	-
27.	TYPICAL INSTALLATION DETAIL FOR FIRE PROTECTION SYSTEM	CEJZ/DAN/99	24/40	10.07.2024	-
28.	TYPICAL INSTALLATION DETAIL FOR FIRE PROTECTION SYSTEM	CEJZ/DAN/99	25/40	10.07.2024	-
29.	AUTOMATIC FIRE ALARM AND DETECTION SYSTEM- GROUND FLOOR	CEJZ/DAN/99	26/40	10.07.2024	-
30.	AUTOMATIC FIRE ALARM AND DETECTION SYSTEM- FIRST FLOOR	CEJZ/DAN/99	27/40	10.07.2024	-
31.	SCHEMATIC PIPING LAYOUT OF FIRE FIGHTING SYSTEM	CEJZ/DAN/99	28/40	10.07.2024	-
32.	SITE PLAN (EXTERNAL FIRE HYDRANT LAYOUT)	CEJZ/DAN/99	29/40	10.07.2024	-
33.	GENERAL NOTES OF HVAC WORK	CEJZ/DAN/99	30/40	10.07.2024	-
34.	TYPICAL HVAC FIXING & SUPPORTING DETAILS	CEJZ/DAN/99	31/40	10.07.2024	-
35.	TYPICAL HVAC FIXING & SUPPORTING DETAILS	CEJZ/DAN/99	32/40	10.07.2024	-
36.	HVAC LAYOUT- GROUND FLOOR	CEJZ/DAN/99	33/40	10.07.2024	-
37.	HVAC SUPPLY AIR LAYOUT- FIRST FLOOR	CEJZ/DAN/99	34/40	10.07.2024	-
38.	HVAC SUPPLY AIR LAYOUT- FIRST FLOOR	CEJZ/DAN/99	35/40	10.07.2024	-
39.	HVAC LAYOUT- ROOF FLOOR	CEJZ/DAN/99	36/40	10.07.2024	-
40.	SCHEMATIC LAYOUT OF HVAC PIPING	CEJZ/DAN/99	37/40	10.07.2024	-

Particular Specifications
LIST OF DRAWINGS

SL No	Description	Ref Drawing No	Sht No	Date of Drg	Revision up-to
1	2	3	4	5	6
41.	MEDICAL GAS SUPPLY LAYOUT-FIRST FLOOR	CEJZ/DAN/99	38/40	10.07.2024	-
42.	MEDICAL GAS SUPPLY LAYOUT- GROUND FLOOR	CEJZ/DAN/99	39/40	10.07.2024	-
43.	MODULAR OT	CEJZ/DAN/99	40/40	10.07.2024	-
	STRUCTURAL DRAWING				
	OFFICE BUILDING				
44.	LAYOUT OF FOUNDATION AND COLOUMN	CEJZ/DAN/ST/99	1/21	10.07.2024	-
45.	RCC DETAILS OF FOUNDATION & COLOUMN SCHEDULE	CEJZ/DAN/ST/99	2/21	10.07.2024	-
46.	RCC DETAILS OF COMBINED FOOTINGS	CEJZ/DAN/ST/99	3/21	10.07.2024	-
47.	RCC DETAILS OF COMBINED FOOTINGS	CEJZ/DAN/ST/99	4/21	10.07.2024	-
48.	RCC DETAILS OF COMBINED FOOTINGS	CEJZ/DAN/ST/99	5/21	10.07.2024	-
49.	RCC DETAILS OF COMBINED FOOTINGS	CEJZ/DAN/ST/99	6/21	10.07.2024	-
50.	RCC DETAILS OF COMBINED FOOTINGS	CEJZ/DAN/ST/99	7/21	10.07.2024	-
51.	RCC DETAILS OF COMBINED FOOTINGS	CEJZ/DAN/ST/99	8/21	10.07.2024	-
52.	LAYOUT AND RCC DETAILS OF PLINTH BEAM & PLINTH LEVEL SLAB & SCHEDULE	CEJZ/DAN/ST/99	9/21	10.07.2024	-
53.	PLINTH BEAM SCHEDULE & MISC RCC DETAILS	CEJZ/DAN/ST/99	10/21	10.07.2024	-
54.	LAYOUT AND RCC DETAILS OF GROUND FLOOR TIE BEAM & SCHEDULE	CEJZ/DAN/ST/99	11/21	10.07.2024	-
55.	LAYOUT AND RCC DETAILS OF FIRST FLOOR SLAB, BEAM & SCHEDULE	CEJZ/DAN/ST/99	12/21	10.07.2024	-
56.	FIRST FLOOR BEAM SCHEDULE, SECTION DETAIL & EXPANSION JOINT DETAIL	CEJZ/DAN/ST/99	13/21	10.07.2024	-

Notes :

1. Contractor to check and verify all dimensions before execution of the work.
2. Figured dimensions shall be followed.
3. All dimensions are given in MM.
4. For respective details refer TD drawings.
5. PCC Cill shall be provided below all Windows/Vents/Jalies, if not shown on drg.
6. Details/Sections shown for one side of symmetrical type of building/portion shall also be applicable for other side with details/sections as mirror image of the same.
7. The details shown in any drawings shall be deemed to be included in the scope of work; even in the Same/connected/related details are not marked in other related drawings.

Signature Of Contractor

Dated: _____

Dy Dir (Contracts)

FOR ACCEPTING OFFICER